



CAPSTONE PROJECT

SOFTWARE DEVELOPMENT

SOCIABLE

Industry: **INFORMATION TECHNOLOGY**

Specialized: **SOFTWARE PROGRAMMING**

Instructor guides: Lê Thị Cẩm Thi

Students perform

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Class: K23WD-02

VTC ACADEMY OF INFORMATION TECHNOLOGY & DESIGN



INSTRUCTOR'S COMMENTS

GRADUATION THESIS

(Supervisor submits this Review to the Department)

1. Full name of student assigned the topic

Students:

Nguyễn Minh Tấn

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Class: K23WD-02

Industry: Information technology

2. Topic name

3. Overview of KLTN:

Page number: Chapter number:

Accompanying physical objects (products):

4. Comments

a) *About the spirit and working attitude of students*

.....

.....

.....

.....

b) *Results achieved by the topic*

.....

.....

.....

.....

c) Limitations of the topic

.....

.....

.....

.....

5. **Suggestion** ☐ Protected ☐ 5. Not protected

Ho Chi Minh City, date october month 31 year 2025

Instructor guides
(Sign and print full name)

[illegible]

VTC ACADEMY OF INFORMATION TECHNOLOGY & DESIGN



REGISTRATION FORM

CAPSTON PROJECT

Courses: K23WD-02

1. Full name of student/student group registering for the topic (group size: 3):

- | | | |
|-----------------------|--------------------|-----------------|
| (1) Nguyễn Thái Hào | MSSV: 124010123077 | Class: K23WD-02 |
| (2) Nguyễn Minh Tấn | MSSV: 124010123004 | Class: K23WD-02 |
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Industry : Information technology

Specialized : Software Engineering

Name of registered topic:

Sociable Application Chat Real-Time and build community.

Students have clearly understood the requirements of the topic and are committed to implementing the topic on schedule and completing it on time.

Ho Chi Minh City, date october month 31 year 2025

Instructor

(Sign and print full name)

Student registration

(Sign and print full name)

Nguyễn Minh Tấn

VTC ACADEMY OF INFORMATION TECHNOLOGY & DESIGN



PROGRESS TRACKING FORM CAPSTON PROJECT

Topic name: Sociable Application Chat Real-Time and build community

2. **Instructor guides:** Lê Thị Cẩm Thi

3. **Students/groups of students carry out the project:**

(1) Nguyễn Thái Hào	MSSV: 124010123077	Class: K23WD-02
(2) Nguyễn Minh Tấn	MSSV: 124010123004	Class: K23WD-02
(3) Nguyễn Trung Kiên	MSSV: 124010122070	Class: K23WD-02

Industry : Information technology

Specialized : Software Engineering

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
1	21/06/2025 Arrive 22/06/2025	Overview Description, Data Flow Diagram, Usecase Diagram (All Member) Software Architecture Design (Nguyễn Minh Tấn)	

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
2	24/06/2025 Arrive 26/06/2025	Sequence Diagram, Activity Diagram (Nguyễn TRung Kiên) Application UI/UX Design, Application Logo - Icon Design (Nguyễn Thái Hào) Data Export Sample (NoSQL), Usecase Diagram Update (NGuyễn Minh Tấn)	
3	25/06/2025 Arrive 27/06/2025	Class Diagram (Nguyễn Thái Hào)	

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
4	29/06/2025 Arrive 30/06/2025	Flutter Project Initialization, Firebase Integration Setup (Auth, Firestore, storage,...), Project Cleanup & Git Hygiene, Git Workflow Definition, Branch Structure Implementation (main, dev, feature/*), Repository Access Control, Initial Version Commit & Push (v1.0.0), Documentation: README & Getting Started (Nguyễn Minh tấn) Commit Message Convention (feat, fix,...), Git Workflow Training, Environment Setup (Flutter + Firebase), (All Member)	

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
5	01/07/2025 Arrive 03/07/2025	Register with Email, Login with Email, Forgot Password, Verify With Email, (Nguyễn Trung Kiên) Register with Phone (OTP), Login with Phone (OTP), Phone Number Formatting (+84) VN, Firebase Firestore (User Data), Username, DateOfBirth (NGuyễn Minh Tấn) Dark Theme UI, Splash Screen, Method Selector (Email/Phone), Privacy Policy (Nguyễn Thái Hào)	

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
6	07/07/2025 Arrive 03/08/2025	Create Server, Edit Server, Delete Server (30 days), Transfer ownership, Create channel (text/voice), Edit channel, Delete channel, Assign role (owner, moderator, member), Role permissions logic, Create category, Sidebar server, Server discovery toggle (public/private), Search server, Server tag + category filter, Display member in server, QR Join Server, Server info page, Server notification settings, (Nguyễn Trung Kiên)	

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
7	07/07/2025 Arrive 03/08/2025	Chat text basic, chat with friend, chat with channel, Real-time stream message, Send image (image picker + upload), Video upload, Send emoji + reaction, Send file, Voice chat Message reply, Edit message, Delete message, UI/UX Chat Form, Channel join/leave logic, Typing UX, Unread message tracking, Notification badge, call with friend, Create voice channel, Join voice call channel (Hight), UI voice/video, Status Online/Offline, (Nguyễn Minh Tấn)	

Session	Day	Content	Instructor's Comments (Sign)
8	07/07/2025 Arrive 03/08/2025	show other people's profiles, Show my profile, Edit Avatar,Edit Info bio - Email, Change Password, List Friends,, personalize profile, Status Online/Offline, Toggle Premium profile (switch), Turn channel notifications on or off, Personal settings interface, Notification setting per server, QR add friend, Devices login, Delete Account , Recovery Email (Nguyễn Thái Hào)	
9	05/08/2025 Arrive 20/08/2025	Premium Feature Development, DefinePremiumProductModel, BuildPremiumPackageScreen, ImplementAvatarFrameSelector, CreateProfileBannerPicker, DevelopProfileIndustryStyler, DisplayOwnedPremiumFeatures, (Nguyễn Thái Hào)	

Session	Day	Content	Instructor's Comments <i>(Sign)</i>
10	05/08/2025 Arrive 20/08/2025	Payment Flow Integration, DefinePurchaseModel, BuildPurchaseOptionScreen, ntegrateMockMomoPaymentGateway, DisplayOrderStatusScreen, SavePurchaseToFirestore, ShowPurchaseHistoryScreen, ImplementMockRefundFunction, AddServerBannerForPremium, ImplementServerMemberLimitUpgrade, (Nguyễn Trung Kiên)	
11	05/08/2025 Arrive 20/08/2025	PolishAllCoreUIs, UnifyAppTheme, ImproveScreenTransitions, AuditComponentSpacing, OptimizeTouchTargets, FixInconsistentUXPatterns, (Nguyễn Minh Tấn)	

Ho Chi Minh City, date october month 31 year 2025

Instructor guides

(Sign and print full name)

ACKNOWLEDGEMENTS

First of all, we would like to express our sincere gratitude to Ms. Cẩm Thi and all the teachers during our time studying at the school. With the dedication to teaching, imparting knowledge and valuable guidance from the teachers, we have had the opportunity to approach, practice and improve many important skills, not only in the professional field but also in thinking, working methods and learning spirit. This is an extremely valuable asset for us to complete the project as well as be more confident in our future career path.

We would like to express our special thanks to the instructor, who has always closely followed, enthusiastically instructed, commented and guided the group throughout the implementation process. The comments and dedicated guidance of the instructor helped the group realize its shortcomings, and at the same time learn more knowledge and practical experience to perfect the best product...

Besides, we would also like to express our deep gratitude to all members of the group. During the working process, everyone has tried their best, shared work, supported each other and overcome difficulties together. It is the spirit of solidarity, cooperation and responsibility of each member that helped our group complete the project on schedule and achieve the expected results.

Although the group has tried its best, the report certainly cannot avoid shortcomings. We look forward to receiving comments, suggestions and guidance from teachers to continue to improve in the future...

Once again, we thank you very much!

Nguyễn Thái Hào

Nguyễn Minh Tấn

Nguyễn Trung Kiên

COMMITMENT

We hereby commit that the project report with the topic "Building Sociable software" is the result of a serious research, study and implementation process of the entire group, under the dedicated guidance of lecturer Cẩm Thi.

During the project, we have made efforts to research and apply the knowledge we have learned along with consulting related documents, data, and research works. All content in the report is done by the group itself, not copied from any source. References and information sources used are cited and clearly stated in the References section to ensure transparency and scientific.

We are fully aware that academic integrity and scientific research are extremely important. Therefore, if any plagiarism, fraud, or copyright infringement is discovered in this report, we will take full responsibility before the instructor and the school.

This report is the result of the cooperation, efforts and high sense of responsibility of each member of the group. We believe that the project is not only an academic exercise, but also a valuable opportunity to practice research skills, teamwork, and practical application.

We respectfully commit and send our sincere thanks to the instructors as well as teachers who have accompanied and created conditions for the group to complete the report in the best way.

Nguyễn Thái Hào

Nguyễn Minh Tấn

Nguyễn Trung Kiên

PROJECT OVERVIEW

TOPIC NAME: Sociable Application Chat Real-Time and build community	
Instructor guides: Lê Thị Cẩm Thi	
Implementation time: 31/10/2025	
Students perform: Nguyễn Minh Tấn - Nguyễn Thái Hào - Nguyễn Trung Kiên	
Topic content: Mobile Application Chat Real Time & build community	
Target	
<i>User-friendly functions</i>	
Object scope	
Implementation	
method	
Expected results	
Theoretical content	
Application	
Confirmation of Supervisor (Sign and print full name)	Ho Chi Minh City, date october month 31 year 2025 Students Nguyễn Minh Tấn Nguyễn Thái Hào Nguyễn Trung Kiên

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PROJECT SUMMARY

The topic "**Building Sociable software**" includes 4 chapters:

Chapter 1: OVERVIEW OF THE TOPIC

The first part will introduce an overview of the research topic, the methods and objects of research as well as the supporting tools to build the system..

Chapter 2: SYSTEM DESIGN ANALYSIS

This section focuses on system analysis and design, including contents such as analysis of functional diagrams, context diagrams, data flow diagrams, etc.

Chapter 3: SYSTEM IMPLEMENTATION AND DEPLOYMENT

This section presents tables, databases and implements the functions in chapter 2 through images from the built software system.

Chapter 4: CONCLUSION AND DEVELOPMENT DIRECTIONS

The final section provides conclusions, an overview of the system, limitations and future development directions..

CHAPTER 1: OVERVIEW OF THE TOPIC

1.1. Ask the question

As technology continues to evolve, the need for online connection and interaction is constantly increasing, especially through chat and community-building platforms. However, while current applications are powerful, they still present some barriers for new users, such as complex interfaces, lack of initial orientation, and lack of control over appropriate content in open communities.

In addition, many platforms are not optimized for a smooth user experience, making it difficult to build an accessible, flexible, and friendly community. Therefore, a solution that focuses on user experience is needed.

The goal of the project is to develop a real-time chat application with strong capabilities in user role delegation (Owner, moderator, member), supporting flexible community organization, ensuring security factors, being friendly to new users, and ensuring appropriate content control.

1.2. Solution

Sociable is designed with outstanding specifications and features:

- Clear interface, smooth experience (Material Design)
- Real-time messaging feature
- Send pictures, videos, files
- Personal profile management
- Optimize interface personalization
- Flexible authorization system: Owner, Moderator, Member
- Topic-based community recommendation system for new users
- Bot automatically sends welcome messages, community rules when joining
- Upload content warning and moderation (planned enhancement)

The application will be deployed using Flutter, Google's cross-platform framework, which allows for the development of a single codebase that runs smoothly across:

Mobile app: iOS, Android

Website: Google Chrome, Safari, Firefox

Desktop: MacOS, Windows (expected to increase)

1.3. System architecture & development tools

1.3.1. Frontend

- **Language:** Dart
- **Framework:** Flutter
- **UI Framework:** Material Design
- **Personalization:** Dark/Light Mode, preferred theme
- **Responsive UI:** Suitable for Mobile, Web
- **Responsive Optimization:** MediaQuery, LayoutBuilder, flutter_screenutil

- **Plugins supporting Web/Mobile:** image_picker, firebase_messaging, flutter_local_notifications

1.3.2. Backend

- **Firebase Firestore:** store and sync messages in real time
- **Firebase Authentication:** Sign up/Login/Authenticate users by email, phone number, Google
- **Firebase Cloud Storage:** store images, videos, documents
- **Firebase Cloud Functions:** advanced event handling (moderation)
- **Firebase Hosting:** deploy website

1.3.3. Development support tools

- **Git & GitHub:** Source code management, branching
- **VS Code:** Main development environment
- **Android Studio** (Emulator – Android)
- **Xcode** (Simulator – iOS) • **Postman:** API testing

1.3.4. Equipment and testing

- **MacBook M2:** build/debug iOS, macOS 26
- **Iphone:** test real device iOS 26
- **Android phone:** test real device Android 13+

1.4. Main functional requirements

- Register/Login/Authentication: Email, phone number
- Manage personal profile: avatar, name, status, cover photo, bio
- Search for other users and make friends.
- Send and receive messages with friends in real time
- Create or search to join “servers” (servers)
- Each server has many categories, each category has many chat channels
- User permissions in the server: owner, moderator, member
- Send and receive text messages in real time chat channels
- Send and display photos, videos, files.
- Voice – video call with friend – channel.

- Suggest servers by topic for new users
- Bot automatically welcomes and sends rules when new members enter the server
- Account management: change password, forgot password, log out, delete account

1.5. Non-functional requirements

- Low data synchronization speed under 1s (quasi real-time)
- Security: user authentication, access authorization
- Ensure smooth UX on all platforms
- Extend for advanced features such as voice call
- Easy to maintain with component architecture
- Responsive UI: suitable for each screen from Mobile → Website – Desktop

1.6. Interface requirements (UI/UX)

- Uniform, modern, intuitive design
- Interface suitable for each platform (mobile/web/desktop)
- High contrast in dark/light mode, theme colors
- Avatar, name, sending time clearly displayed in chat room
- Easy navigation between server/channel/Profile
- UX is easy to access and familiarize

1.7. Hardware & Software Requirements

1.7.1. Client – User device

Android devices:

- Android OS 8.0 or higher
- Minimum 3GB RAM
- Stable internet connection (Wi-Fi or 4G)

iOS devices:

- iOS 12 or later
- Minimum 3GB RAM
- Stable internet connection

Web (browser):

- Computer with 4GB RAM or more
- Google Chrome, Safari, or Firefox browser latest version
- Stable internet connection

Desktop (Expected to be advanced):

- Operating System: Windows 10+ or macOS 11+
- CPU Intel i5 or higher
- RAM 8GB or higher • Internet connection

1.7.2. Client

- Latest updated browser (for Web)
- Allow access to photos, files, camera (for mobile)
- Allow push notifications (Firebase Cloud Messaging – mobile/web) • No heavy installation required, just need internet connection to use

1.7.3. Developer – Hardware

MacBook (Apple Silicon – M1/M2):

Used to build and debug iOS, macOS

Install Xcode 15 or higher

RAM from 8GB or higher

Windows (or Linux) Laptop:

Used to build Android, Web

Windows 10+ or Ubuntu 22.04 operating system

Minimum RAM 8GB

CPU from Intel i5 or higher

Actual test equipment:

iPhone with iOS 14 or later

Android phones with Android 8.0 or later

1.7.4. Developer – Software

Flutter SDK $\geq 3.22.0$

Dart SDK ≥ 3.2

Android Studio + Flutter plugin

Xcode 15+ (for iOS/macOS)

Visual Studio Code

Git + GitHub CLI

Firebase Console (backend management)

Figma (UI design)

Discord (team management, documentation, ideas)

Postman (API testing)

1.8. Report layout

In addition to the introduction and conclusion, the content of the thesis is structured into 4 chapters as follows::

Chapter 1: OVERVIEW OF THE TOPIC

Chapter 2: APPLICATION DESIGN ANALYSIS

Chapter 3: SYSTEM IMPLEMENTATION AND DEPLOYMENT

Chapter 4: CONCLUSION AND DEVELOPMENT

DIRECTIONS

CHAPTER 2: SYSTEM DESIGN ANALYSIS

2.1. System requirements analysis

2.1.1. Functionality requirements

STT	Function name	Describe
1	Sign In with Email	Users log in to their Email account in the App.
2	Sign In with Phone	Users log in to their Phone Number account in the App.
3	Sign Up with email	Allow users to register new accounts using email
4	Verify Email	Authenticate user account by sending verification to email
5	Forgot password	Send a new password verification email when the user submits a forgot password request
6	Change password	Allows users to change their password by entering their old and new passwords.

7	Sign Up with phone	Allow users to register new accounts using phone number
8	Verify Phone	Authenticate user account by sending verification to phone
9	Forgot password	Function to help users reset password when they forget password login information by sending verification to email
10	Change password	Function allows users to change to a new password while still logged in
11	User profile	Show your friends' profiles
12	Make friends & mesage	The function allows users to make friends and chat directly with each other in the system.
13	Create server (Owner)	Function allows users to create new servers, become owners to manage members, channels and settings of that server
14	Update server	The function allows the owner or administrator to edit server information such as photos, descriptions, or membership settings.
15	Delete server	The function allows the owner to completely delete the server along with all data, channels and members in it.
16	Server & room manament	Function to manage the entire server and chat rooms inside, including creating, editing, deleting and assigning permissions to each room or member.

17	Rights and roles	The ability to assign roles and permissions to room members, defining who can send messages, manage, or edit room settings
18	Customize the interface	The feature allows owners to customize the server interface such as colors, icons, channel layouts and display styles to create their own identity.
19	Server boost	The feature allows users to upgrade their servers to increase performance, unlock additional premium features like sound quality, animated icons, and larger upload capacity.
20	Premium	Paid upgrades give users more perks like interface customization, increased upload limits, and access to premium in-app features.
21	Notifications and search	The function helps users receive notifications about new activities and quickly search for friends or servers in the system.
22	Bot	The function allows adding automatic programs to the server that will automatically greet
23	Payment frames	Allows users to select and pay for frames in the store, then the purchased frame is downloaded to the user's profile and the transaction is recorded in the database.
24	Message	The function of sending or receiving messages between users or systems.
25	Channel message (server)	The message function is sent in a specific channel of the server.

26	Voice & call (with friends)	voice or video call function directly with friends.
27	Voice & call (channel server)	voice or video calling functionality within a channel or server.

2.1 Functional requirements description table

2.1.2. Non-functional requirements

2.1.2.1. System requirements

- Performance requirements: Fastest possible response time.
- Requirements for database logic: Data must be guaranteed to be complete and consistent..
- Design constraints:
 - The system is built on closed source code..
 - The database management system is SQL..
 - Analysis and design are performed according to UML standards..
 - Non-copyrighted support tools, other support libraries must be is open source.
 - The system is designed to be able to develop in the future by adding or removing modules, or integrating the system into another system easily.

2.1.2.2. Interface requirements

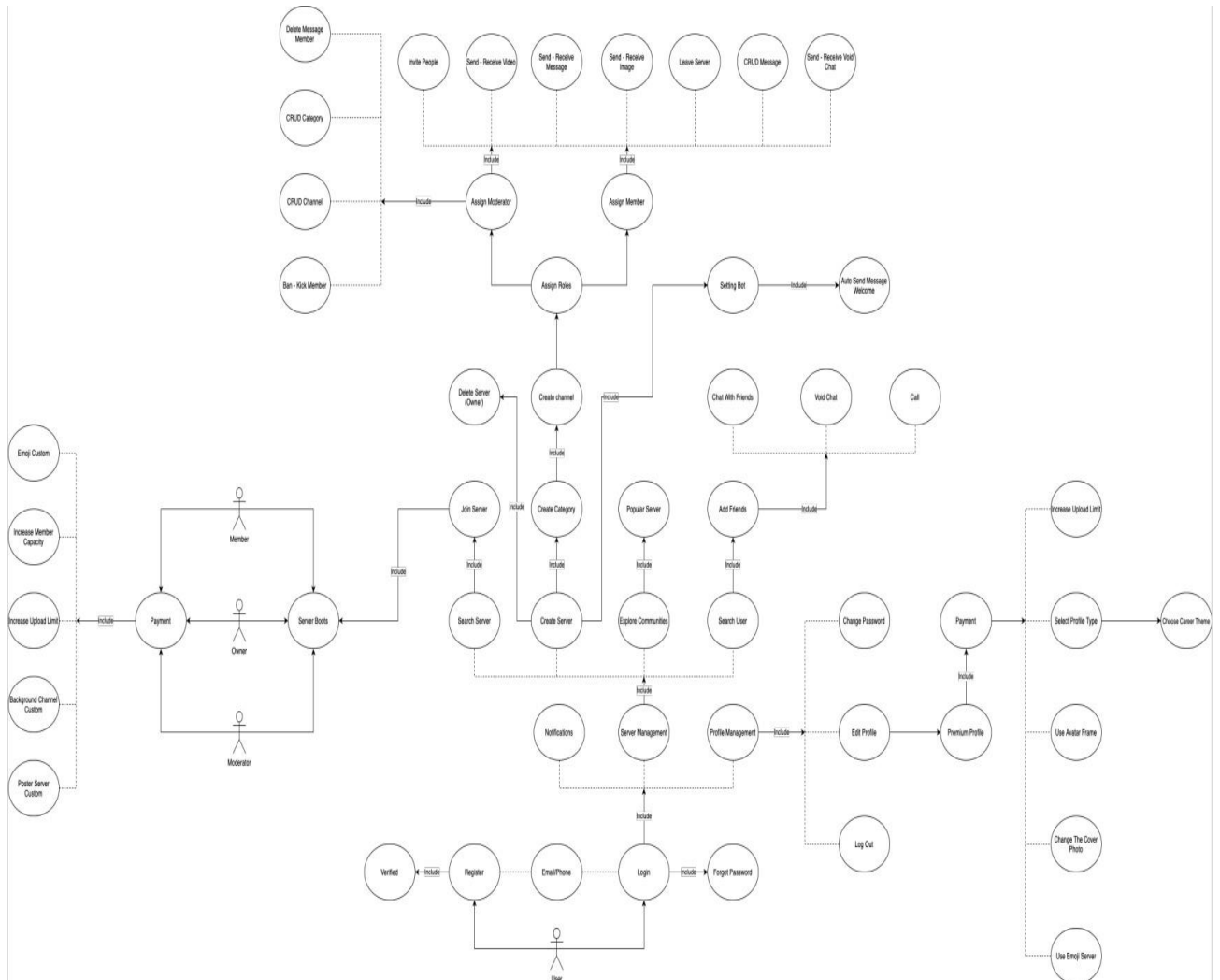
- Font: Tahoma 10
- Form Size: Always try to ensure a 16:9, 19.5:9 ratio to ensure symmetry with the screen.
- The system meets the requirements:
 - Professional, aesthetic interface
 - Unicode font 6909
 - Minimum screen resolution: 360 x 640 pixels (lowest standard for current Android/iOS smartphones),Multi-screen compatibility: The interface must be responsive and display well on different screen sizes (small smartphones, large screens, tablets),Language:The entire system uses English as the default language

○ The date format used in the system is: DD/MM/YYYY ○

The number format used in the system is: 000.000.000

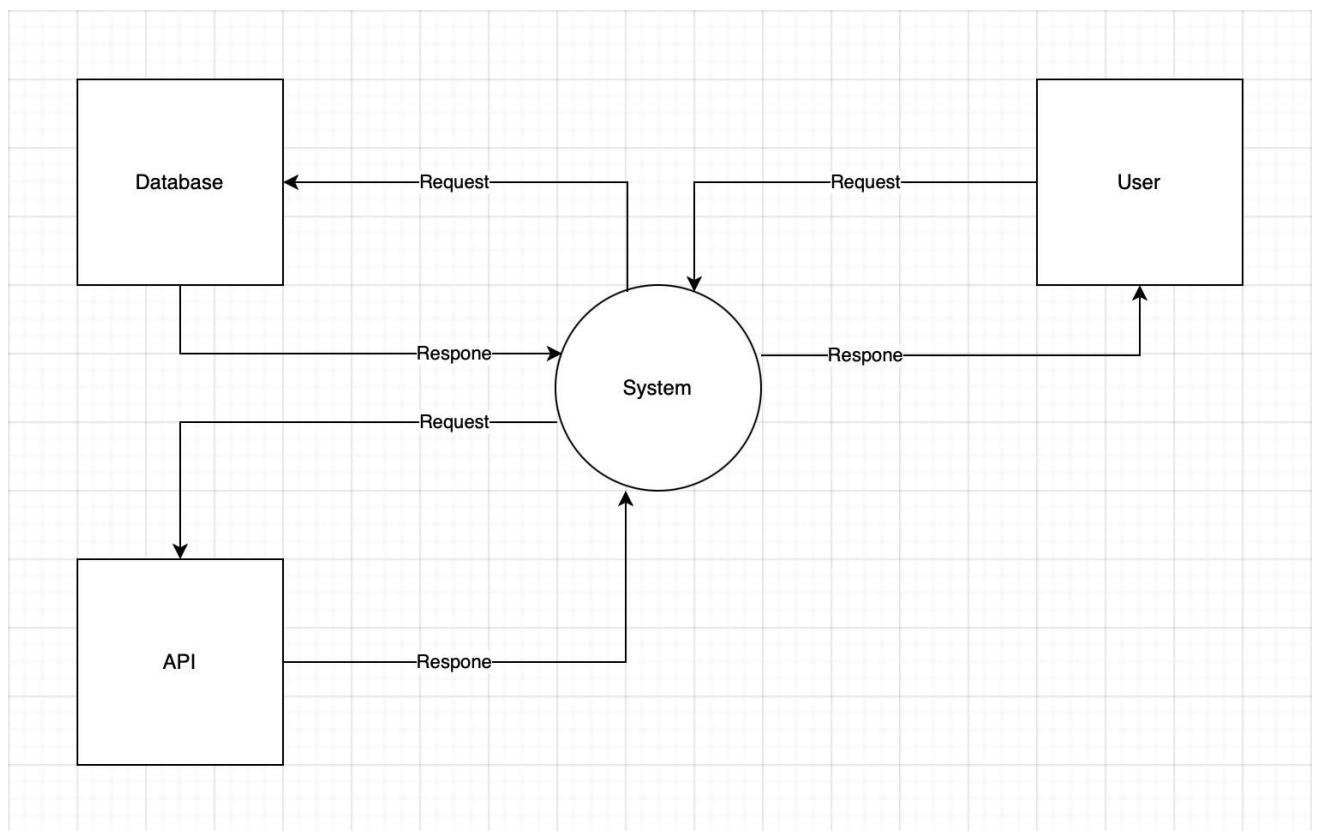
2.2. Build a usecase diagram 2.2.1.

Main usecase diagram

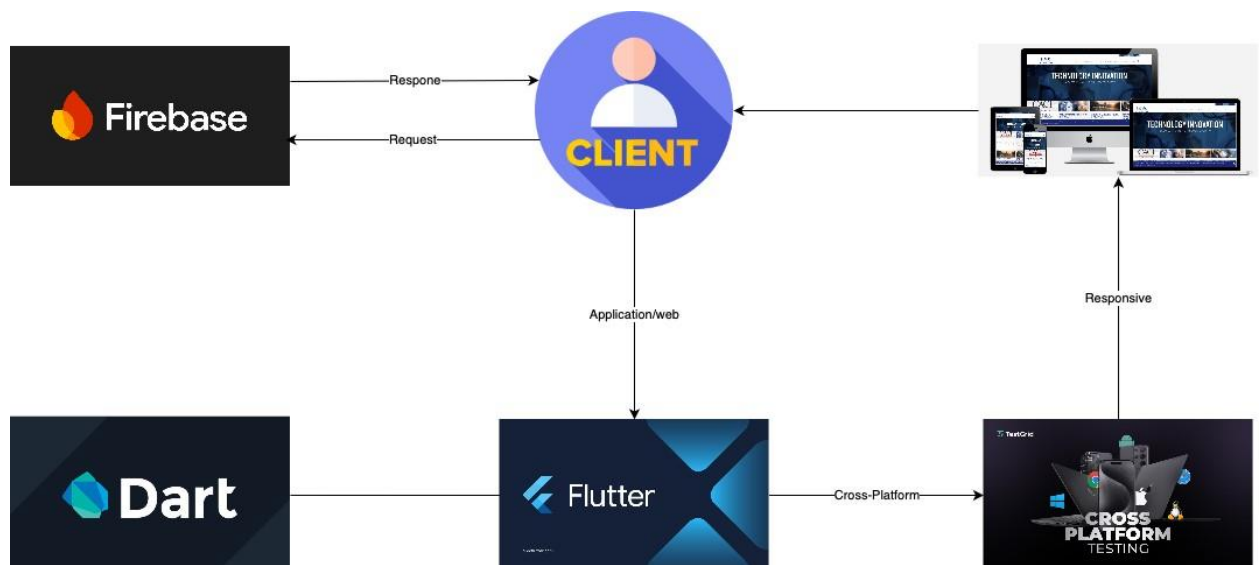


Hình 2-1 Main usecase diagram

2.2.2. Data Flow Diagram



2.2.3. Software Architecture Design



2.2.4. Sign In with Email

ID and Name:	Sign In with Email		
Main actors:	Client		
Describe:	Users log in with their Email account via Firebase Auth to access access resources and data.		
	User is not logged in		
	Success: User successfully registered account Failure: The system displays an error message.		
Main stream	Success: User is authenticated and redirected to the home page with resource access Failure: The system displays an error message.		

2.2.5. Sign by phone number

ID and Name:	Sign by phone number		
Main actors:	Client		
Describe:	Users log in with their phone number account via Firebase Auth to access access resources and data.		
	User is not logged in		
	Success: User successfully registered account Failure: The system displays an error message.		
Main stream	Success: User is authenticated and redirected to the home page with resource access Failure: The system displays an error message.		

2.2.6. Sign up with email

ID and Name:	Sign up with Email		
Main actors:	Client		
Describe:	New user, no account yet, register for an account by Email		
	User has not registered an account		
	Success: User successfully registered account Failure: The system displays an error message.		
Main stream	Success: User is authenticated and redirected to the home page with resource access Failure: The system displays an error message.		

2.2.7. Sign up by phone number

ID and Name:	Sign up by phone number		
Main actors:	Client		
Describe:	New user, no account yet, register for an account by phone number		
	User has not registered an account		
	Success: User successfully registered account Failure: The system displays an error message.		
Main stream	Success: User is authenticated and redirected to the home page with resource access Failure: The system displays an error message.		

2.2.8. *verify email*

ID and Name:	Verify email		
Main actors:	Client		
Describe:	User sends email verification request		
	User has not verified email		
	Success: User successfully verified email Failure: The system displays an error message.		
Main stream	My phone: User enters email and password to log in. The controller receives the login information and sends it to the component. authenticate The authentication component will check the login information, confirm with database If authentication is successful, it will notify and grant access to user If authentication fails, an error message will be displayed to the user.		

2.2.9. *verify phone number*

ID and Name:	verify phone number		
Main actors:	Client		
Describe:	User sends phone number verification request		
	User has not verified phone number		
	Success: User successfully verified phone number Failure: The system displays an error message.		
Main stream	My phone: User enters phone number and password to log in. The controller receives the login information and sends it to the component. authenticate The authentication component will check the login information, confirm with database If authentication is successful, it will notify and grant access to user If authentication fails, an error message will be displayed to the user.		

2.2.10. Forgot Password

ID and Name:	Forgot Password		
Main actors:	Client		
Describe:	Users interact with the system by entering email , phone number , sending requests. Forgot password request, the system sends a forgotten password email to the user. use reset password..		
	User can't remember password		
	Success: User is authenticated and redirected to Home page with resource access. Failure: The system displays an error message.		
Main stream	My phone: Users can enter their email and request a password reset. Users can submit password reset requests. The system will check the email, return the results and perform the operations. relate to. The system will send a verification code and display a success message. The system will check the authentication code and update the new password.		

2.2.11. change password

ID and Name:	change password		
Main actors:	Client		
Describe:	The user interacts with the system by receiving the current password. download, import		
	User can't remember password		
	Success: User is authenticated and redirected to Home page with resource access. Failure: The system displays an error message.		
Main stream	My phone: Users can enter their email and request a password reset. Users can submit password reset requests. The system will check the email, return the results and perform the operations. relate to. The system will send a verification code and display a success message. The system will check the authentication code and update the new password.		

2.2.12. User Profile

ID and Name:	User Profile		
Main actors:	Client		
Describe:	Users interact with the system by clicking on the person-shaped icon.		
Main stream	User visits profile page Click “Edit profile” to update information (name, description, avatar...) Click “Save” to update With profile upgrade (Prime): select plan → payment → update status		

2.2.13. Make friends

ID and Name:	Make friends		
Main actors:	User(client) ⇄ Friends (client)		
Describe:	Allows users to search and add friends to their Sociable Friends List, establish relationships for messaging (DM), phone/video calling, or interact in shared servers		
Main stream	Access the Make Friends Feature: The "Add Friend" interface opens, allowing the User to enter a name or tag to find the user. Search and Send Friend Requests: Friend request sent, Friend receives friend request in their notification Accept or Reject Friend Requests: If accepted, User and Friend become friends, opening up DM messaging, voice/video calling capabilities. If declined, Request is cancelled. Manage Your Friends List: Friends list is updated, User has control over		
	relationships.		

2.2.14. create server (Owner)

ID and Name:	create server (Owner)		
Main actors:	User(client ->Owner)		
Describe:	Users interact with the system by creating servers.		
Main stream	Go to “Create Server” interface Enter server name, description, choose avatar Select privacy settings (public/private) Click “Create” → the system saves and takes you to the new server interface		

2.2.15. update server

ID and Name:	update server		
Main actors:	User(Owner->Moderator)		
Describe:	Users interact with the system by editing the server		
Main stream	In the server interface → click “Server Settings” Maybe: • Rename server • Update description • Change profile picture Enable/disable features like moderation, user authentication,... Click “Save changes” → the system updates new data		

2.2.16. Delete server

ID and Name:	Delete server		
Main actors:	Owner		
Describe:	The user interacts with the system by deleting the server.		
Main stream	In “Server Settings” → select “Delete Server” The system displays warnings and requests confirmation. Re-enter server name or confirm with password Server is completely deleted from the system		

2.2.17. Server & Room Management

ID and Name:	Server & Room Management		
Main actors:	User(Owner/Moderator)		
Describe:	Allows Owners to create, edit, and manage Discord servers (communities or groups), as well as create/configure rooms/channels within the server to organize communication (text, voice, or stage). This feature supports Moderators in managing members and permissions, ensuring the server runs smoothly.		
Main stream	Owner has the highest permissions; Moderators are granted through roles. Use "Channel Overrides" to customize per-room permissions Create and Configure Room (Channel): Room is created, system applies permissions immediately, displayed in server sidebar Edit/Delete Room: To delete a room: Right click on the room > Delete Channel (confirm to avoid data loss)		

2.2.18. Rights and roles (room)

ID and Name:	Rights and roles (room)		
Main actors:	User(owner->moderator->member)		
Describe:	User authorization system, Owner has the highest authority. Can ban/kick members		
Main stream	Server creator becomes Owner Owner permission: add Moderator from member list Moderator can manage content, ban/kick if needed The remaining member is Member – has basic rights (chat, participate)		

2.2.19. Customize the interface (server)

ID and Name:	Customize the interface (server)		
Main actors:	User(Owner)		
Describe:	Users customize server interface, wallpaper, banner..		
Main stream	Go to “Customize Appearance” Choose theme, background or upload image Upload banner server System applies changes → updates interface immediately		

2.2.20. Server Boost

ID and Name:	Server Boost		
Main actors:	User(Owner/ Moderator/ Member)		
Describe:	Allow Owners or contributors to boost the server to unlock premium features such as: adding custom emojis, uploading large files, increasing audio/video quality limits, server backgrounds...		
Main stream	Access the Boost interface • User selects the server to upgrade • Click on “Boost Server” or the corresponding icon View Boost package information • The system displays Boost Server • Describe the corresponding benefits: members, upload capacity, custom backgrounds, custom banner... Make payment • User selects the amount of Boost they want to buy or give away • Confirm payment method → payment successful Server status update system • Increase server boost • Unlock new features for all members now		

2.2.21. Premium profile

ID and Name:	Premium		
Main actors:	User(client)		
Describe:	Pay for profile upgrade		
	User is logged in and has a valid payment method		
Main stream	User selects upgrade item → selects package (Prime Profile Đ Payment information display system User confirms and completes payment Corresponding service processing and upgrade system Users with higher profile or servers with more functionality (e.g. more emojis, larger file uploads...)		

2.2.22. Notifications and Search

ID and Name:	Notifications and Search		
Main actors:	User(client)		
Describe:	Receive system notifications and search for users		
	Active users in the system		
Main stream	When there is related activity (new message, invitation, server update...), the system sends a notification. Users can open, view and process notifications. In the search screen, the user enters the name or user ID → results are displayed. You can make friends, chat, and invite people to the server from search results. Users never miss an update and easily find and connect with others		

2.2.23. Auto Message

ID and Name:	Auto Message		
Main actors:	New Member (server)		
Describe:	Send automatic welcome messages		
	Server Has at least 1 chat room designated as “welcome channel”		
Main stream	User Joins Server Members receive an invitation or automatically find and join the server After confirmation, the system will add the new member to the server list. Enable Auto Message System detects "New User Joined" event Send Welcome Message Auto Message will send a template message to the #welcome channel " Welcome @username to the server!" The message can include emojis, images of the user when they first join		

2.2.24. Payment Frames

ID and Name:	Payment Frames		
Main actors:	User(client)		
Describe:	Pay Frames		
	User is logged in and has a valid payment method		
Main stream	User selects the frame they want to buy and makes payment Payment successful returns to profile page Select edit profile to page containing frame and use		

2.2.25. Message (Chat with friends)

ID and Name:	Message (Chat with friends)		
Main actors:	User(client) ↔ Friends (client)		
Describe:	Chat with friends		
	Provides an interface and communication flow for users to text, send pictures, videos or other multimedia to friends in the contact list.		
Main stream	The system loads the friend list from the server, displaying it in order (e.g. most recent message, online status). The user enters text into the input box. Press the “Send” button or the Enter key to send the message. The system sends the message to the server (or queue if offline) and updates the status “sent” The user selects the attachment button to send a photo, video, or file. The system checks the file size (limit, e.g. 20MB for photo/video, if premium, the maximum that can be sent is).		

2.2.26. Chat with channel (server)

ID and Name:	Chat with channel (server)		
Main actors:	User(Owner/Moderator/Member)		
Describe:	Chat with channel		
	Allows users to join a server, enter channels (text or voice) within the server, and perform interactions such as sending messages, reactions, or participating in voice chat, based on permissions granted by the Owner.		
Main stream	Join Server: The system adds a User to the server, assigns a default role (usually @everyone), and displays a list of channels (text/voice) in the server. Go to Channels: User can enter the channel, see the content (messages or list of people in voice chat). Interaction in Text Channel: Messages are sent, displayed in real time to all members with channel viewing rights. The system saves messages to history.		

2.2.27. *Voice/ Call Message (with friends)*

ID and Name:	Voice/ Call Message (with friends)		
Main actors:	User(client) ⇄ Friends (client)		
Describe:	Voice/ Call Message (with friends)		
	Allows two users (friends) to make 1:1 voice or video calls via Direct Message on Sociable, supporting direct communication or sending text messages in parallel.		
Main stream	Open Direct Message (DM): The system displays a DM window with message history (if any) and call control buttons (Voice Call/Video Call) Initiate Call: Call is initiated, Friend receives notification (audio/visual) on their device Join the Call: If Friend accepts, both are connected to a 1:1 call; the interface displays mic status, webcam, and screen sharing options Interactions in Calls: Calls are smooth; audio/video is streamed in real time; DM messages are saved in history End Call: 1:1 call stops; DM window returns to normal chat state. Voice/video content is not saved, but DM text history is retained.		

2.2.28. *Voice/ Call (channel/ server)*

ID and Name:	Voice/ Call (channel/ server)		
Main actors:	User(Owner/Moderator/Members)		
Describe:	Voice/ Call (channel/ server)		
	Allows users to join a voice channel in a Discord server to perform voice chat, video chat, or screen sharing, while also helping admins manage interactions in the channel		

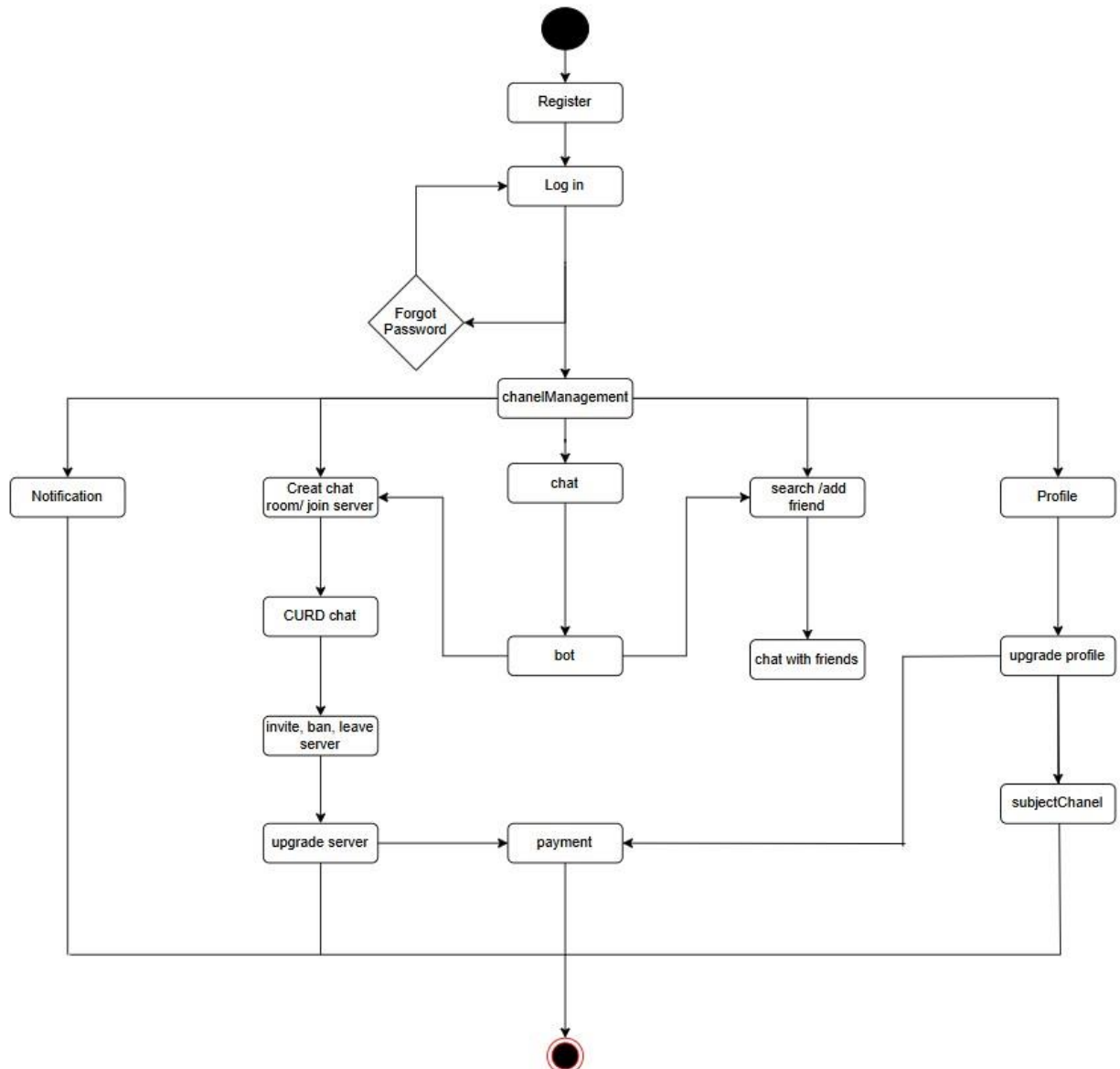
Main stream	Join Server: The system adds a User to the server, assigns a default role (e.g., @everyone), and displays a list of channels, including voice channels. Visit Voice Channel: The user is connected to the voice channel, sees the list of members in the channel, and can turn on the mic/speaker. Interaction in Voice Channel: Audio/video is transmitted smoothly; other members hear/see User's content. System displays status (mic on/off, video on/off) Managed by Admin/Moderator: Voice channels are managed according to server rules, ensuring a stable communication experience. Leave Voice Channel: User is disconnected from voice channel, can no longer hear/see channel content. System updates channel member list
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2.2.29. *Video call with friend*

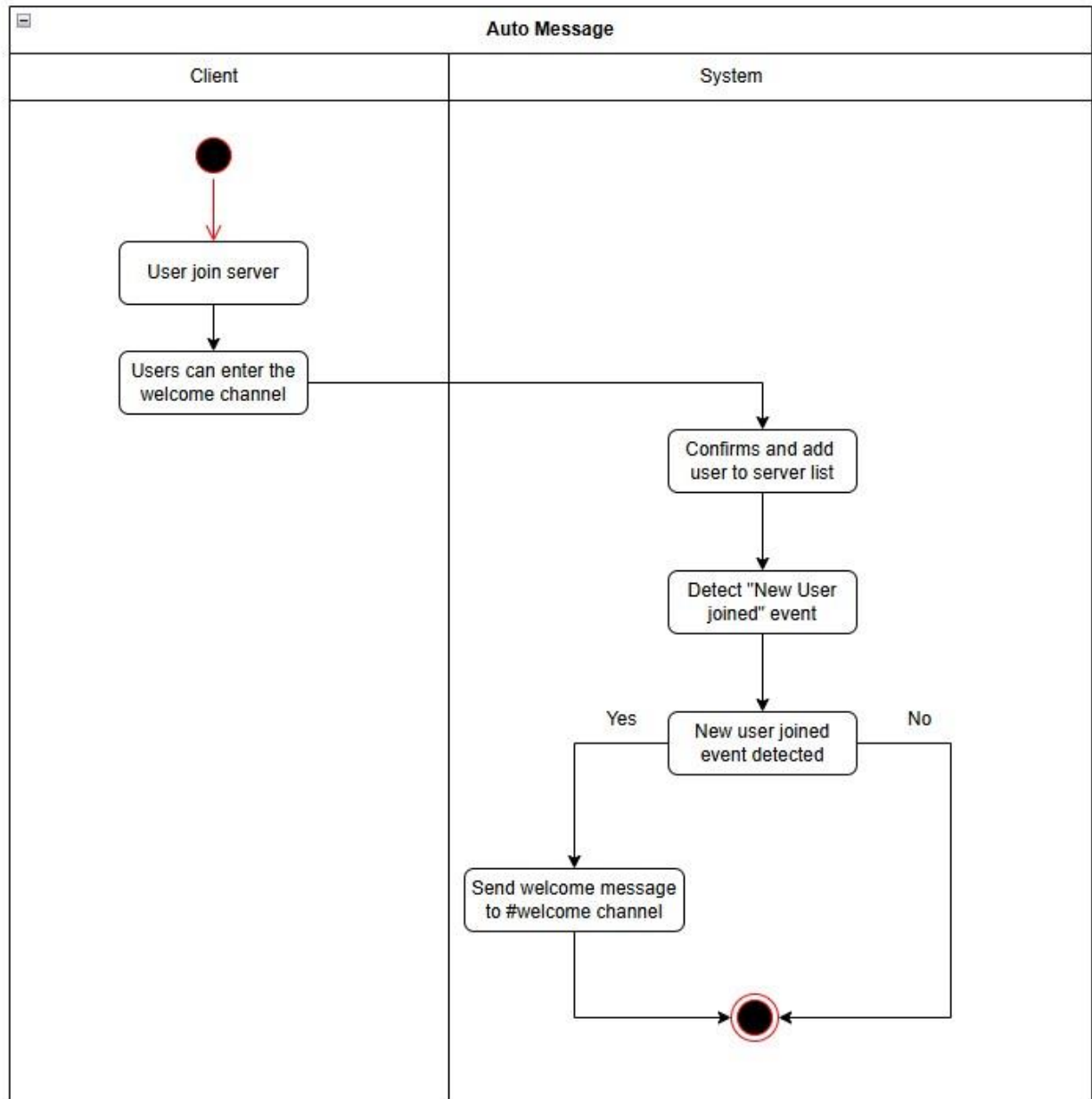
ID and Name:	Video call with friend		
Main actors:	User(client) ⇄ Friends (client)		
Describe:	Video call with friend		
	Allows two users (friends) to make 1:1 video calls via DM on Sociable, supporting direct communication with images and audio		
Main stream	Open Direct Message (DM): The system displays the DM interface with controls, including "Video Call" (camera icon) Initiate a Video Call: Video invitation sent, Friend receives notification with Accept/Decline option Join a Video Call: If accepted, both are connected by video; the interface displays each other's video feed, with control buttons (mute, camera on/off) Interact in Video Calls: Video calls are smooth; quality adjusts based on bandwidth End Video Call: Call ends; DM window returns to normal chat state. Video content is not saved, only DM text history is saved.		

2.3. Sequence Diagram and Class Diagram

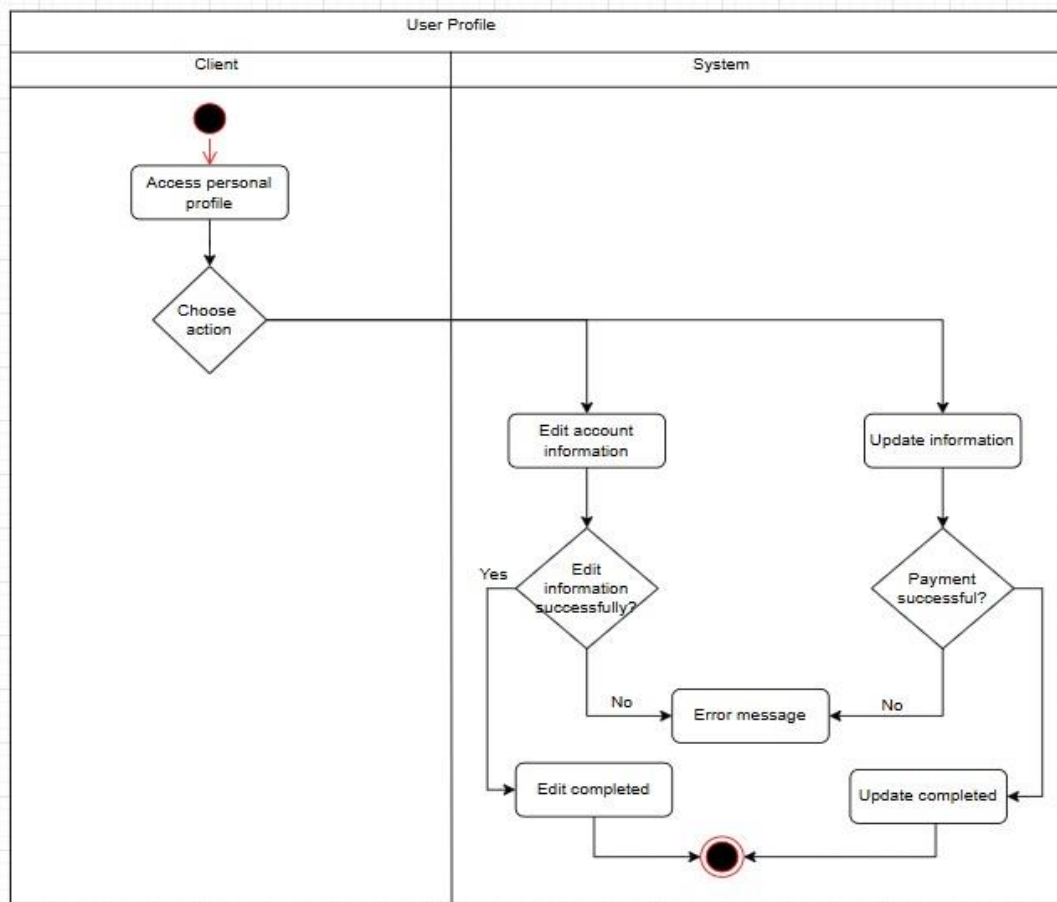
2.3.1. Activity diagram



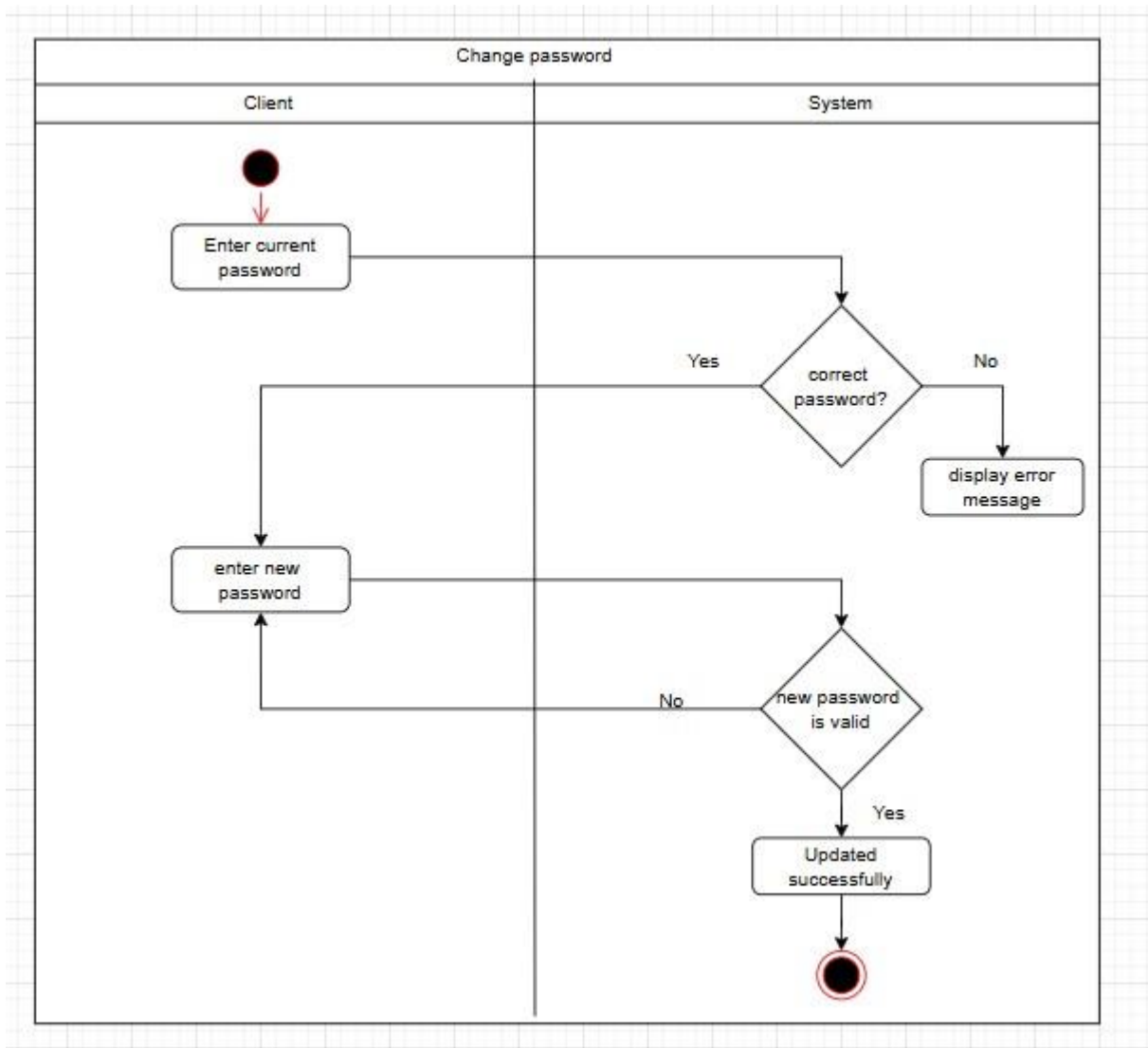
2.3.2. Activity Auto Message



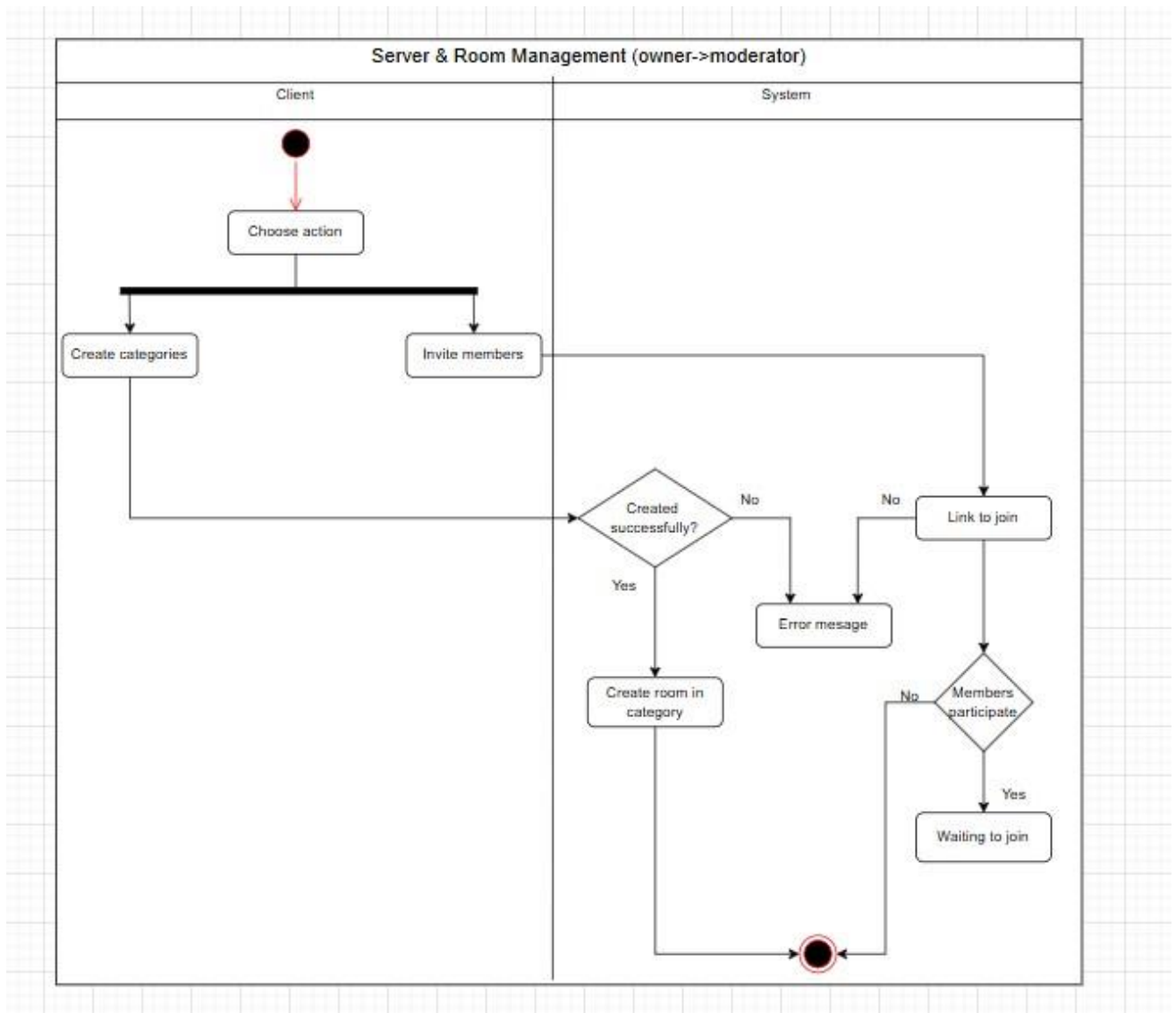
2.3.3. Activity user profile



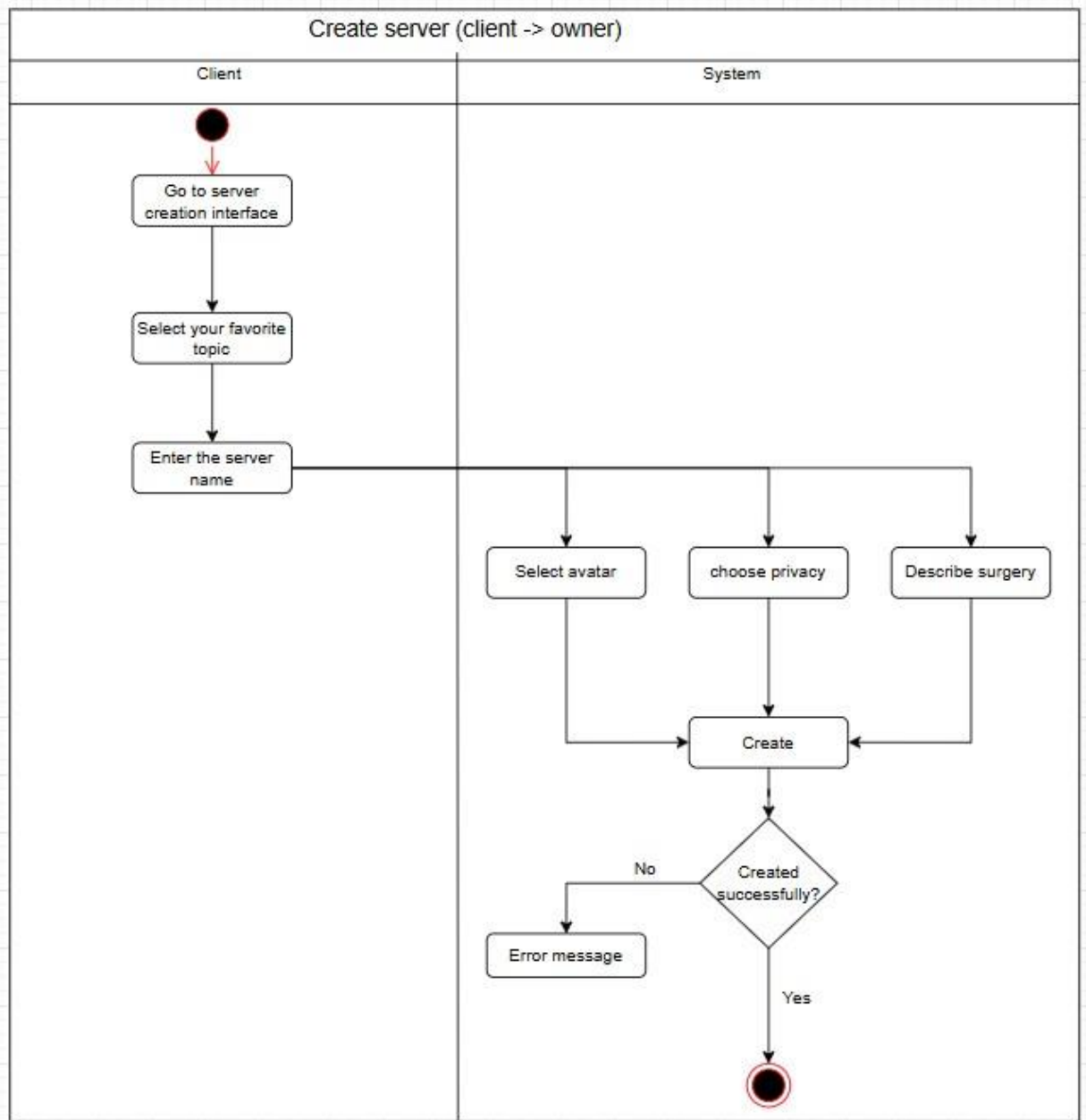
2.3.4. Activity change password



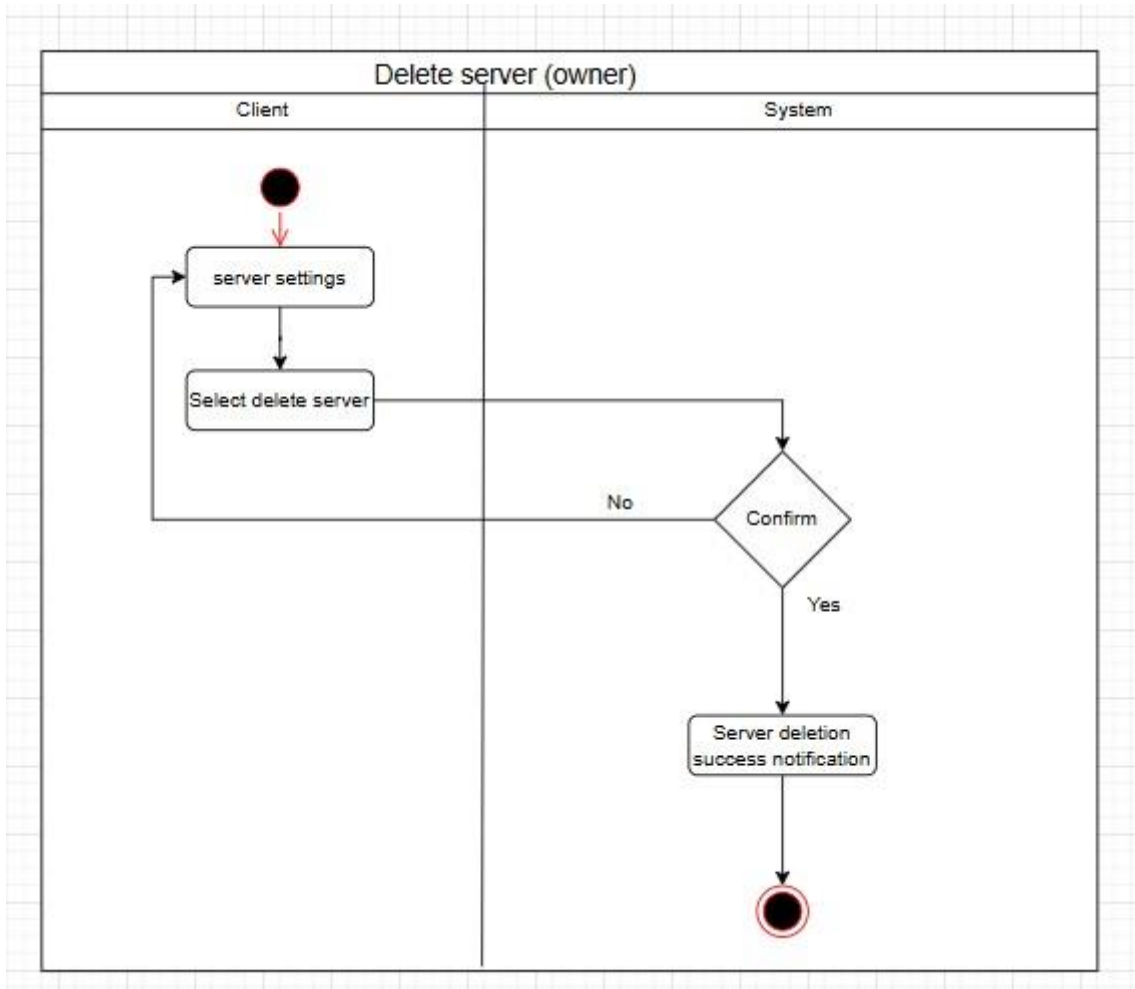
2.3.5. Activity Create room chat



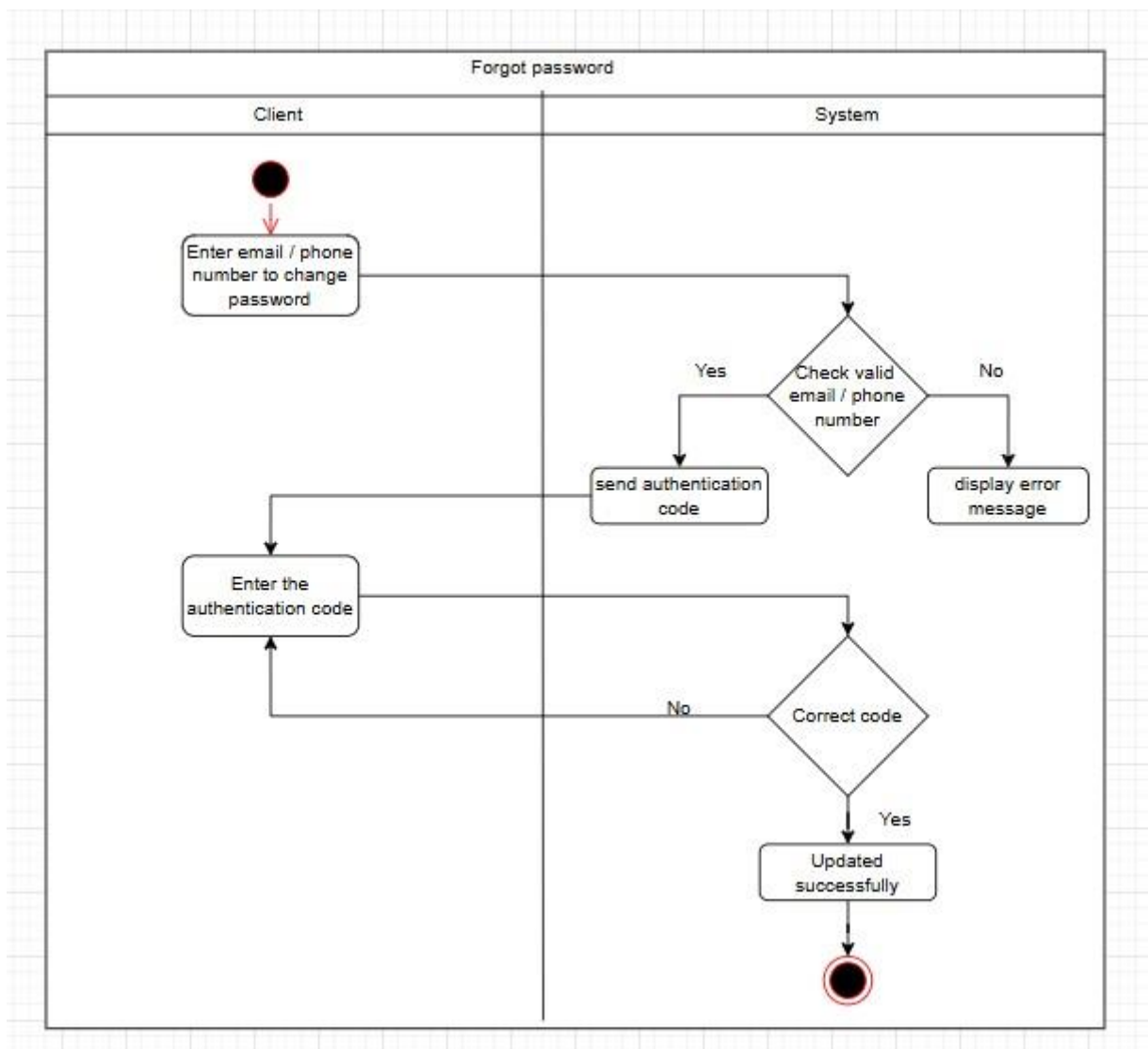
2.3.6. Activity Create Server



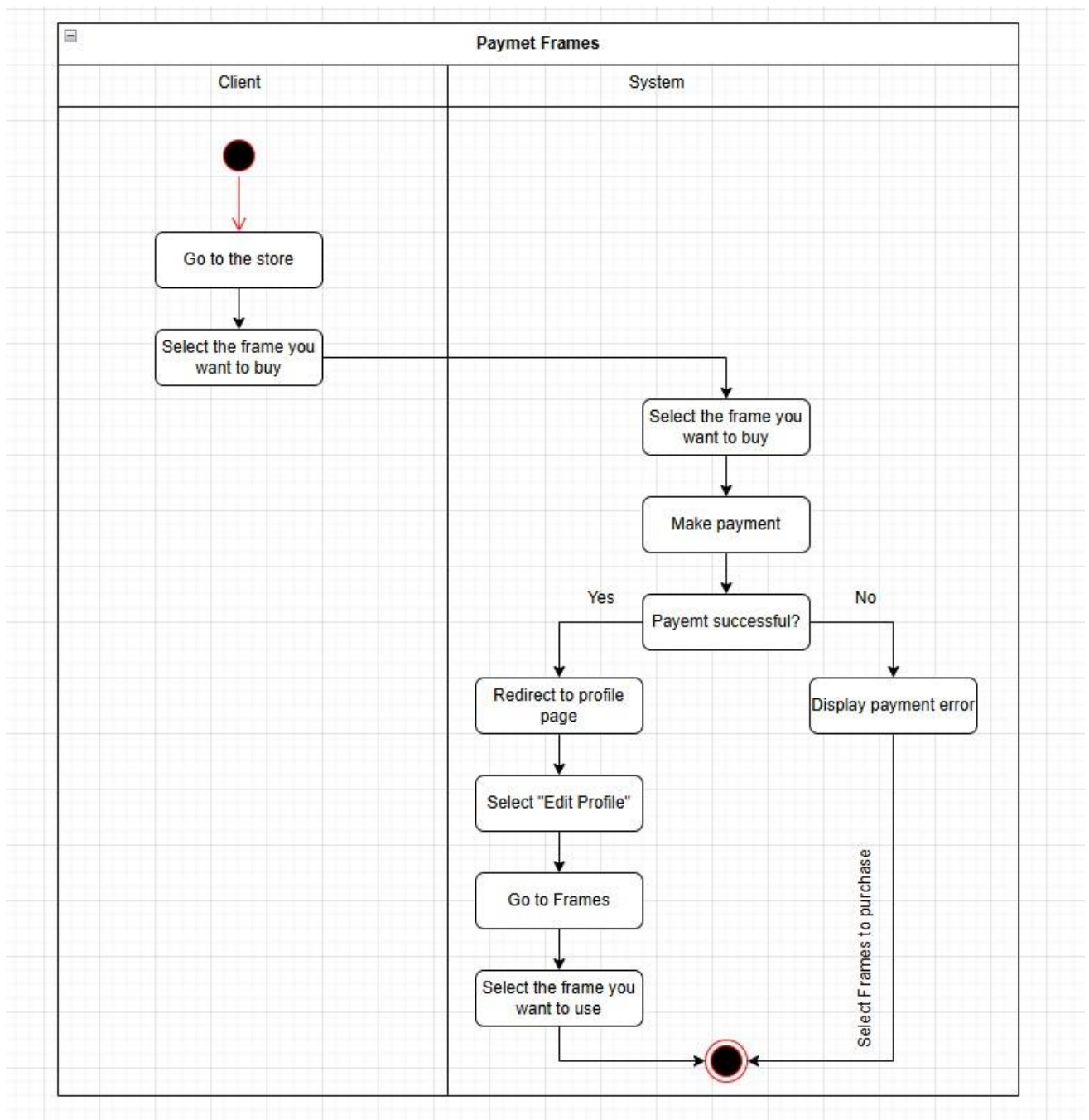
2.3.7. Activity delete server



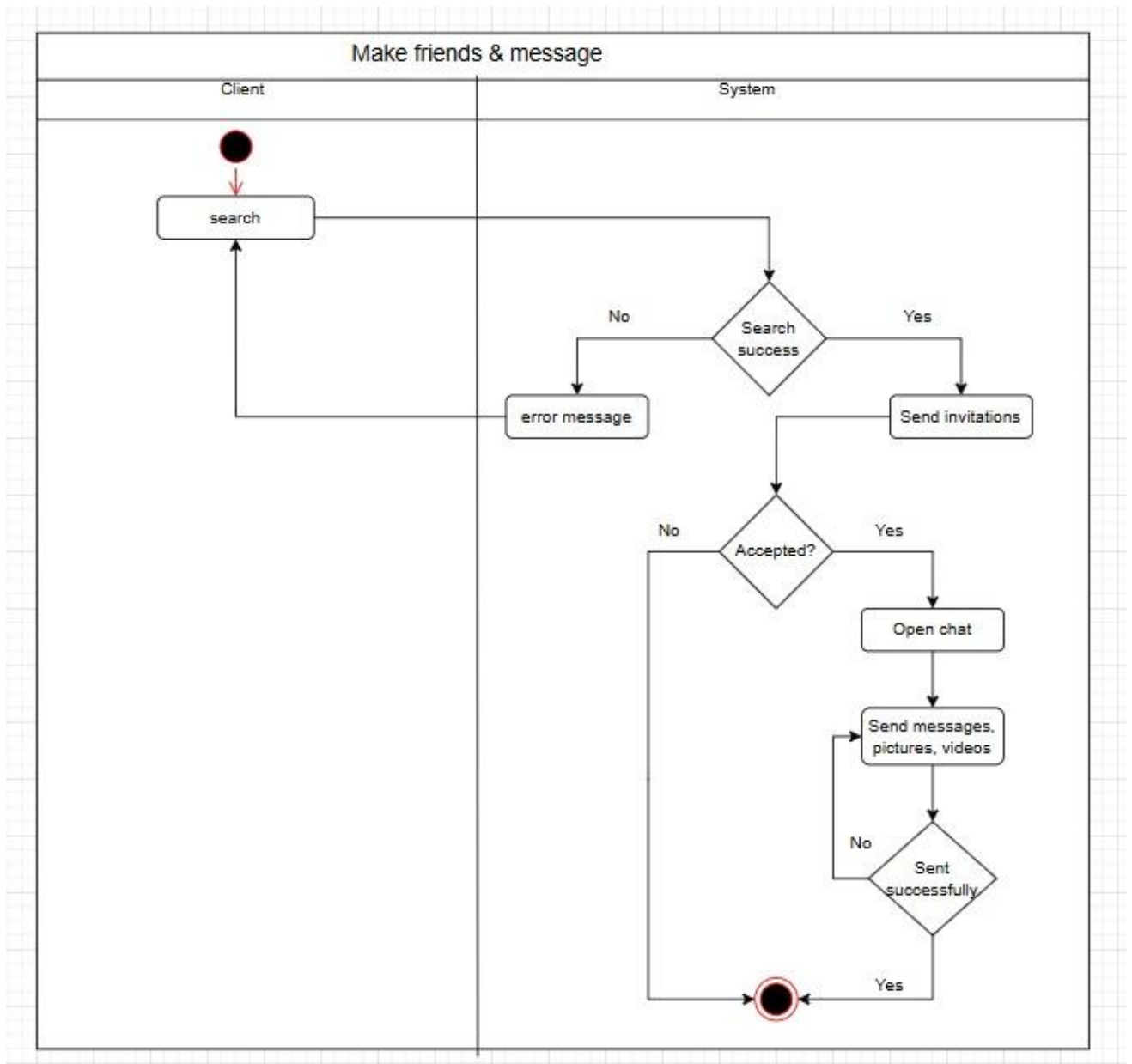
2.3.8. Activity forgot Password



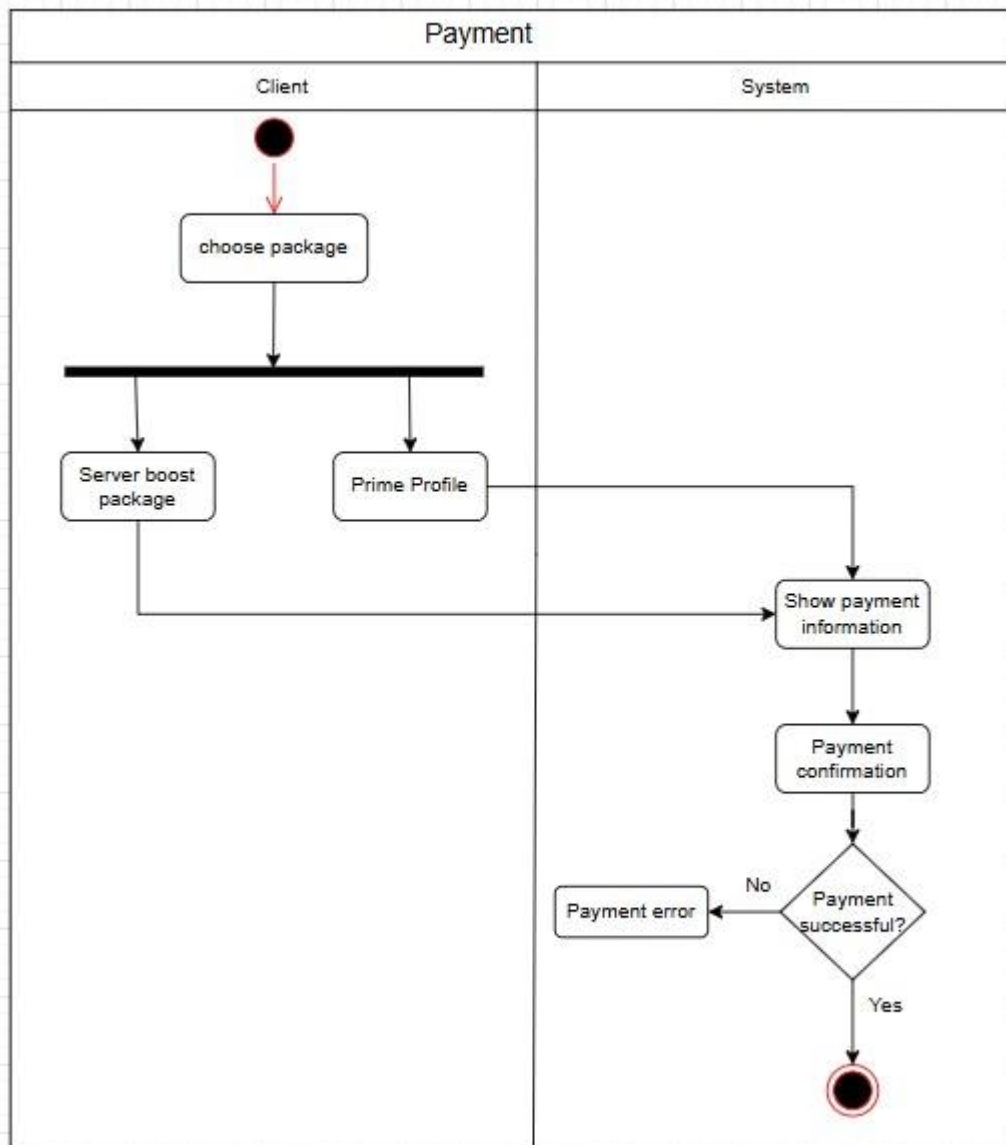
2.3.9. Activity Frames



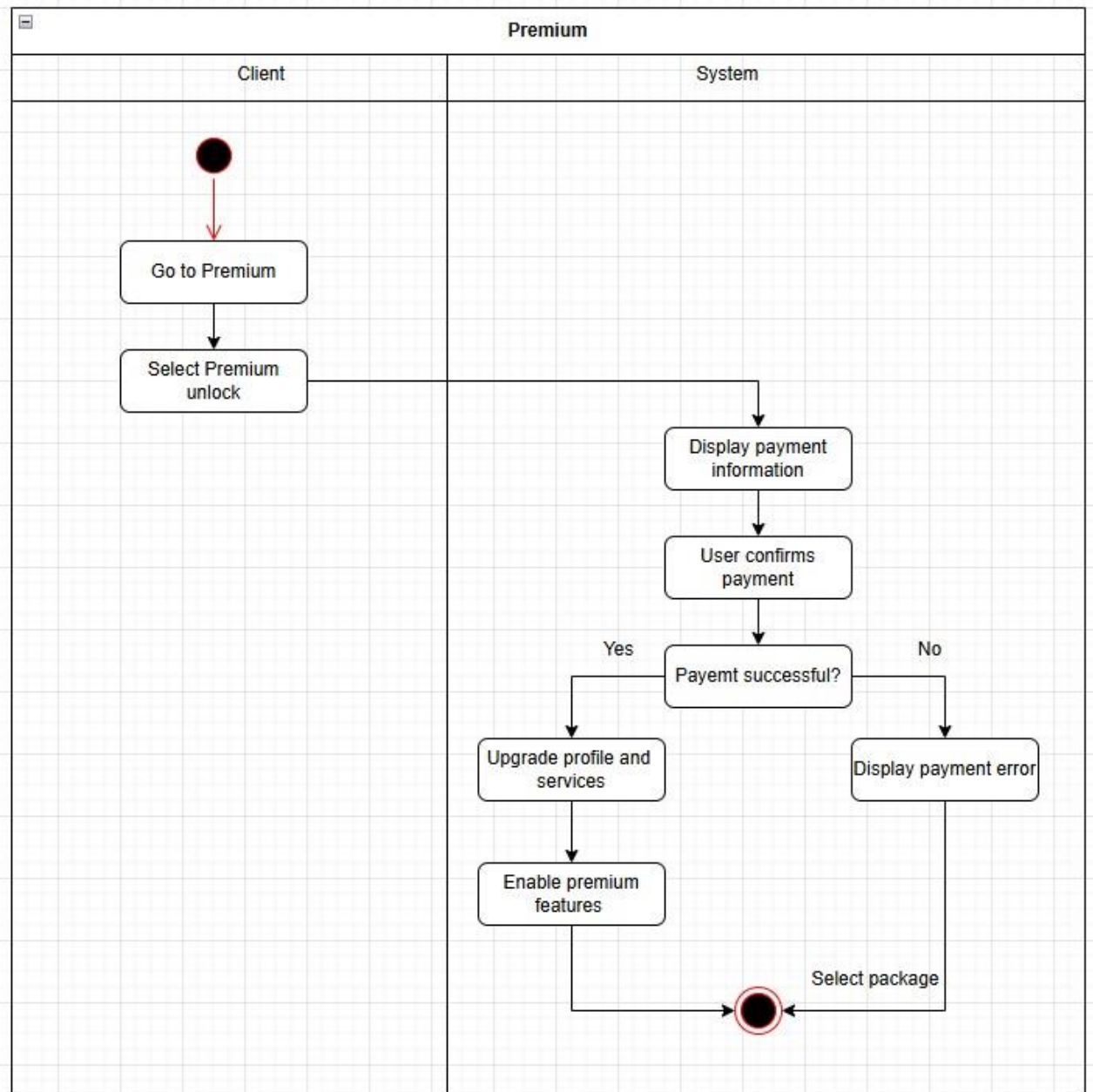
2.3.10. Activity make friends & message



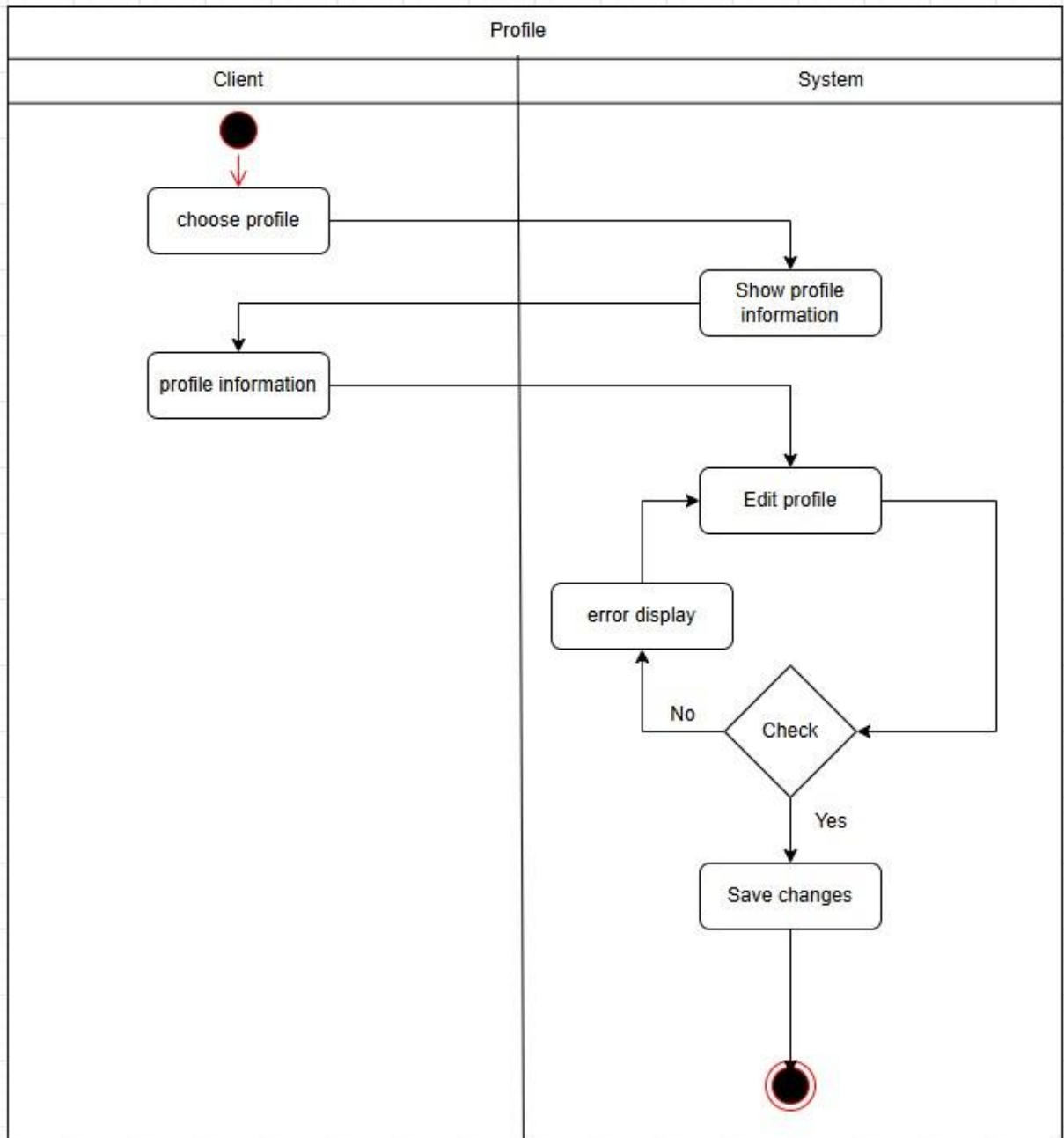
2.3.11. Activity Payment



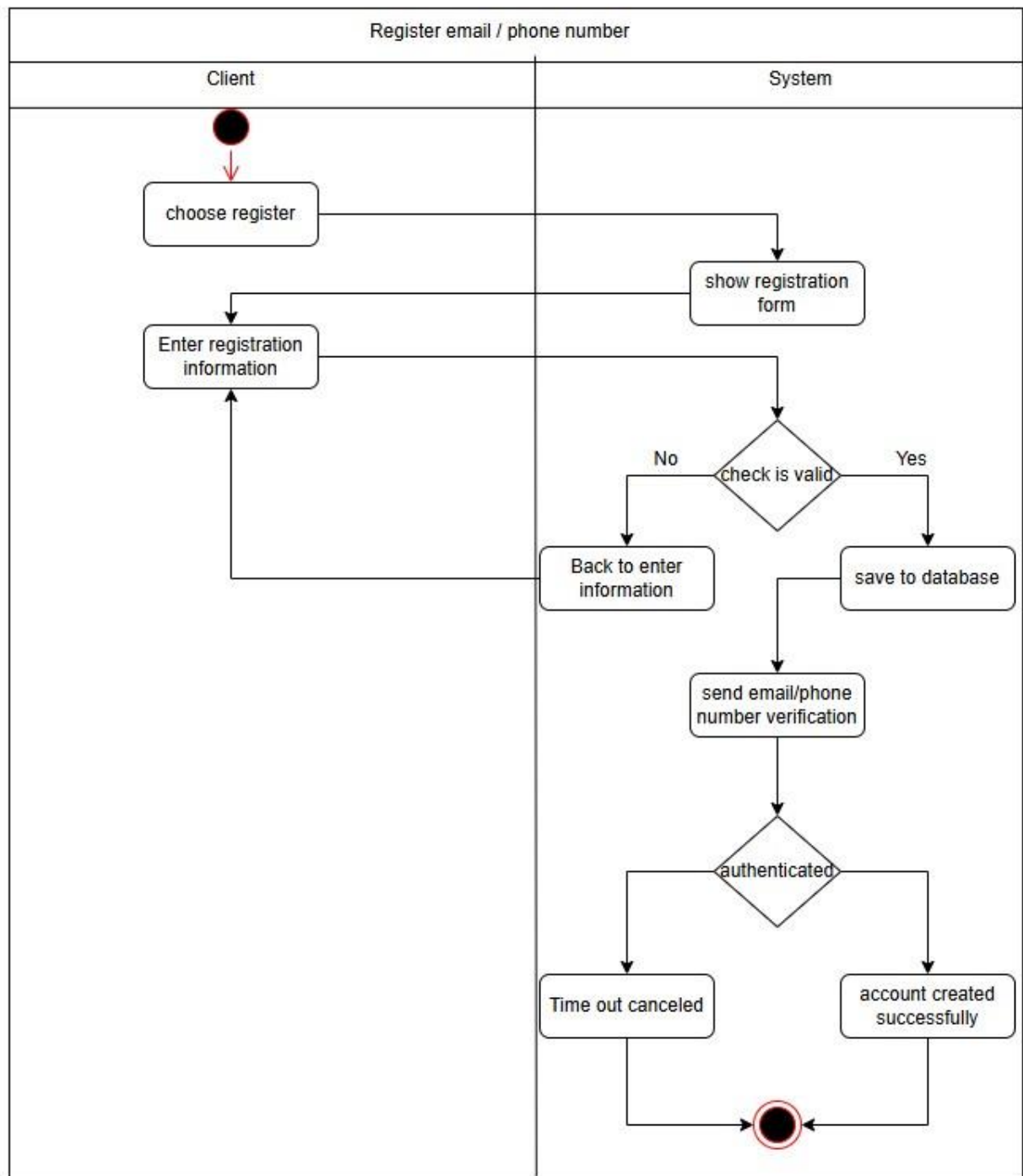
2.3.12. Activity Premium



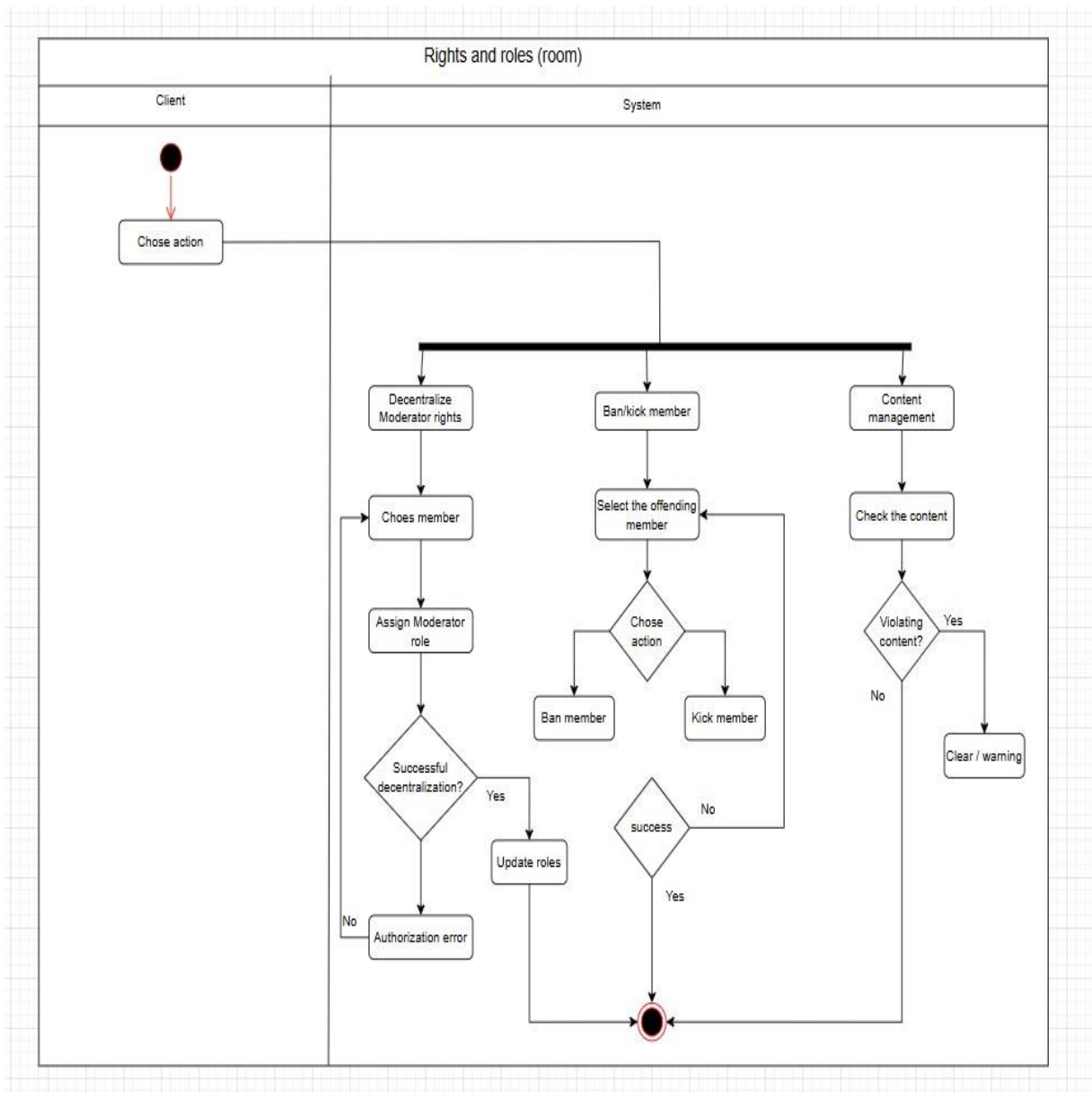
2.3.13. Activity Profile



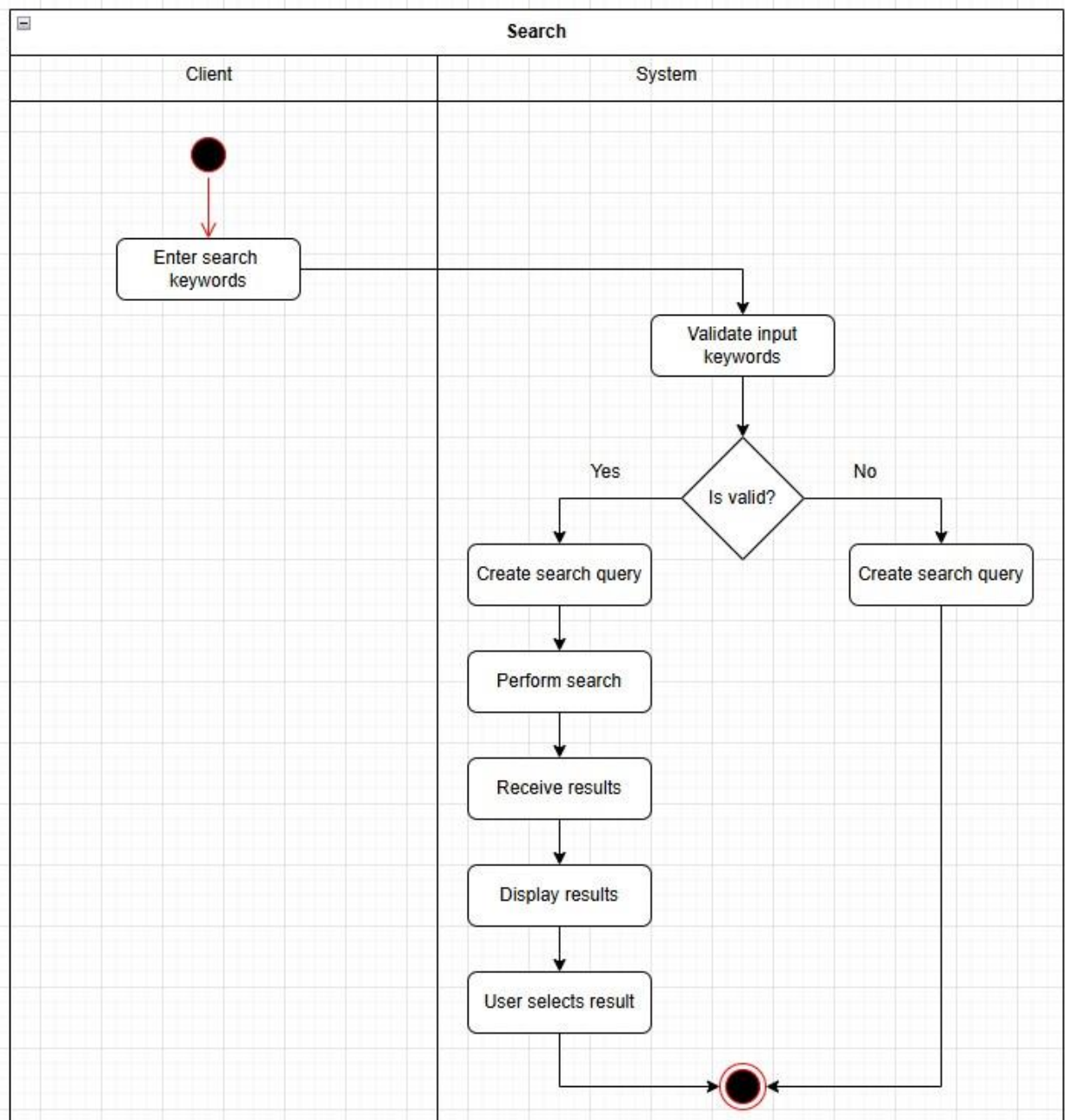
2.3.14. Activity register email & phone number



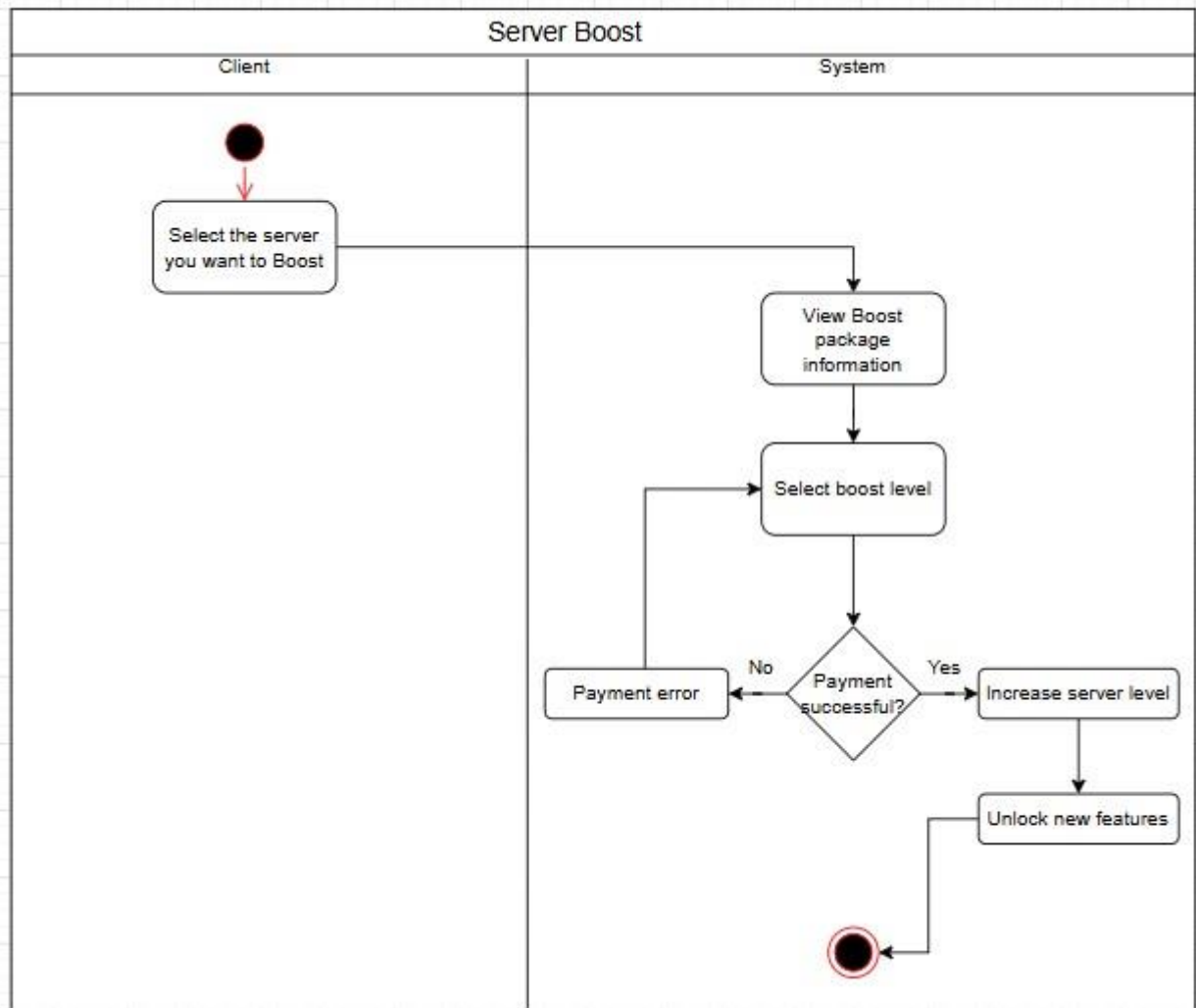
2.3.15. Activity Rights and roles



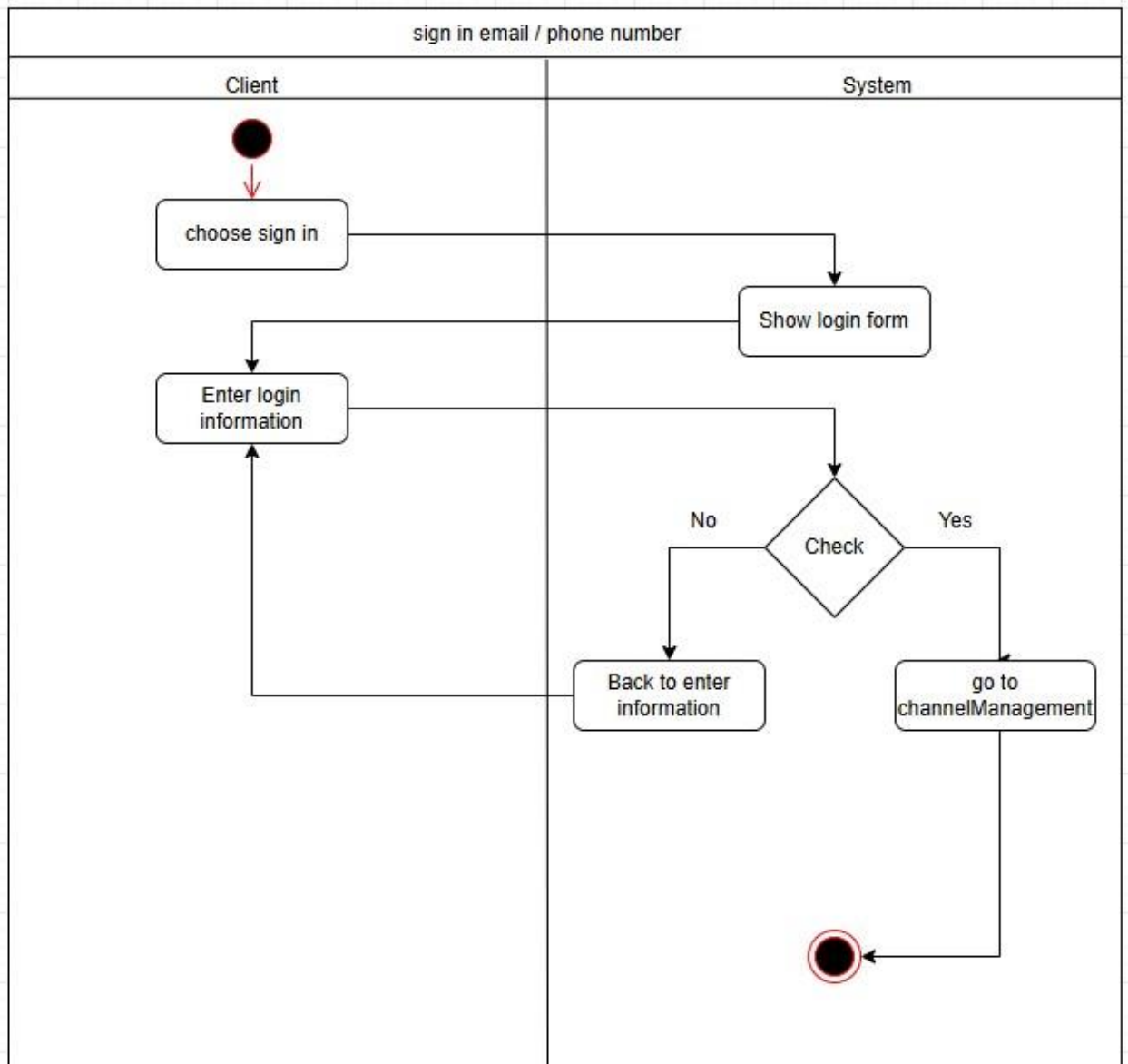
2.3.16. Activity Search



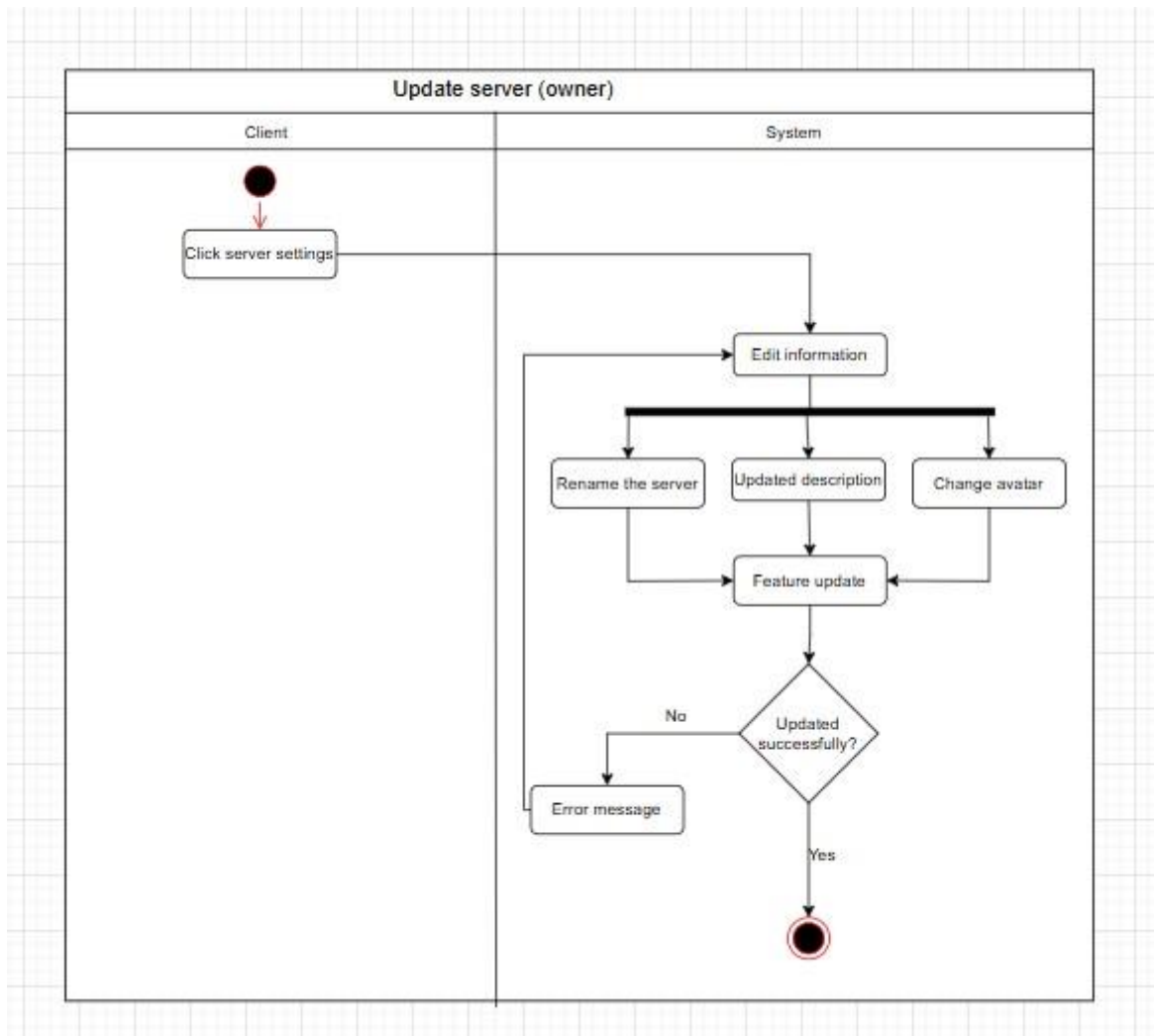
2.3.17. Activity server boost



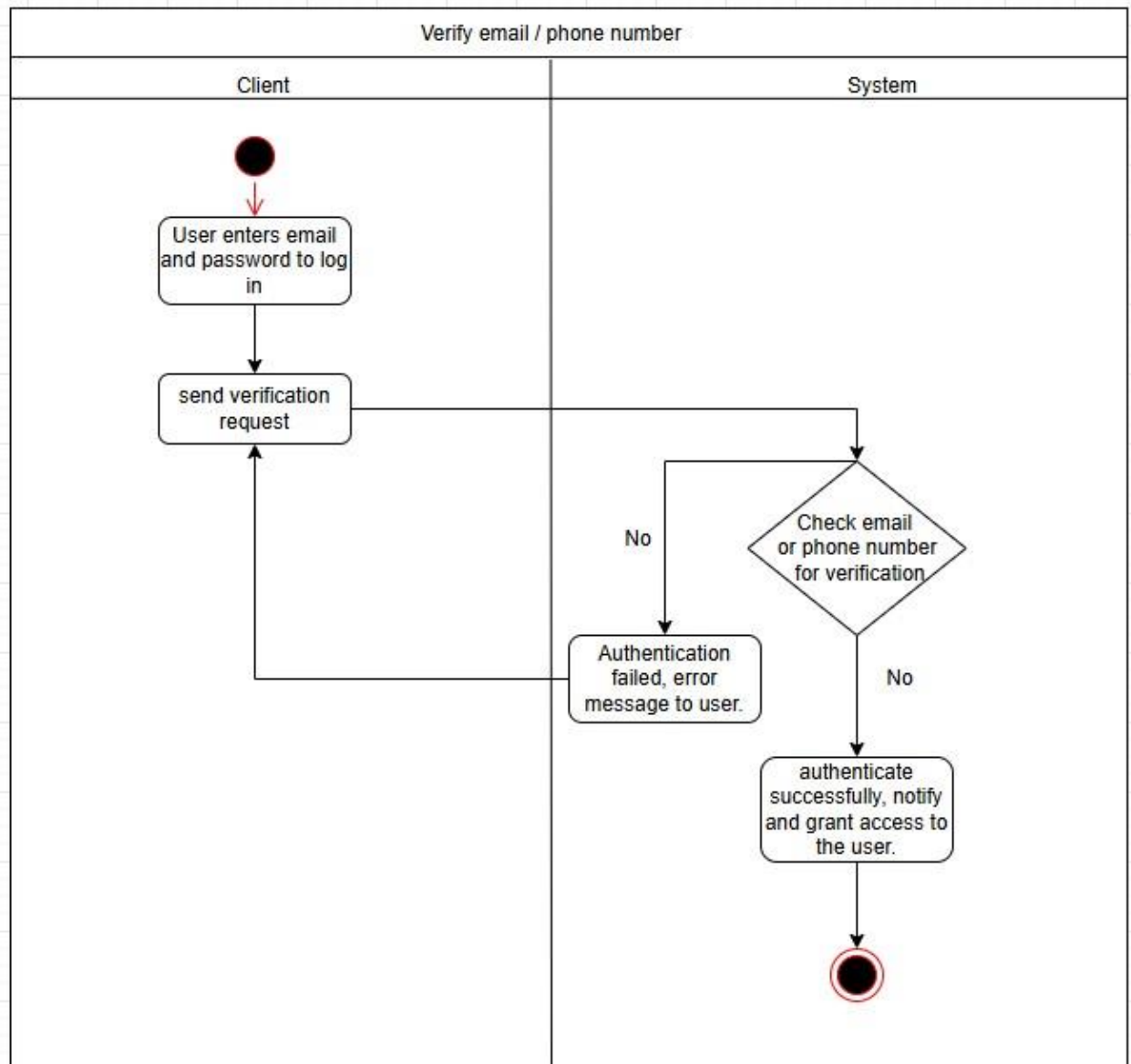
2.3.18. Activity sign in email & phone number



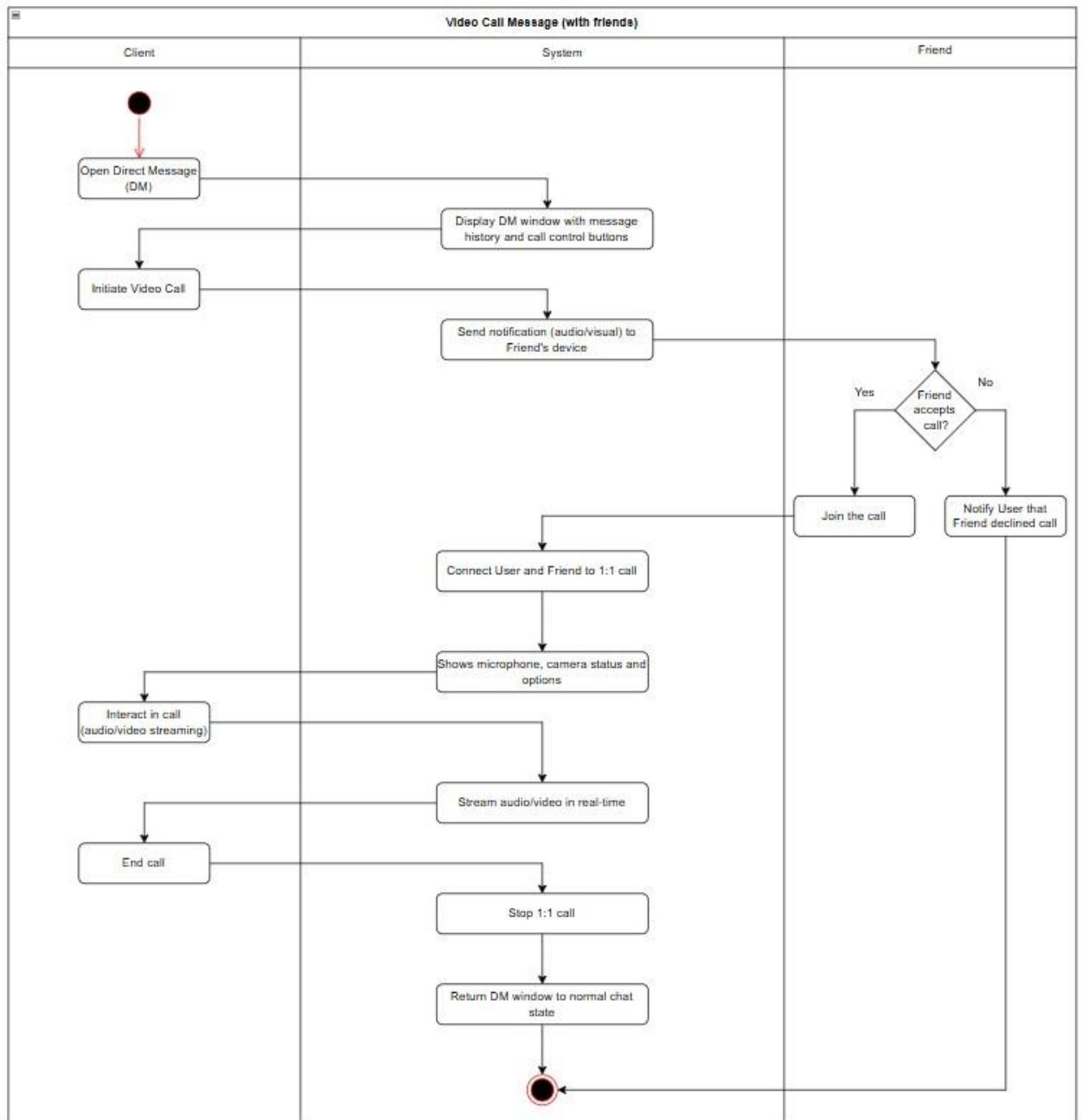
2.3.19. Activity update server



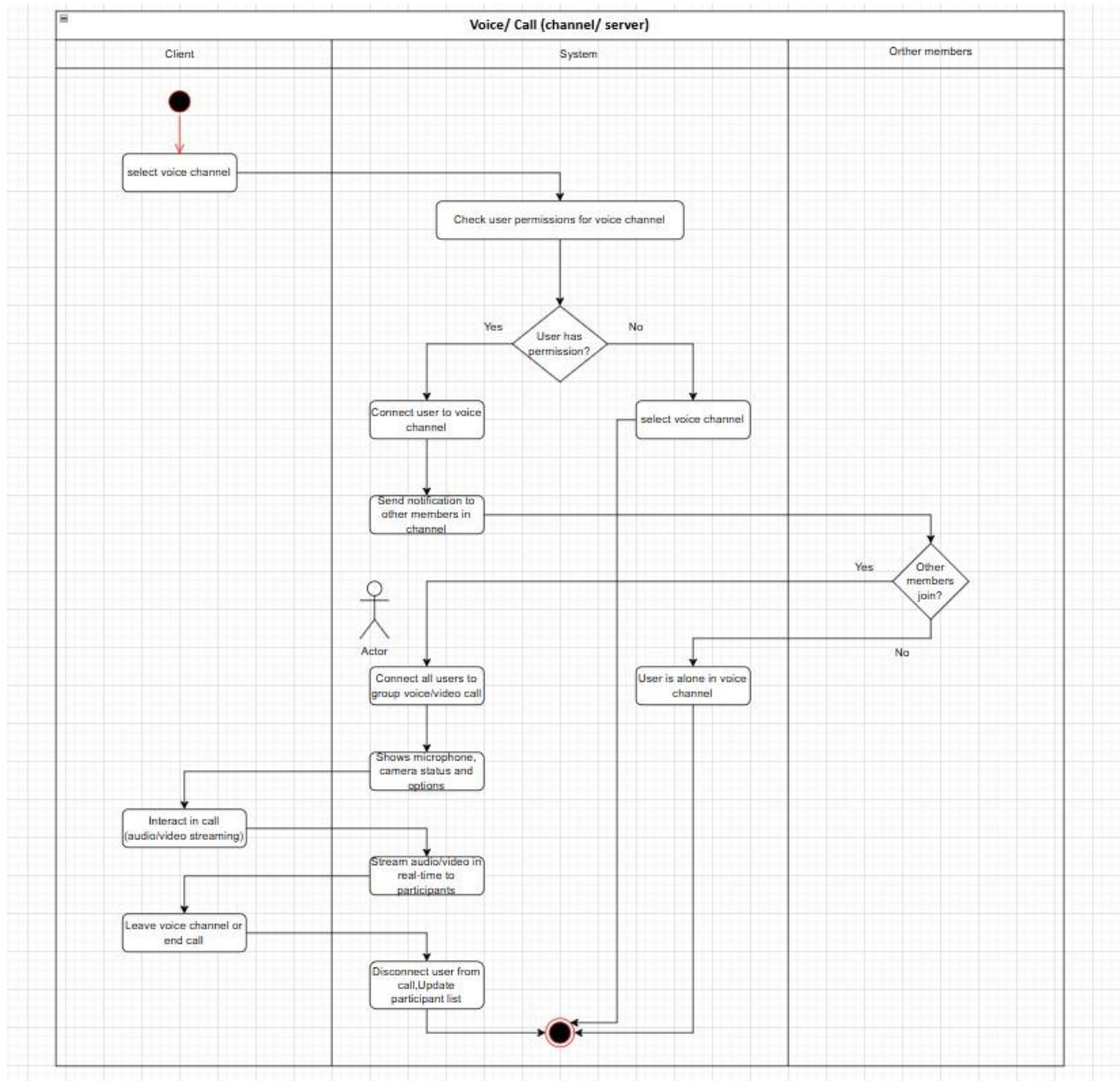
2.3.20. Activity verify email & phone number



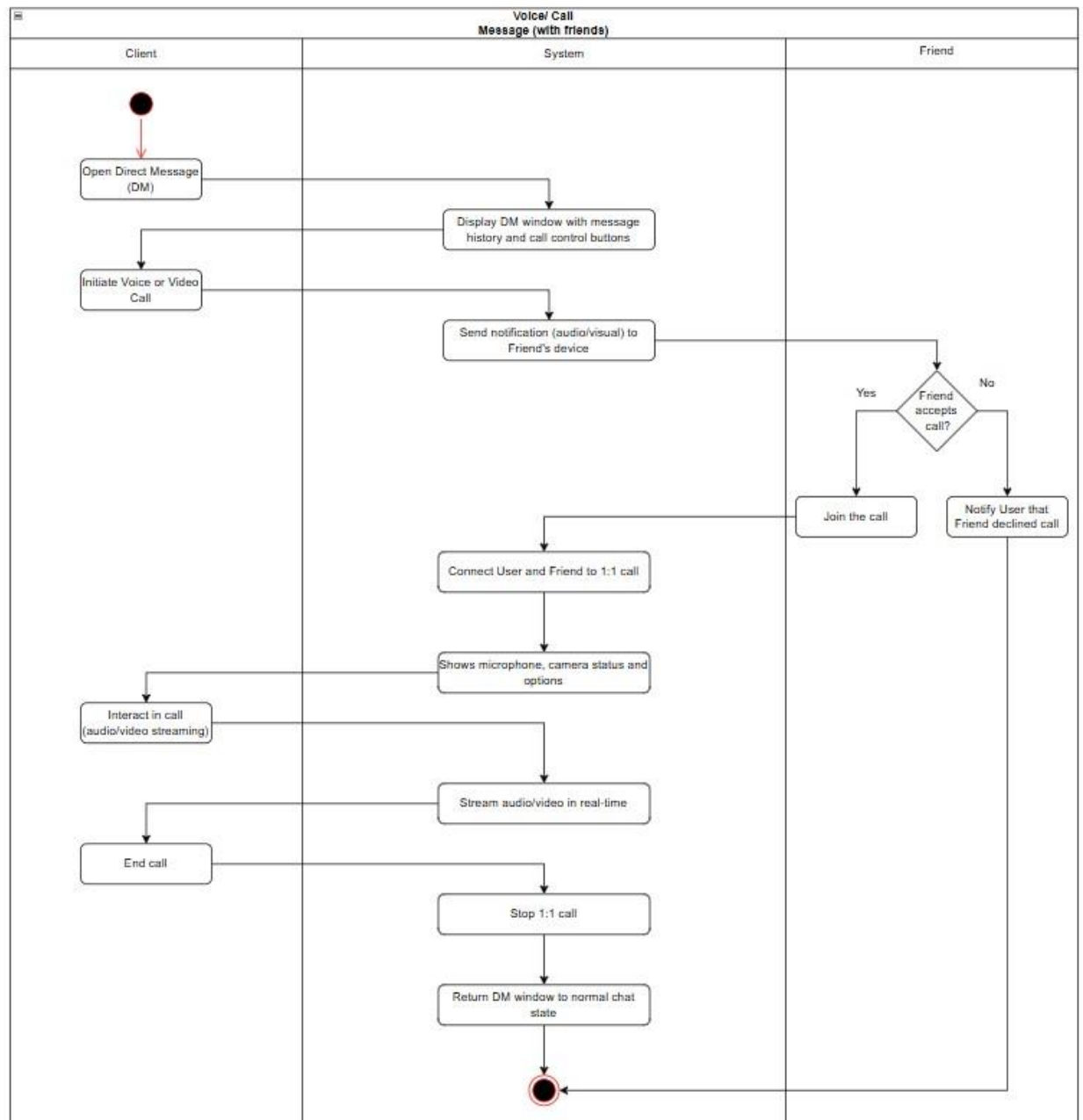
2.3.21. *video call friend*



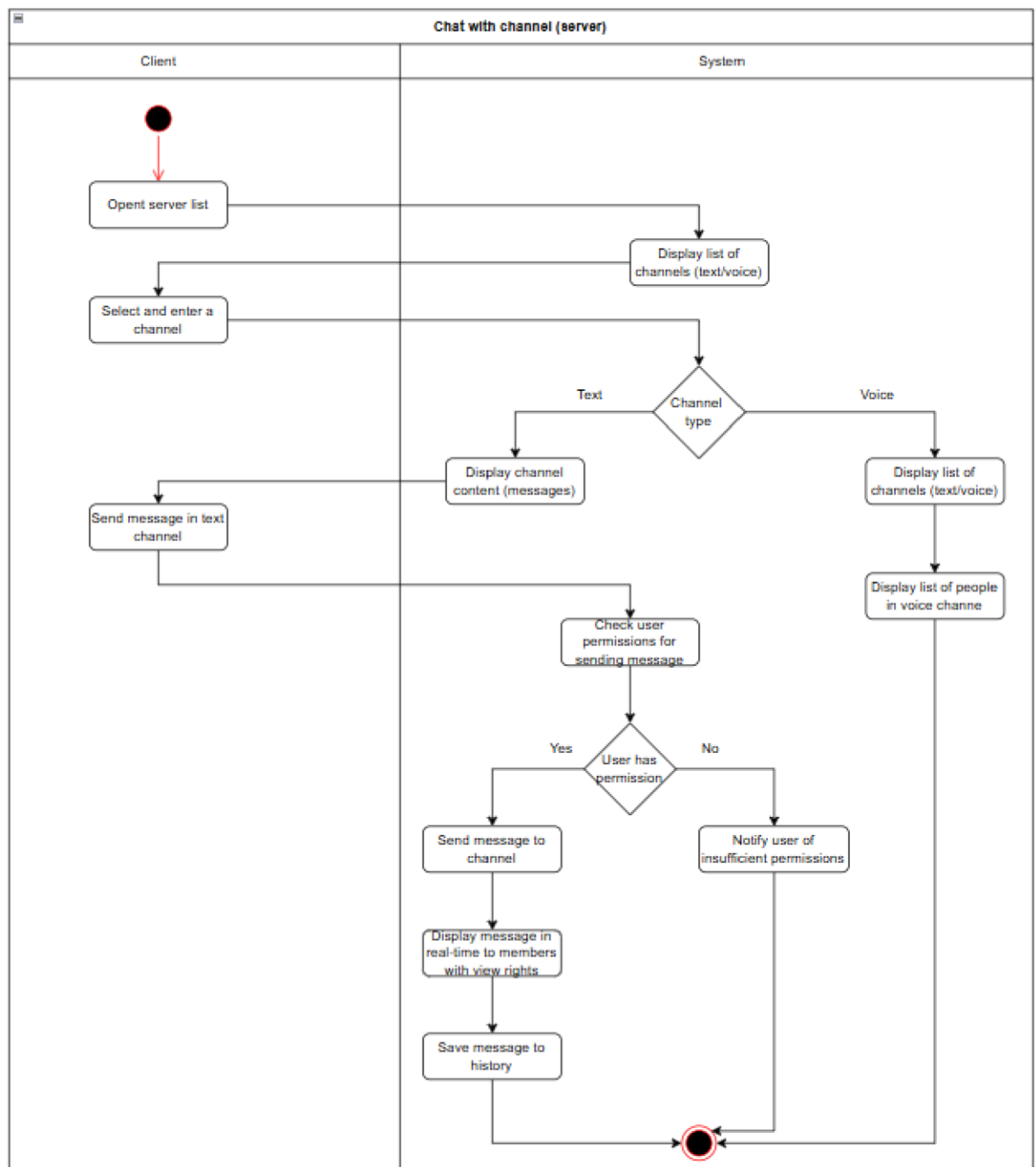
2.3.22.voice/call(channel/ server)



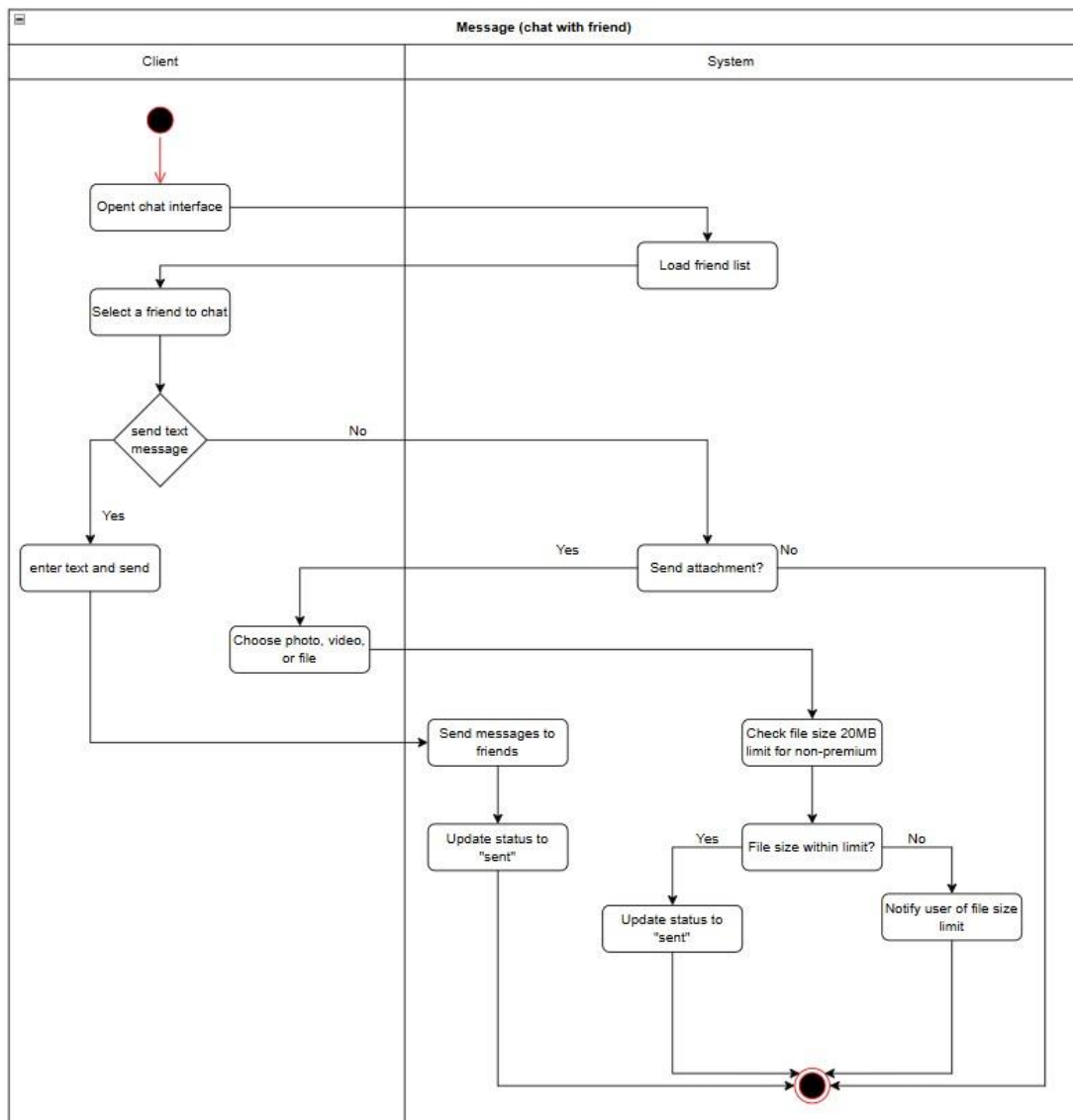
2.3.23.voice/call message(with friends)



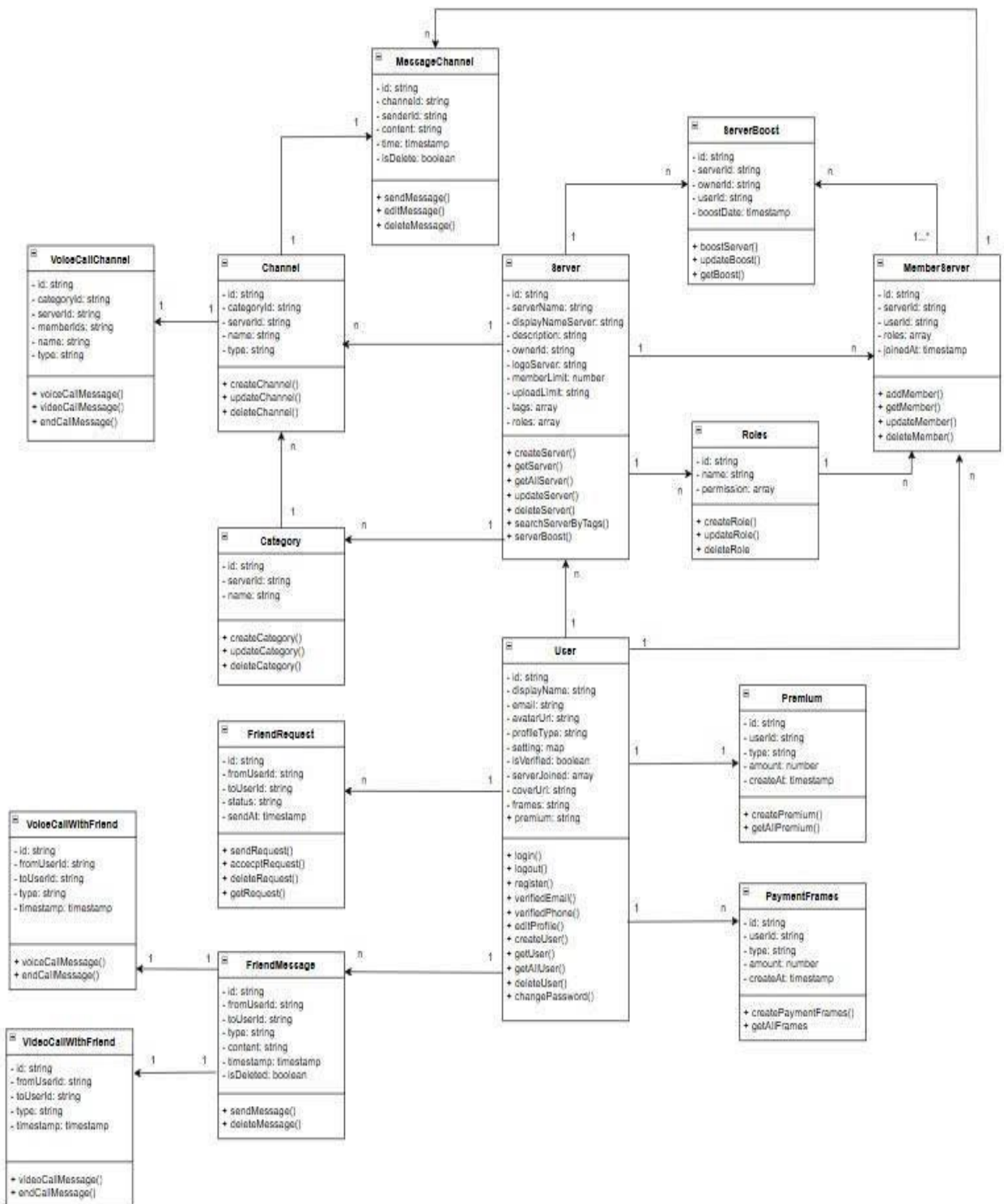
2.2.24 . Activity Chat with channel



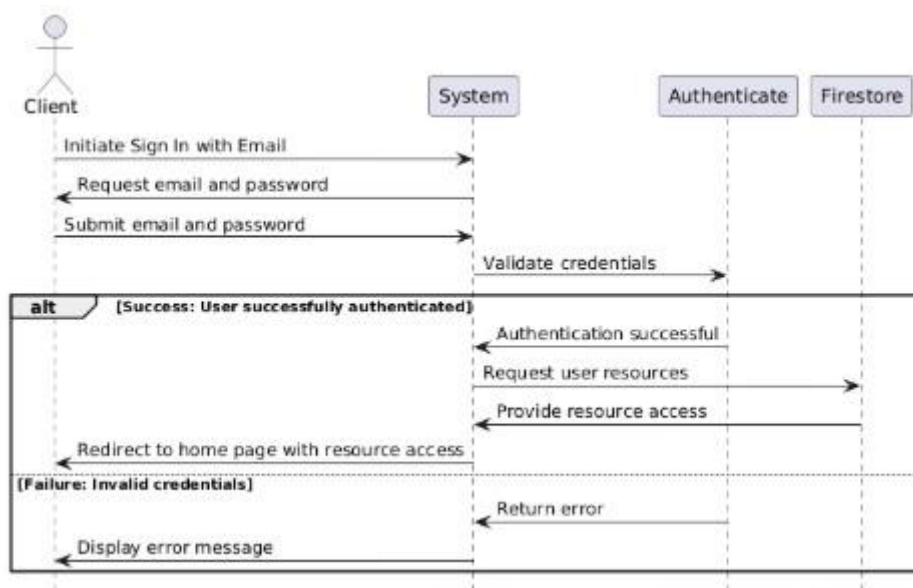
2.2.25. Activity Chat with channel



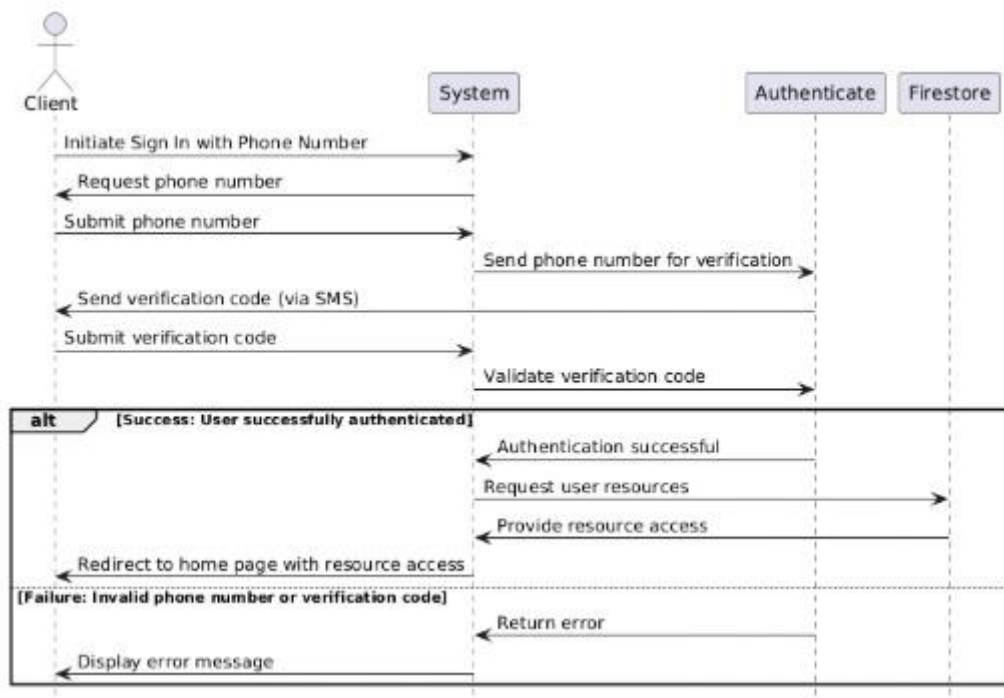
2.1.1. class diagram



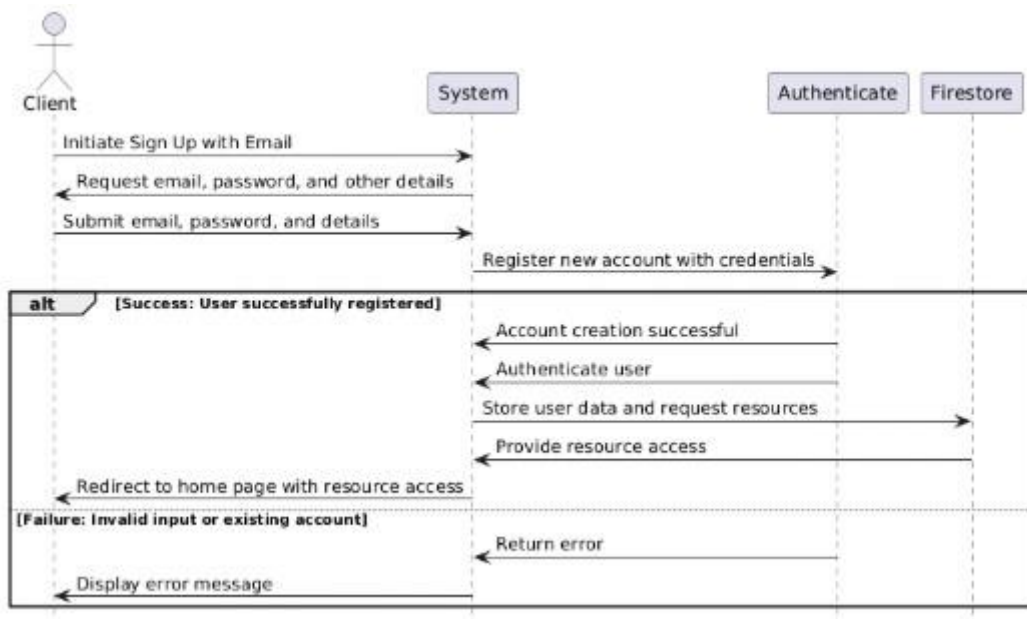
2.1.2. Sign In with Email



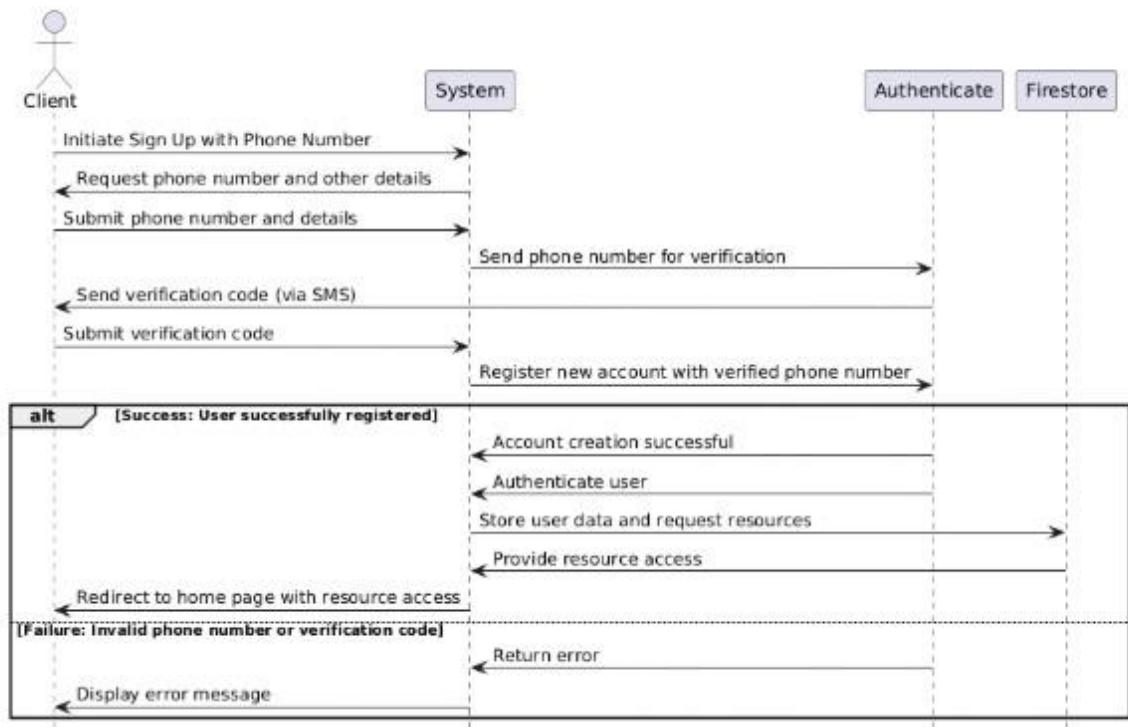
2.1.3. Sign by phone number



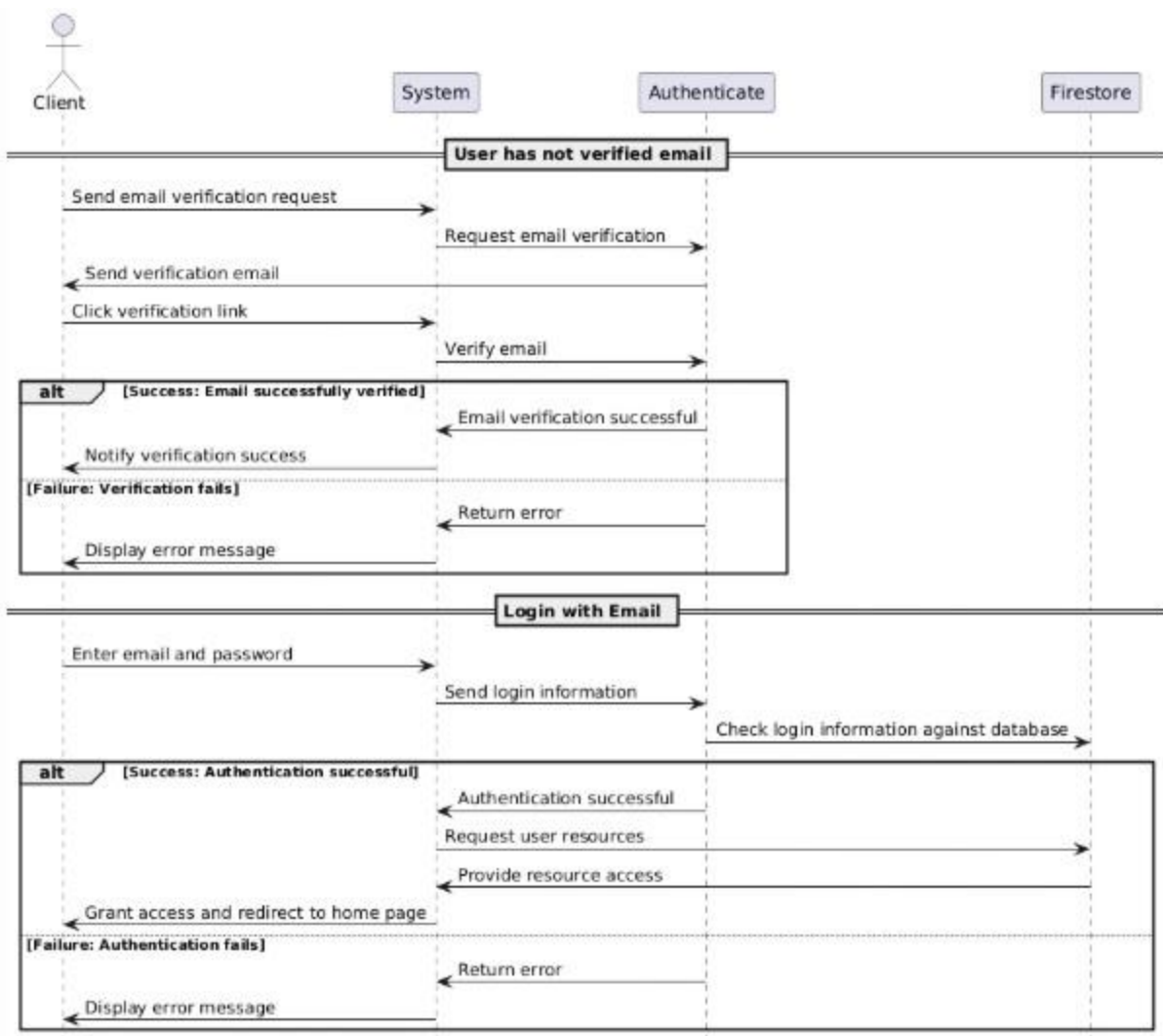
2.1.4. Sign up with email



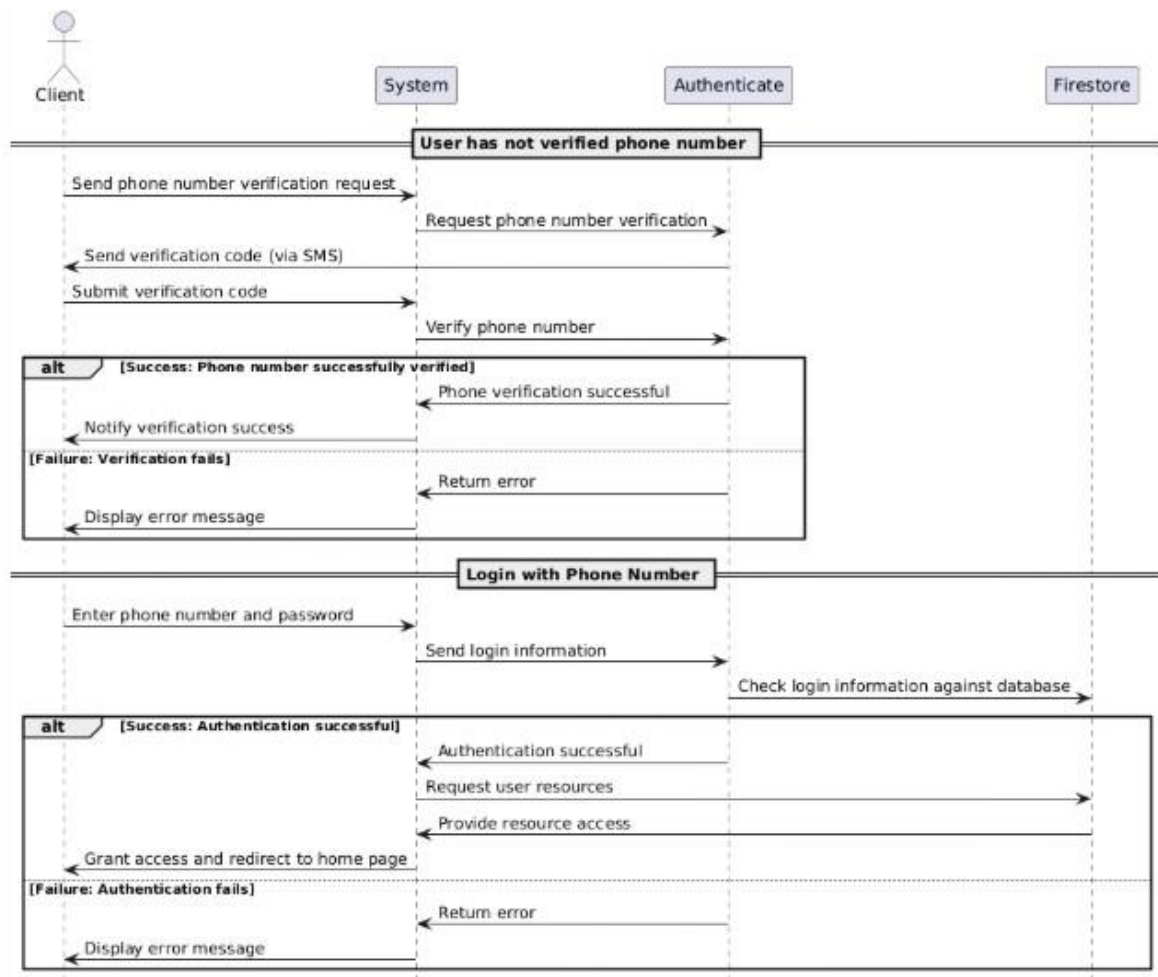
2.1.5. Sign up by phone number



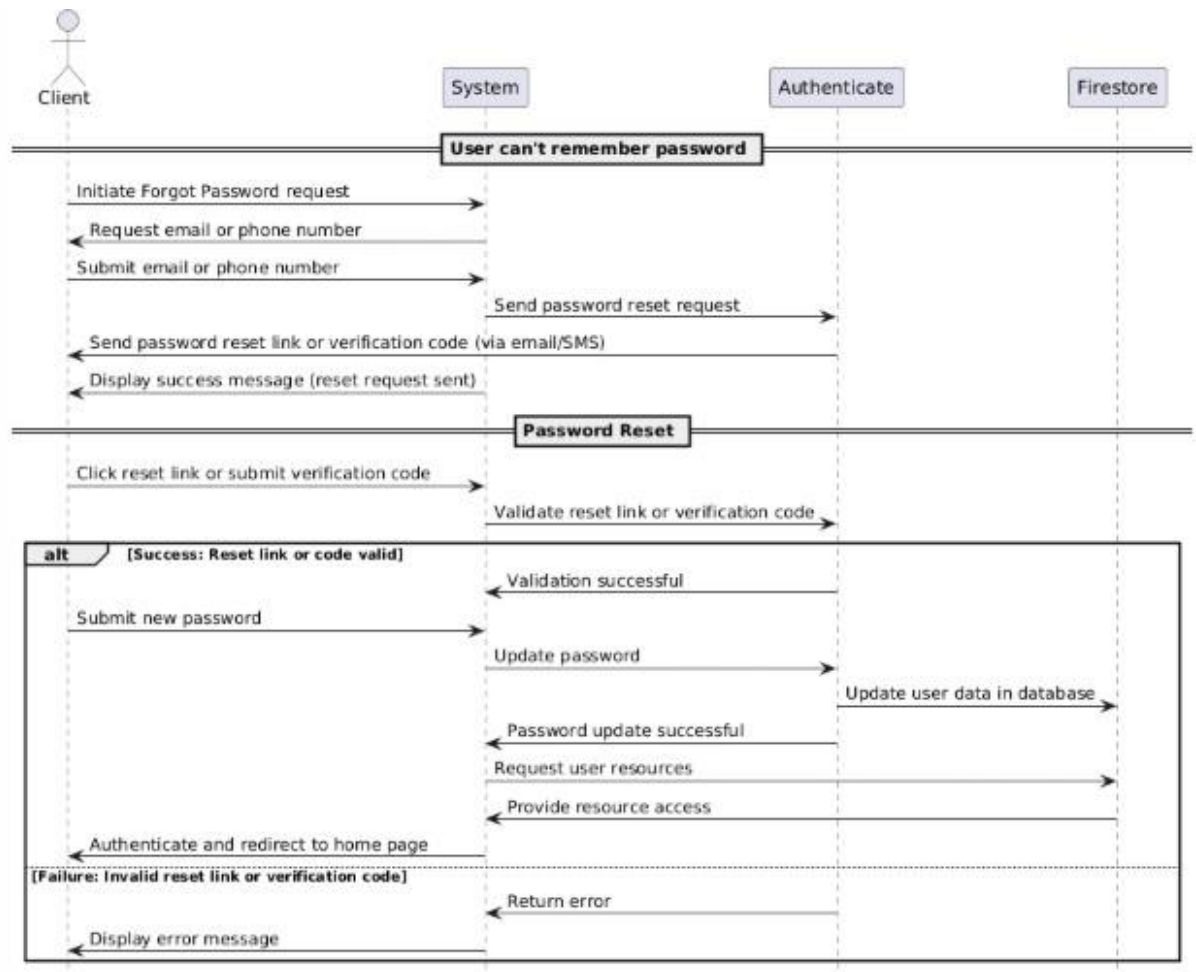
2.1.6. verify email



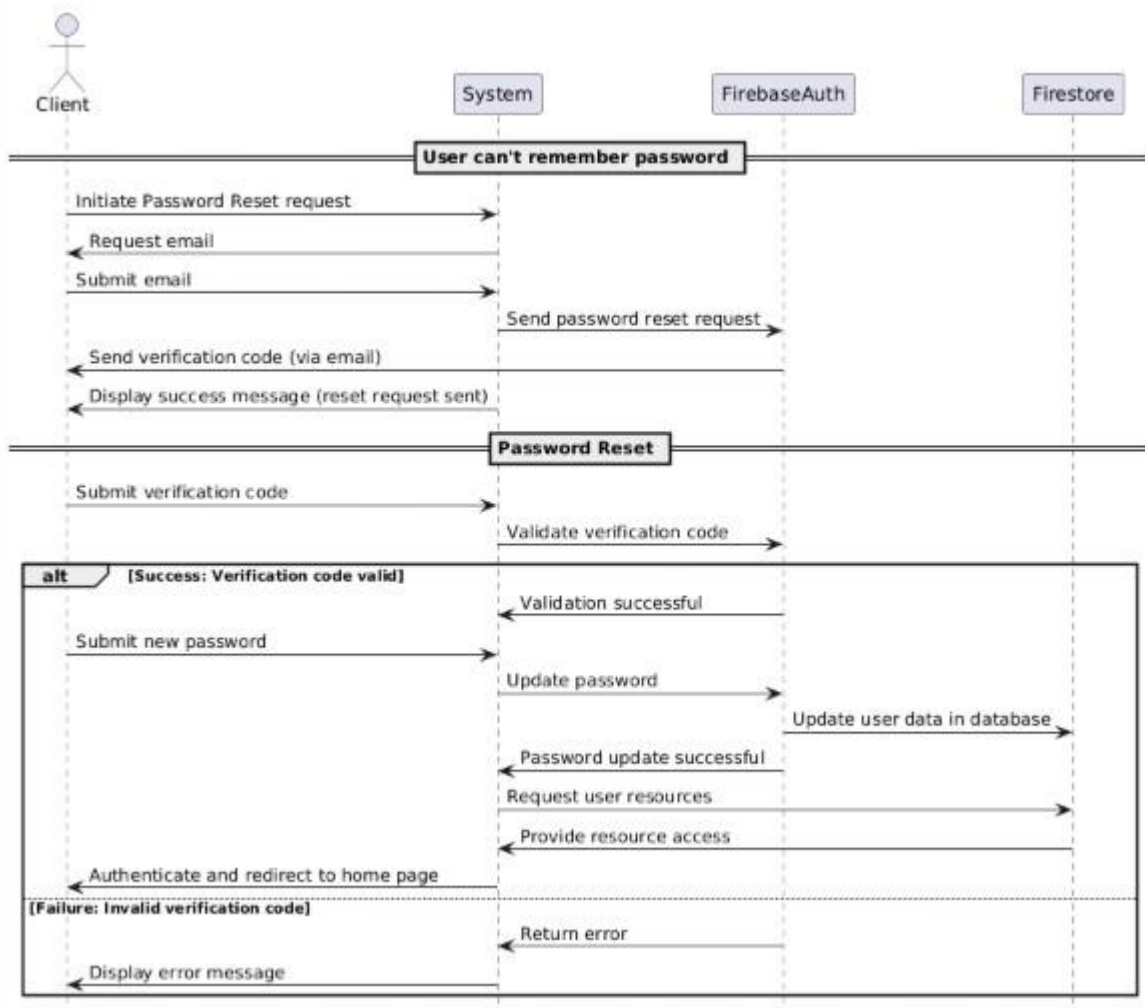
2.1.7. verify phone number



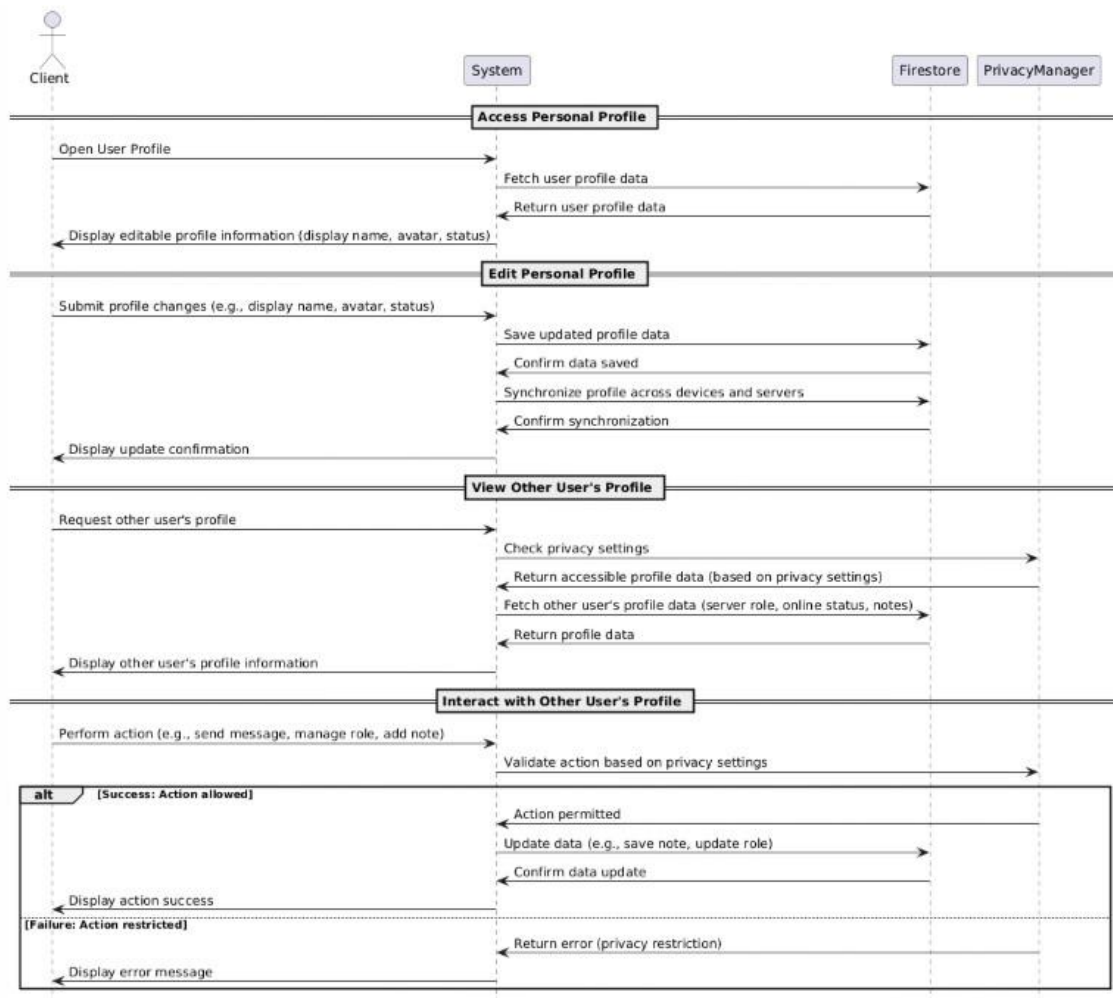
2.1.8. Forgot Password



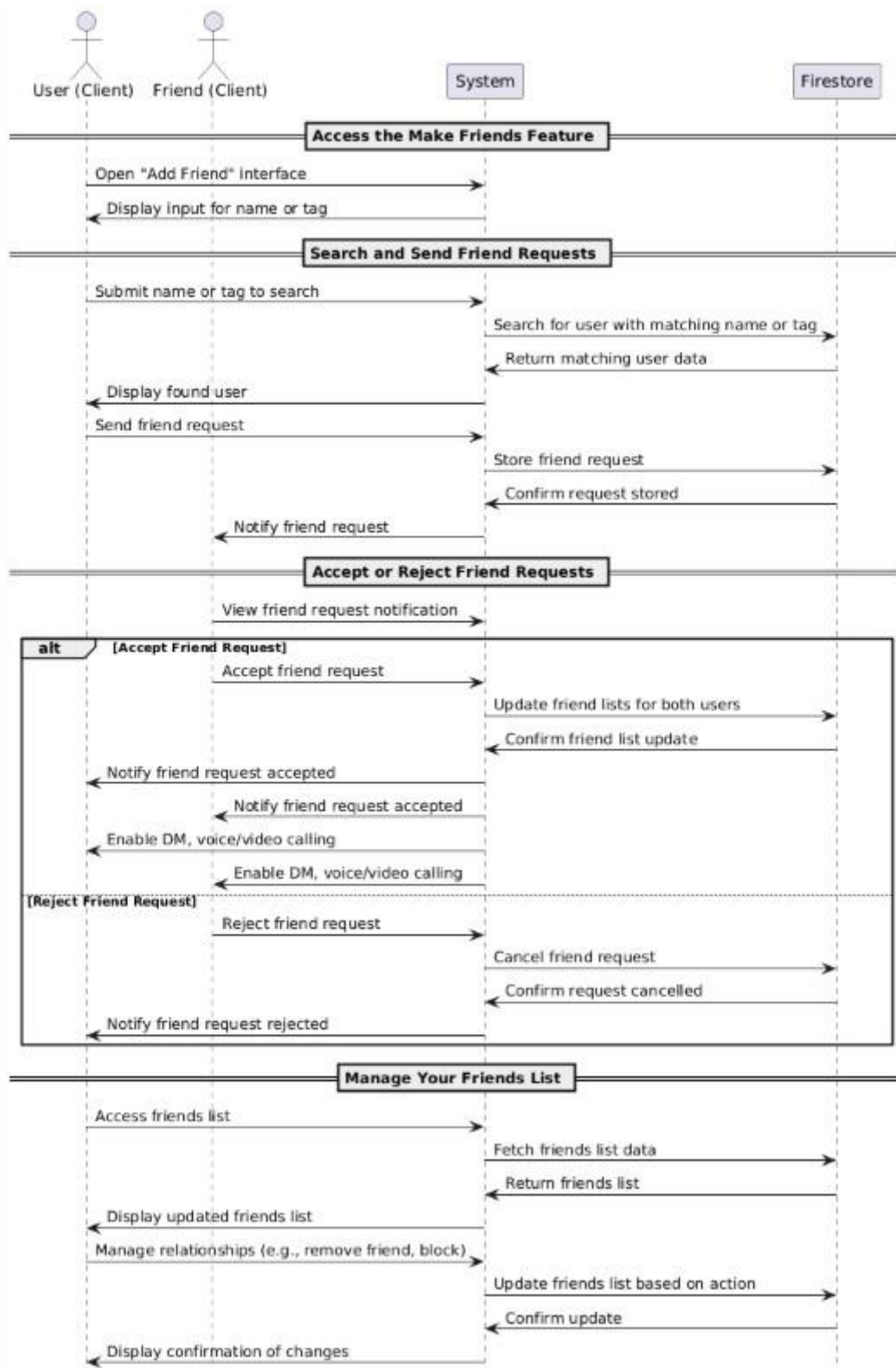
2.1.9. change password



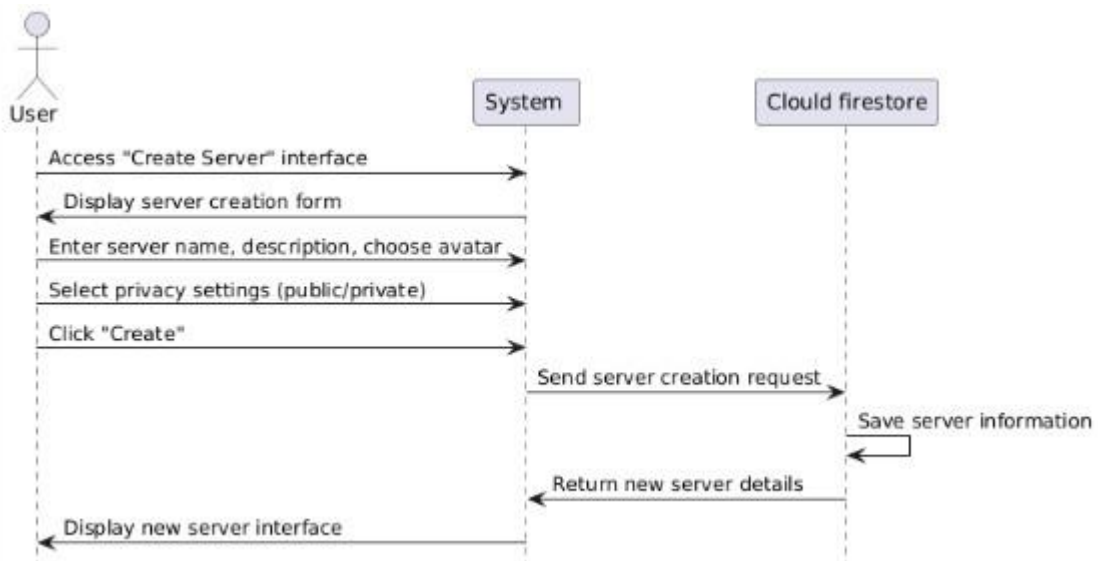
2.1.10. User Profile



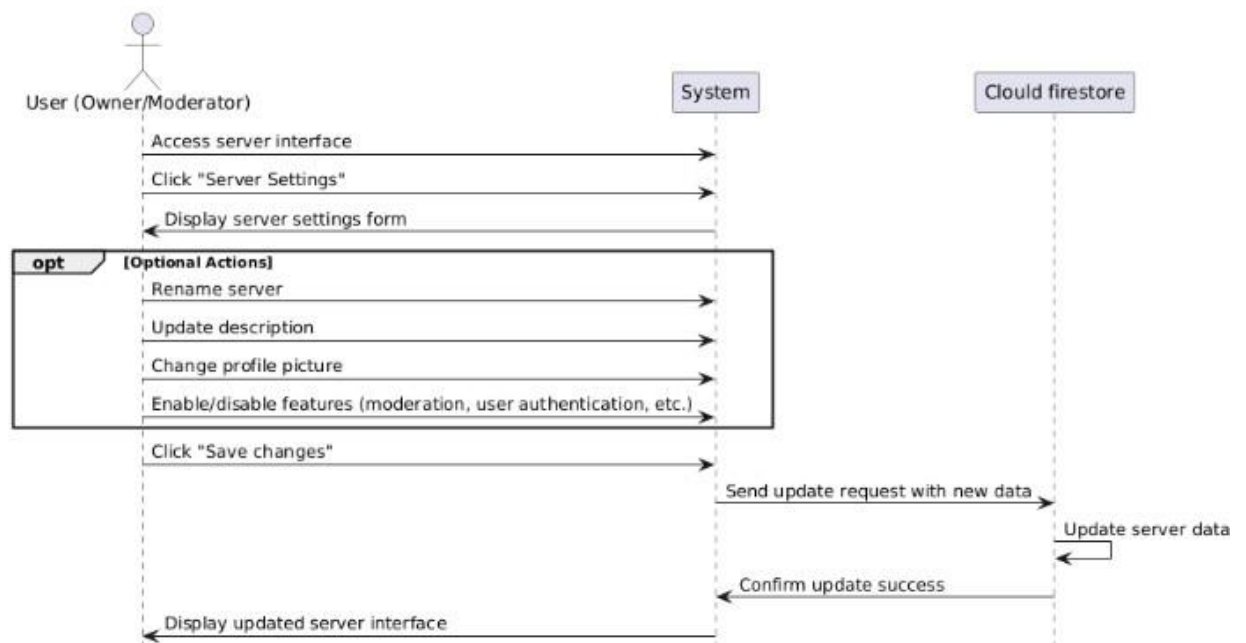
2.1.11. Make friends



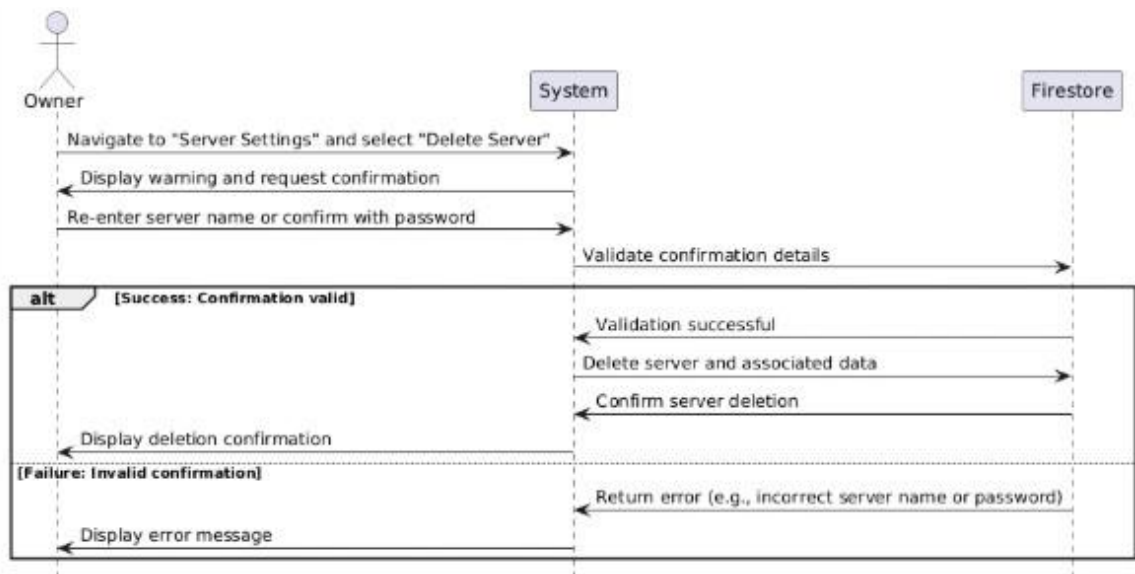
2.1.12. create server (Owner)



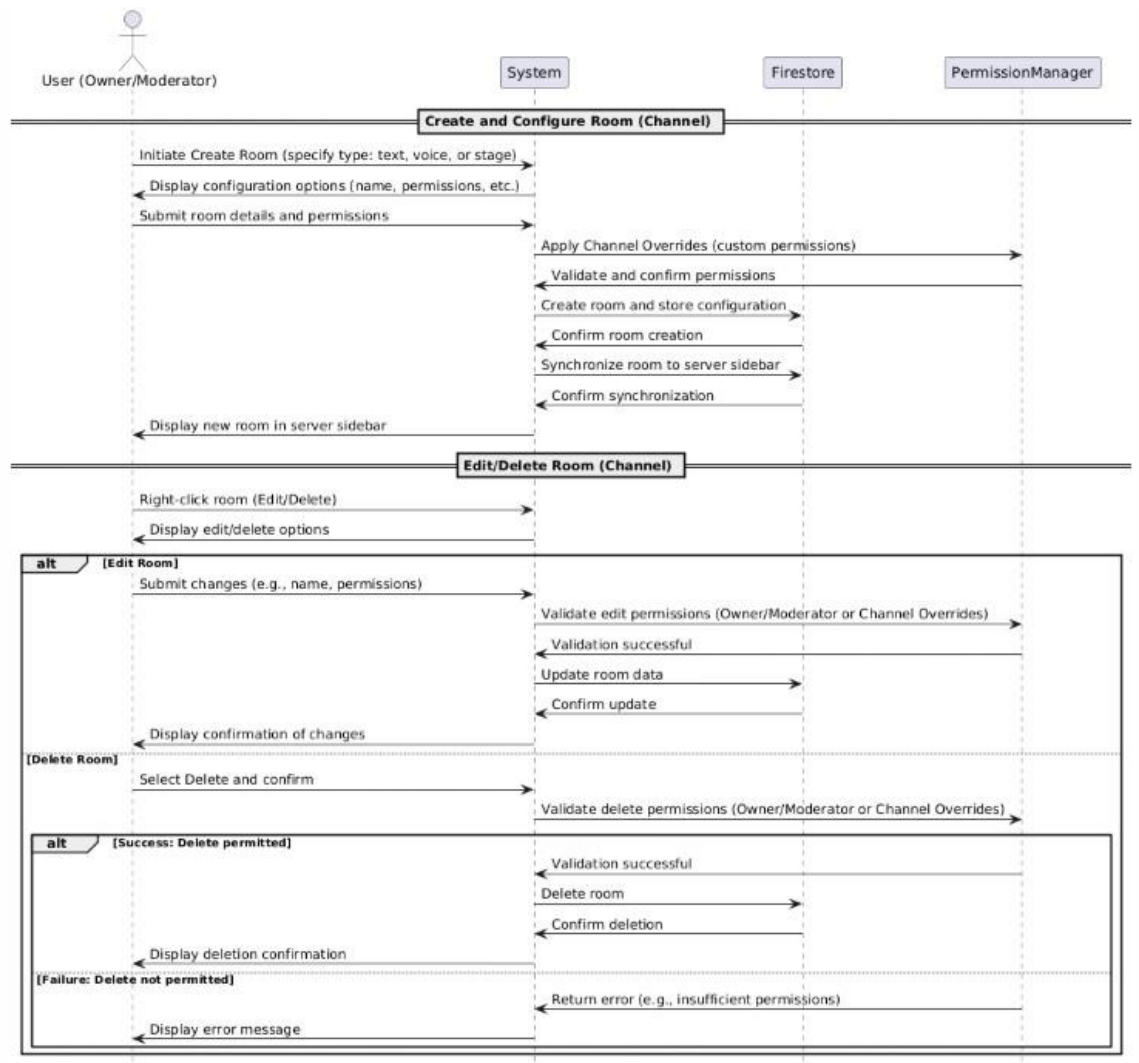
2.1.13. update server



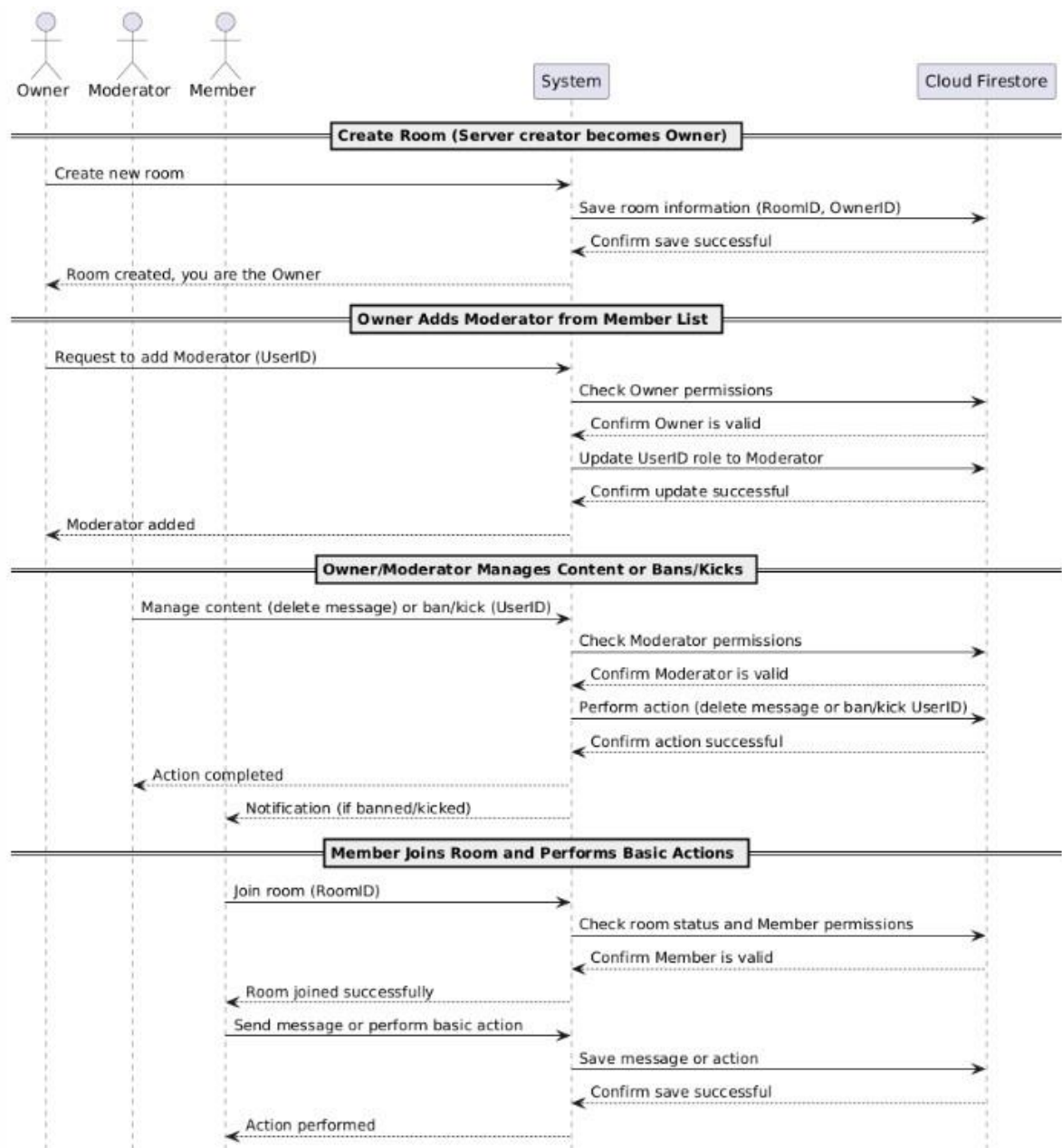
2.1.1. delete server



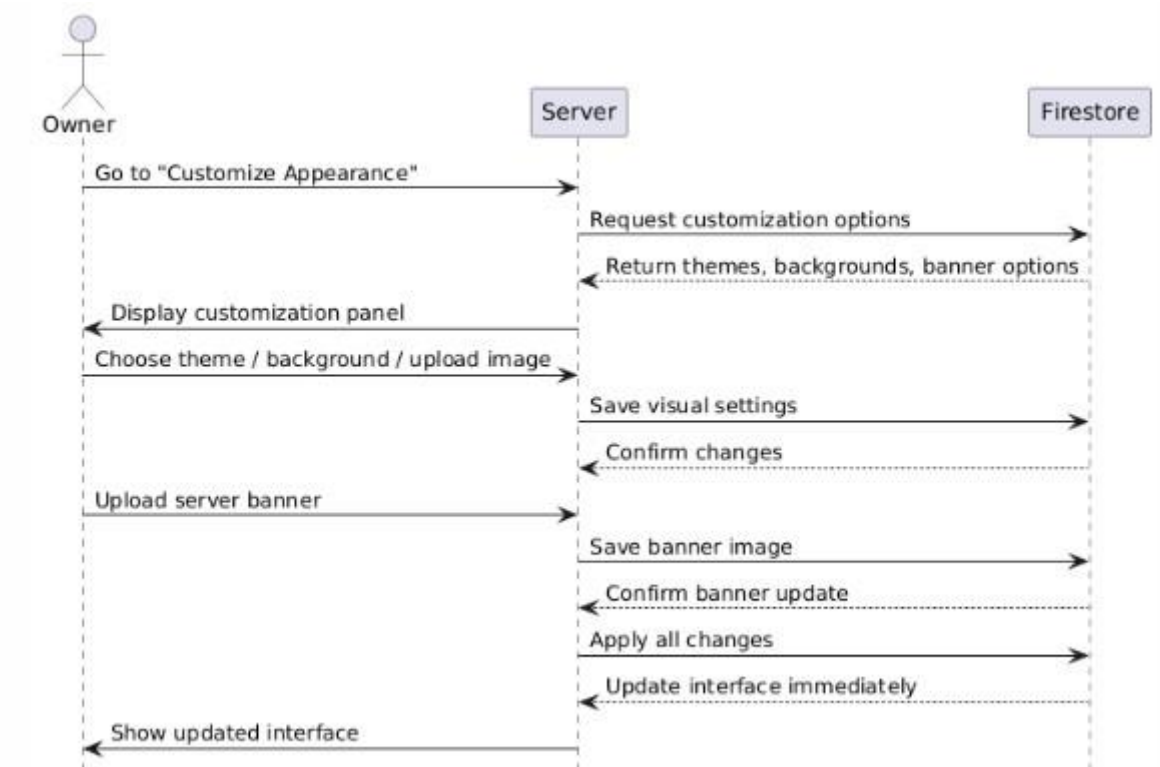
2.1.2. Server & Room Management



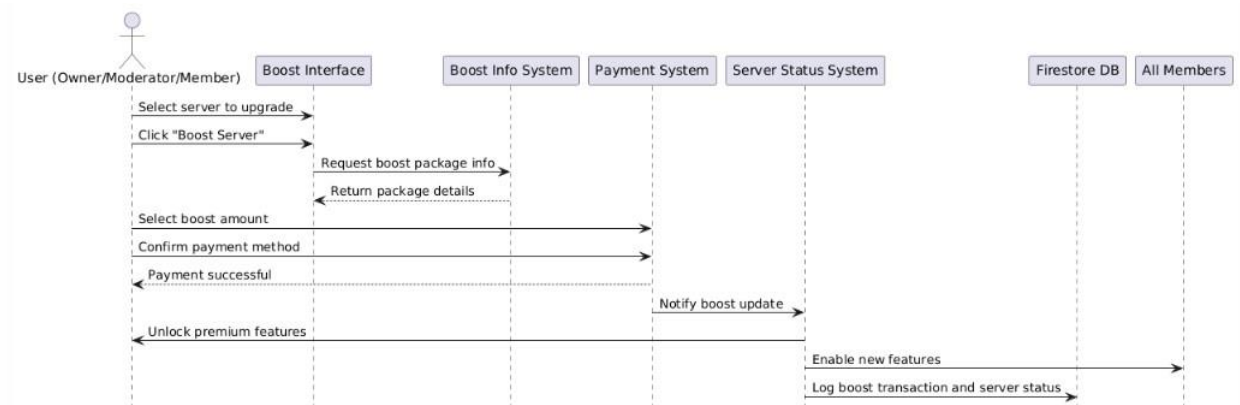
2.1.3. Rights and roles (room)



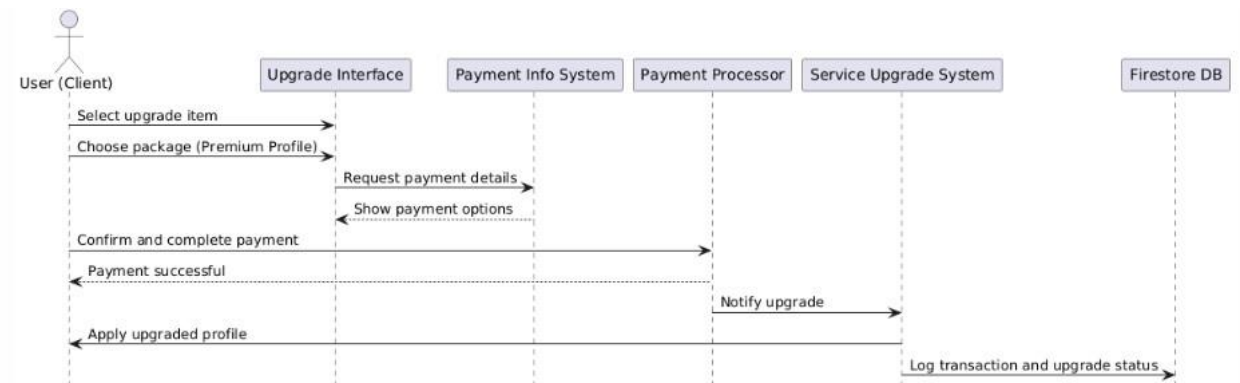
2.1.4. Customize the interface (server)



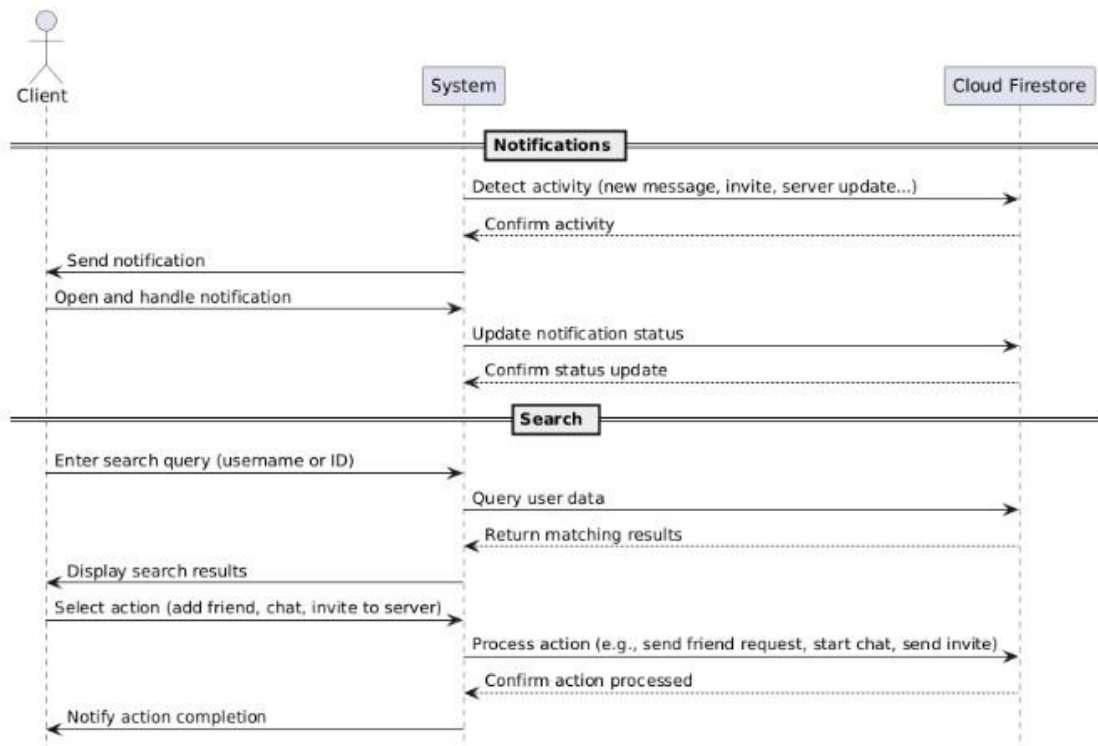
2.1.5. Server Boost



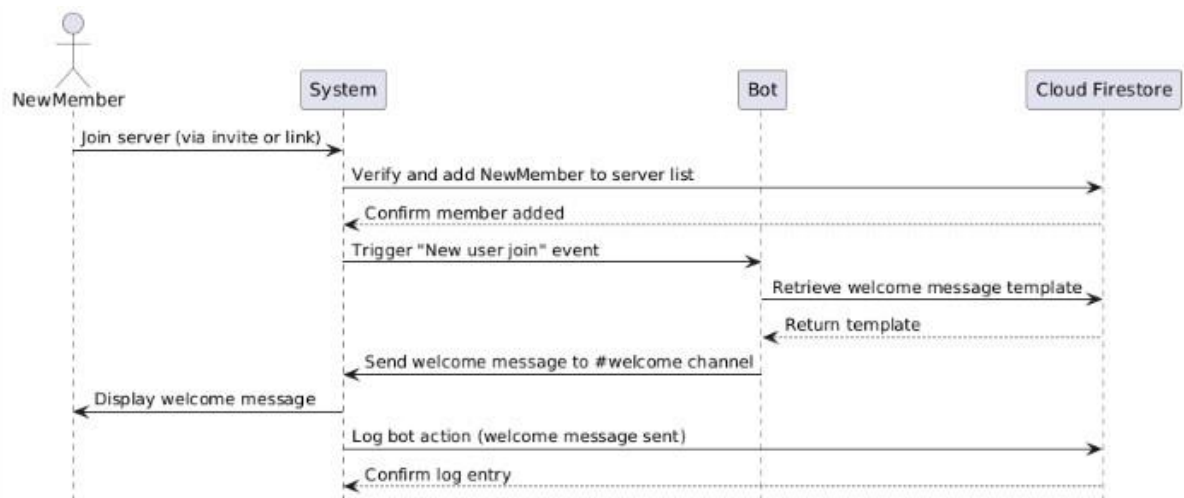
2.1.6. Premium



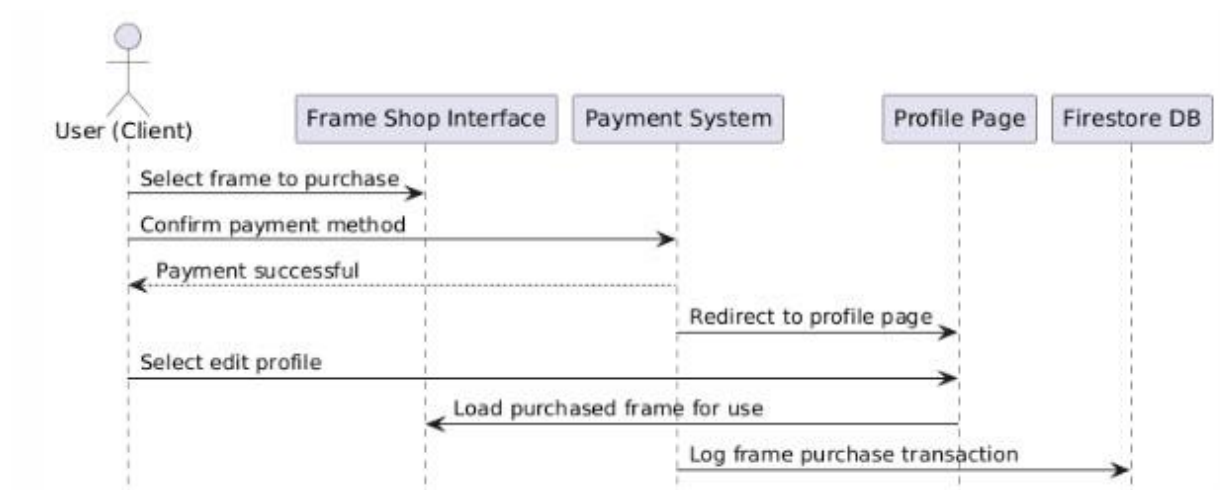
2.1.7. Notifications and Search



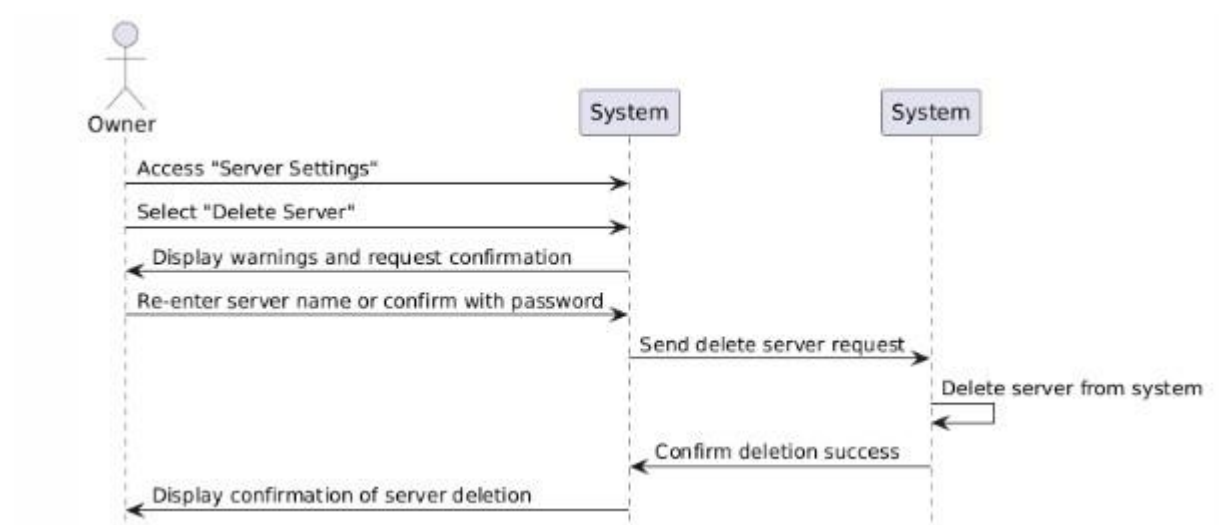
2.1.8. Auto message



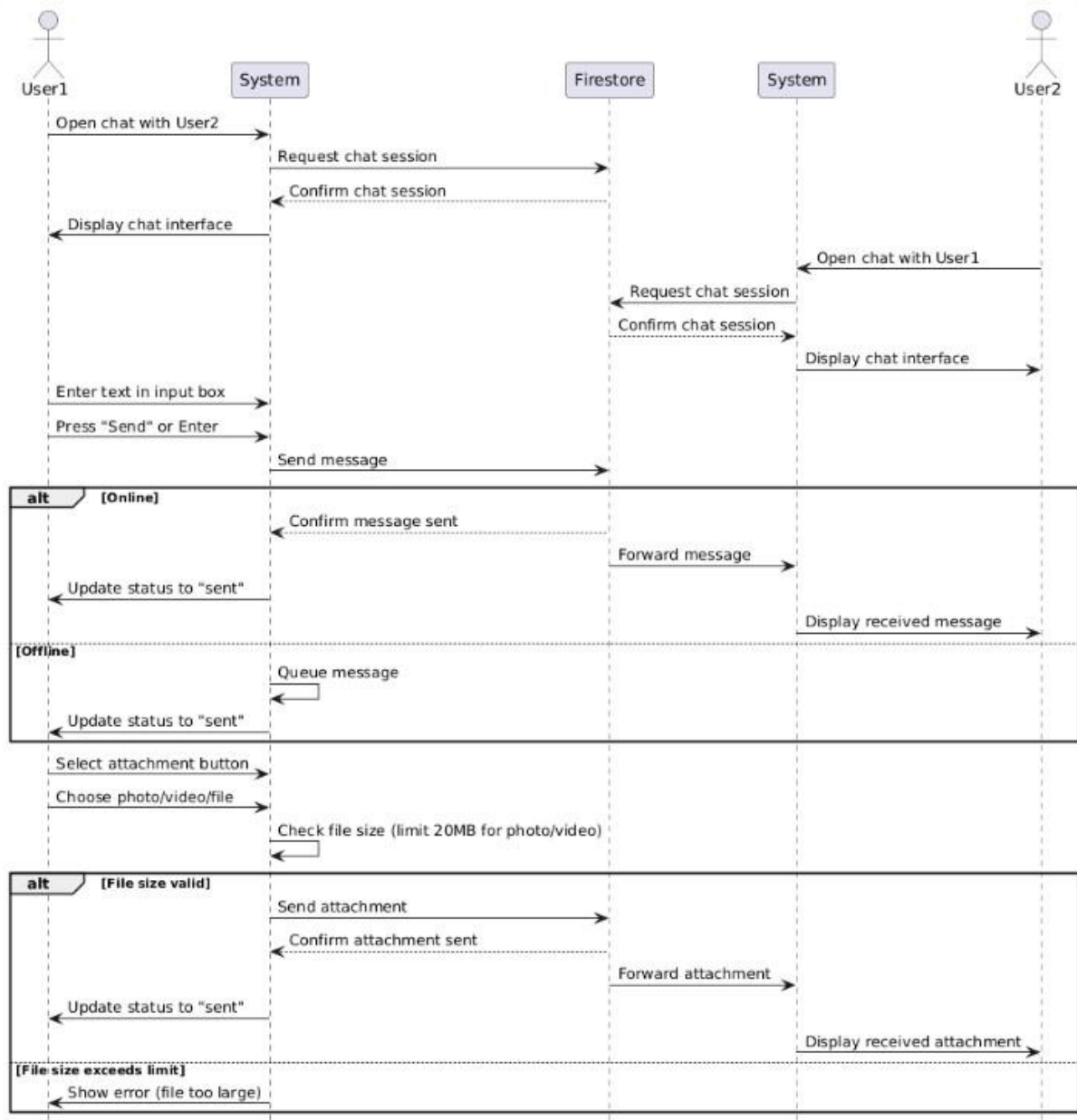
2.1.9. Payment Frames



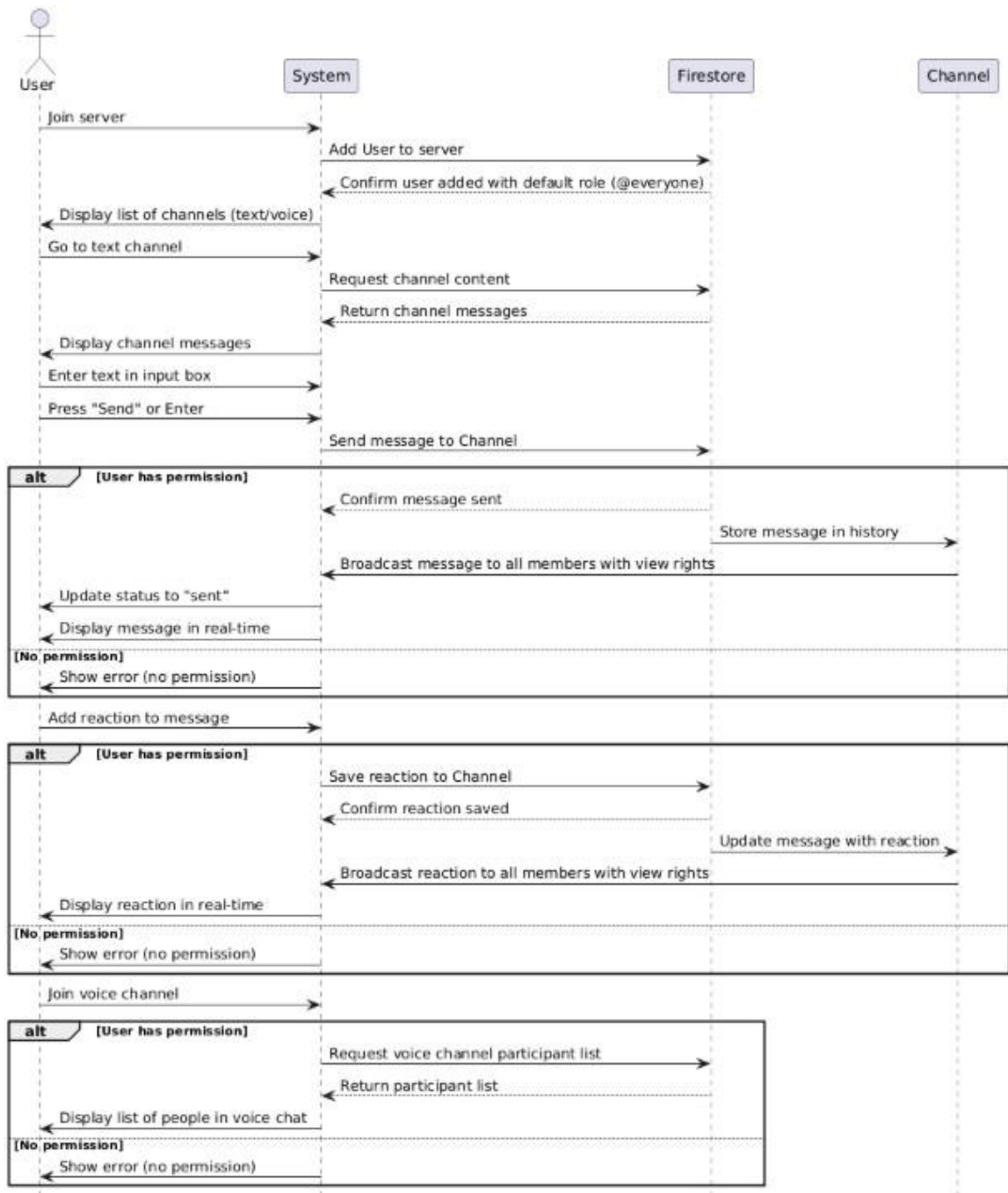
2.1.10. Delete server



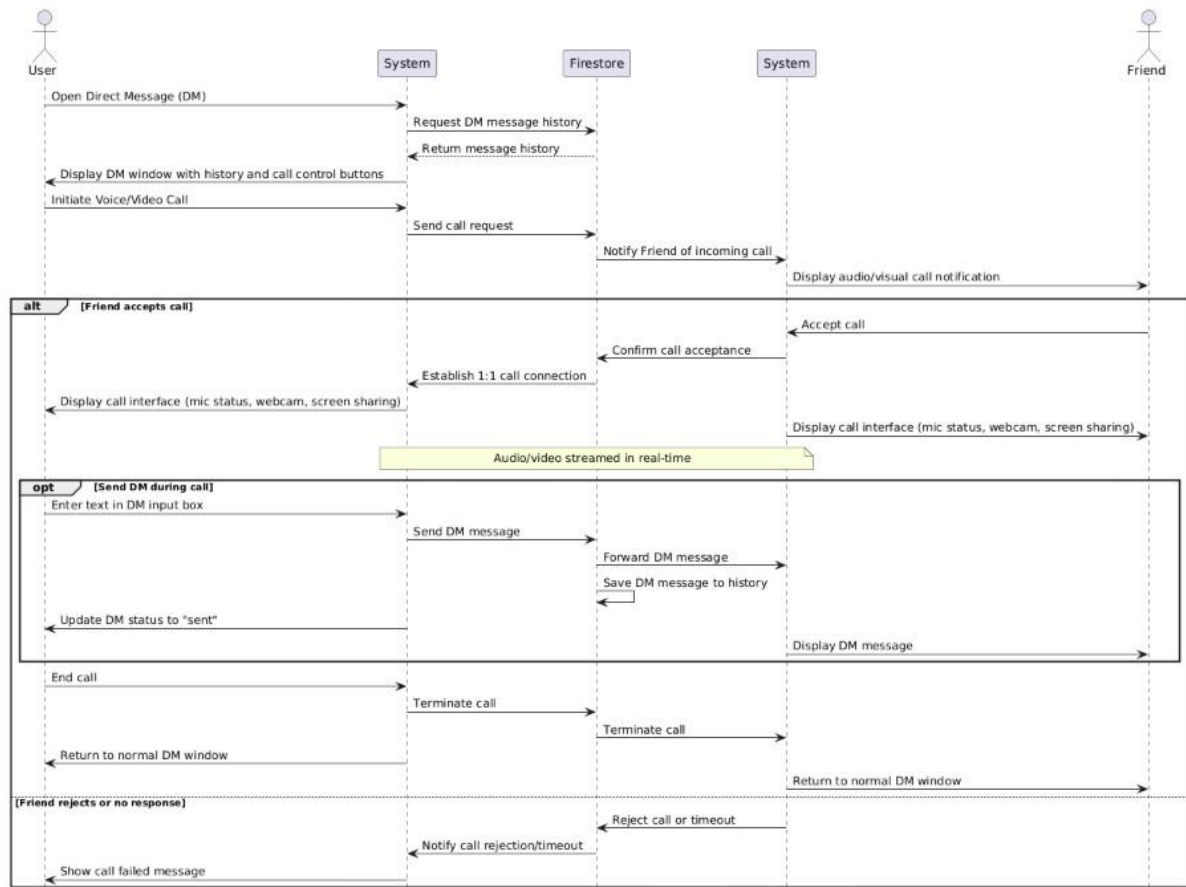
2.1.11. Message



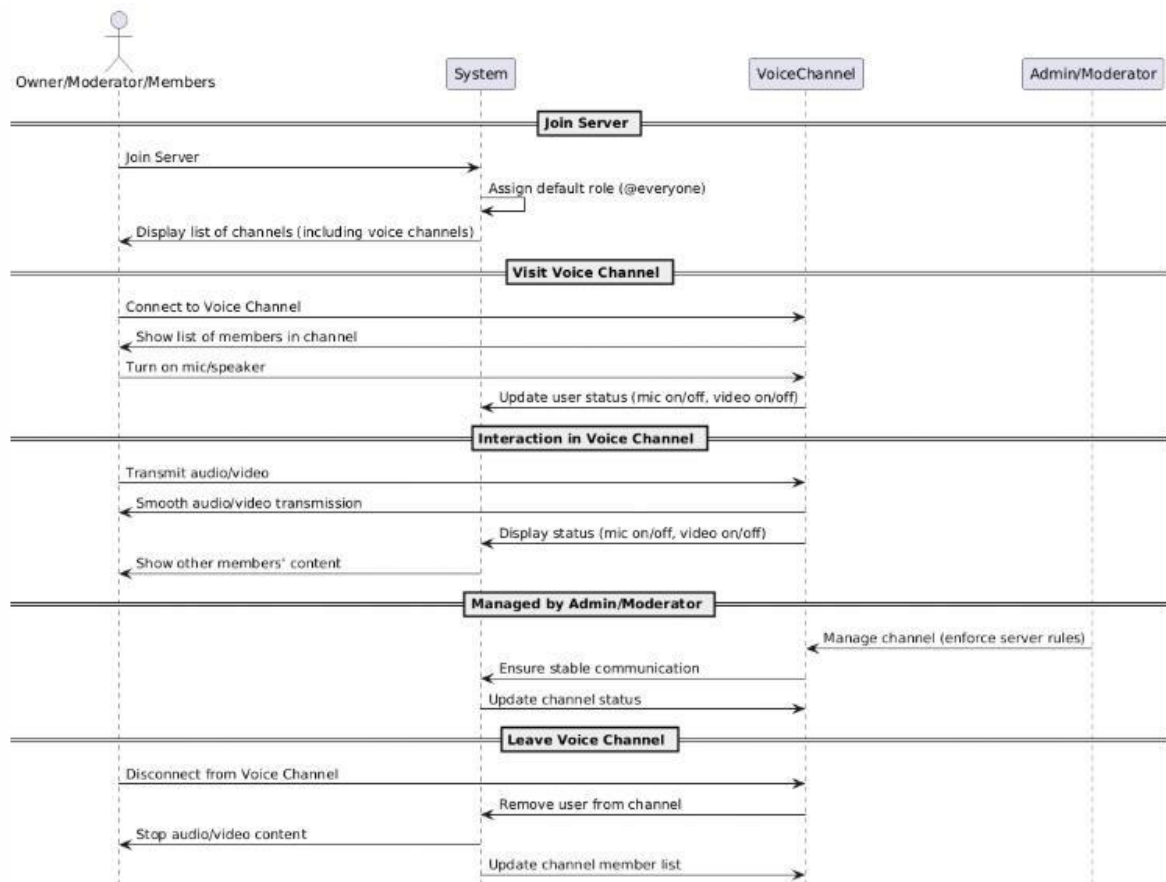
2.1.12.Channel Message (server)



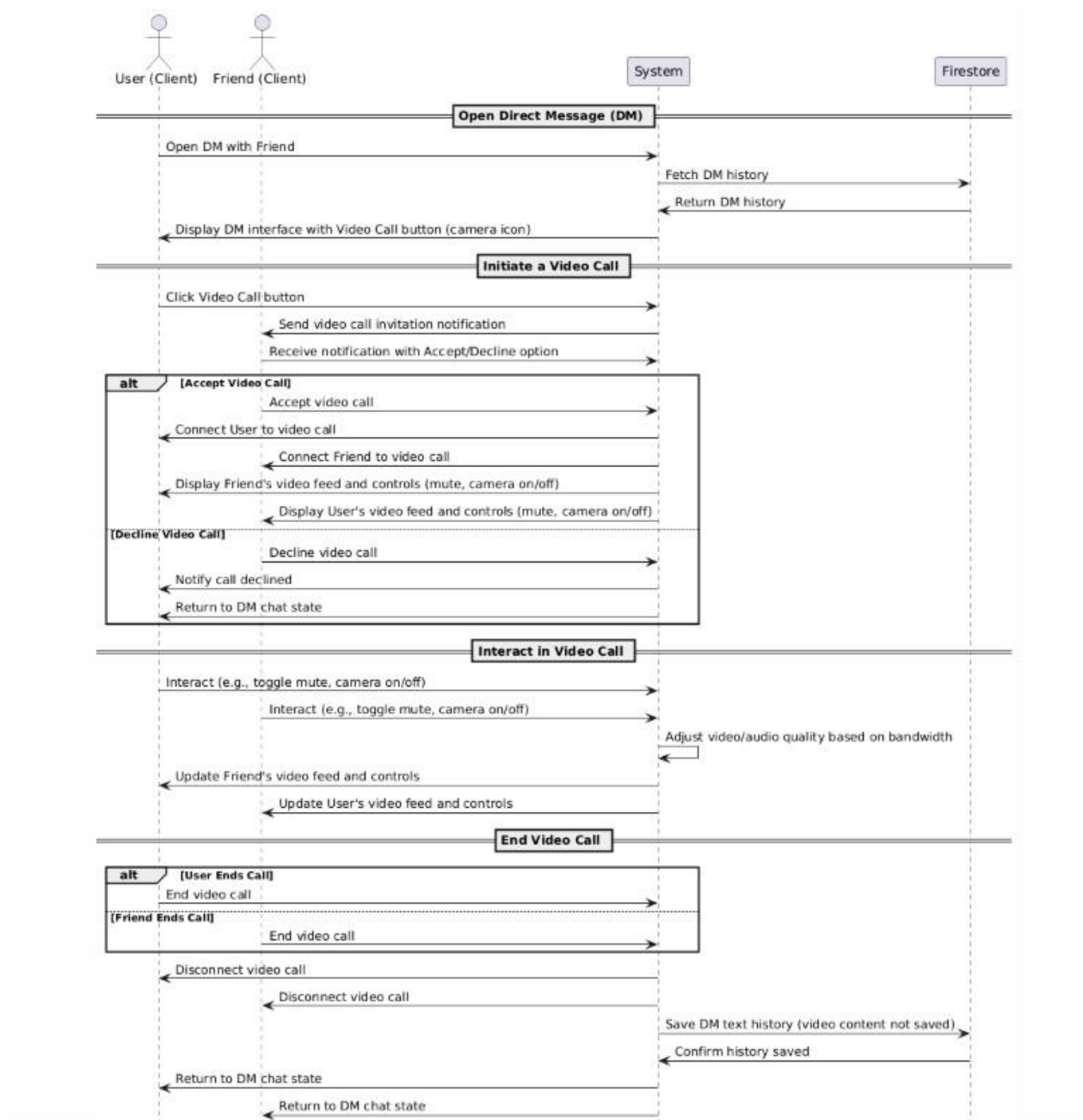
2.1.13. Voice/ Call (with friends)



2.1.14. Voice/ Call (channel/ server)



2.1.15. Video call with friend



2.2. Database design

2.2.1. Database table design

2.2.1.1. Table: Users

- Description: Stores User information • List of schools:

STT	Field	Type	Description
1	displayName	string	Display Name
2	email	string	Email or Phone Number
3	avatarUrl	string	Avatar Link
4	coverUrl	string	Profile Cover Photo
5	profileType	string	basic / premium
6	serversJoined	array	Server List Joined
7	settings	map	Personal Settings
8	isVerified	boolean	Verified
9	About me	string	User bio
10	Authmethod	string	Login Method (email, phone, Google...)
11	createdAt	datetime	Account Creation Date
12	dateOfBirth	date	Date of Birth
13	isEmailVerified	boolean	Email Verified
14	isOnline	boolean	Online Status
15	isPhoneVerified	boolean	Phone Verified
16	lastActive	datetime	Last Time User Was Active
17	lastSeen	datetime	Last Seen Time
18	phoneNumber	String	User's Phone Number

19	Pronouns	String	User's Preferred Pronouns
20	seletedFrameId	String	Selected Profile Frame ID
21	seletedFrameImageUrl	String	Profile Frame Image URL
22	seletedFrametile	string	Title/Name of Selected Frame
23	ServerJoined	array	List of Joined Servers (<i>duplicate?</i>)
24	Status	String	Custom Status or Availability
25	Username	String	Unique Username

2.2.1.2. Table: server •

Description: Stores Server information

- List of schools:

STT	Field	Type	Description
1	name	string	Server Name
2	ownerId	string	Server Owner ID
3	description	string	Description
4	logoServerUrl bannerUrl	string string	Avatar
5	isVerified	boolean	Cover Photo
6	memberIds	number	Verified
7	uploadAt	number	Membership Limits
8	Tags	array	Capacity Limits
9	channelIds	array	List of Channel IDs in the Server
10	createdAt	datetime	Server Creation Date
11	DisplayName	String	Public Display Name of Server
12	iconUrl	String	Alternative Icon URL (if different)

13	inviteCode	string	Server Invite Code
14	Ispublic	Boolean	Whether Server is Public or Private
15	Roles	array	List of Roles in Server
16	userRoles	map	Mapping of User ID to Their Role(s)

2.2.1.3. Table: Channel

- Description: Stores Channel information
- List of schools:

STT	Field	Type	Description
1	categoryId	string	Group ID
2	name	string	Channel Name
3	type	string	Text / Voice / Video
4	createdAt	timestamp	Time When Channel Was Created
5	Description	string	Description of the Channel
6	Icon	string	URL or Icon Name for the Channel
7	isPrivate	boolean	Whether the Channel is Private
8	Position	number	Position/Order in the Channel List
9	ServerId	string	ID of the Server This Channel Belongs To
10	updatedAt	timestamp	Time When Channel Was Last Updated

2.2.1.4. Table: MessageChannel

- Description: Stores MessageChannel information
- List of schools:

STT	Field	Type	Description
1	channelId	string	Channel ID
2	senderId	string	Sender
3	type	string	text / image / file
4	content	string	Content
5	timestamp	timestamp	Time
6	isDeleted	boolean	Deleted
7	channelName	string	Name of the Channel (optional, for reference)
8	isRead	boolean	Whether the Message Has Been Read

9	MessageId	string	Unique ID of the Message
10	messagePreview	string	Short Preview or Summary of the Message
11	recipientIds	array	List of Recipient User IDs (for private/group)
12	senderName	string	Display Name of the Sender
13	SenderId	string	ID of the User Who Sent the Message
14	senderName	string	Display Name of the Sender
15	ServerId	string	ID of the Server (if applicable)
16	ServerName	string	Name of the Server (optional, for reference)

2.2.1.5. Table: Category •

Description: Stores Category information

- List of schools:

STT	Field	Type	Description
1	serverId	string	Server ID
2	name	string	Group Name

2.2.1.6. Table: FriendResquest •

Description: Stores FriendRequest information

- List of schools:

STT	Field	Type	Description
1	fromUserId	string	Sender
2	toUserId	string	Recipient
3	status	string	pending / accepted / rejected
4	sentAt	timestamp	Sending time
5	Timestamp	Timestamp	Last Updated Time (optional, or can be removed if same as sentAt)
6	reponseType	string	Type of Response (e.g., manual, auto, timeout)

2.2.1.7. Table: FriendMessage •

Description: Stores FriendMessage information •

- List of schools:

STT	Field	Type	Description
1	fromUserId	string	Sender
2	toUserId	string	Recipient
3	type	string	text / image / video / file
4	content	string	Content
5	timestamp	timestamp	Time sent
6	isDeleted	boolean	Deleted

2.2.1.8. Table: Roles

- Description: Stores information of Roles
- List of schools:

STT	Field	Type	Description
1	name	string	Role Name
2	permissions	array	Permission List

2.2.1.9. Table: MemberServer

- Description: Stores MemberServer information • List of schools:

STT	Field	Type	Description
1	serverId	string	Server ID
2	userId	string	User ID
3	roles	array	Role List
4	joinedAt	timestamp	Join Time

2.2.1.10. Table: ServerBoost

- Description: Stores ServerBoost information • List of schools:

STT	Field	Type	Description
1	serverId	string	Boosted Server
2	userId	string	Booster
3	timestamp	timestamp	Time
4	Amount	number	Boost Amount / Price
5	autoRenew	boolean	Whether the Boost Will Auto Renew
6	cancelReason	string	Reason for Cancelling Boost (if any)

7	cancelledAt	timestamp	Time When Boost Was Cancelled
8	createAt	timestamp	Time When Boost Record Was Created
9	Currency	string	Currency Code (e.g., USD, VND)
10	Duration	string	Boost Duration (e.g., 1 month, 3 months)
11	EndDate	timestamp	Boost Expiration Date
12	Matadata	map	Additional Metadata
13	Status	string	Boost Status (active, cancelled, expired...)
14	Tier	string	Boost Tier (e.g., Tier 1, Tier 2, etc.)
15	updateAt	timestamp	Last Time Boost Info Was Updated

2.2.1.11. Table: call friends •

Description: Stores call friends information

- List of schools:

STT	Field	Type	Description
1	callerId	string	Caller
2	receiverId	string	Recipient
3	type	string	Voice / Video
4	startedAt	timestamp	Start
5	endedAt	timestamp	End
6	status	string	Missed / Completed / Rejected

2.2.1.12. Table: PaymentFrames • Description:

Stores information of PaymentFrames

- List of schools:

STT	Field	Type	Description
1	userId	string	Performer
2	type	string	profile / server
3	amount	number	Amount
4	createdAt	timestamp	Time
5	completedAt	timestamp	Time When Transaction Was Completed
6	Currency	String	Currency Used (e.g., VND, USD)
7	Description	String	Description of the Payment
8	errorMessage	string	Error Message (if Payment Failed)

9	expiresAt	Timestamp	Expiration Time (if applicable)
11	Id	String	Unique Transaction ID
12	Metadata	Map	Additional Metadata Related to Payment
13	momoRequestId	String	Momo Request ID (if using Momo Gateway)
14	orderId	String	Merchant Order ID
15	paymentMethod	String	Method Used (e.g., Momo, ZaloPay, CreditCard)
16	paymnetUrl	String	Redirect URL for Payment (if applicable)
17	Status	String	Payment Status (e.g., pending, success, failed)
18	stransactionId	String	ID Returned by Payment Gateway
19	updateAt	timestamp	Last Time Transaction Was Updated

2.2.1.13. Table: automessage •

Description: Stores information of Bots

- List of schools:

STT	Field	Type	Description
1	name	string	Bot Name
2	trigger	string	Bot Call Keyword
3	action	string	Response Content

2.2.1.14. Table: serverbans

- Description: Stores information of serverbans
- List of schools:

STT	Field	Type	Description
1	bannedAt	timestamp	Time When the User Was Banned
2	bannedByUserId	string	ID of the User Who Issued the Ban
3	bannerUserId	string	ID of the User Who Was Banned
4	expiresAt	timestamp	Time When the Ban Expires (if temporary)
5	isActive	boolean	Whether the Ban is Still Active

6	Reason	String	Reason for the Ban
7	ServerId	String	ID of the Server Where the Ban Was Issued

2.2.1.15. Table: joinRequest

- Description: Stores information of joinRequest
- List of schools:

STT	Field	Type	Description
1	createdAt	timestamp	Time When the Report or Feedback Was Created
2	Message	string	Message Content Submitted by the User
3	ServerId	string	ID of the Server (if feedback is serverrelated)
4	Status	string	Status of the Report: pending, resolved, etc.
5	userId	string	ID of the User Who Submitted the Feedback

2.2.1.16. Table: friendsRequest

- Description: Stores information of friendsRequest
- List of schools:

STT	Field	Type	Description
1	createdAt	timestamp	Time When the Friend Request Was Created
2	fromUserAvatar	string	Avatar URL of the Sender
3	fromUserId	String	ID of the User Who Sent the Request
4	fromUserName	String	Username or Display Name of the Sender
5	respondedAt	Timestamp	Time When the Request Was Responded To
6	Status	String	Status
7	toUserAvatar	String	Avatar URL of the Receiver
8	toUserId	String	ID of the User Who Received the Request

9	toUserName	string	Username or Display Name of the Receiver
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2.2.1.17. Table: direct_message

- Description: Stores information of direct_message
- List of schools:

STT	Field	Type	Description
1	Content	String	Message content
2	conversationId	string	ID of the conversation this message belongs to
3	deliveredAt	timestamp	Timestamp when the message was delivered to the recipient(s)
4	editedAt	timestamp	Timestamp when the message was last edited
5	isDelete	boolean	Whether the message has been deleted
6	Metadata	map / object	Extra data like attachments, reactions, or custom info
7	readAt	timestamp	Timestamp when the message was read (if individual message view)
8	Readby	array	List of user IDs who have read the message
9	receiverId	string	ID of the recipient (for 1-on-1 messages)
10	replytoId	string	ID of the message being replied to
11	senderId	string	ID of the sender
12	Timestamp	timestamp	Time when the message was sent
13	Type	string	Voice

2.2.1.18. Table: conversatus

- Description: Stores information of coversatus
- List of schools:

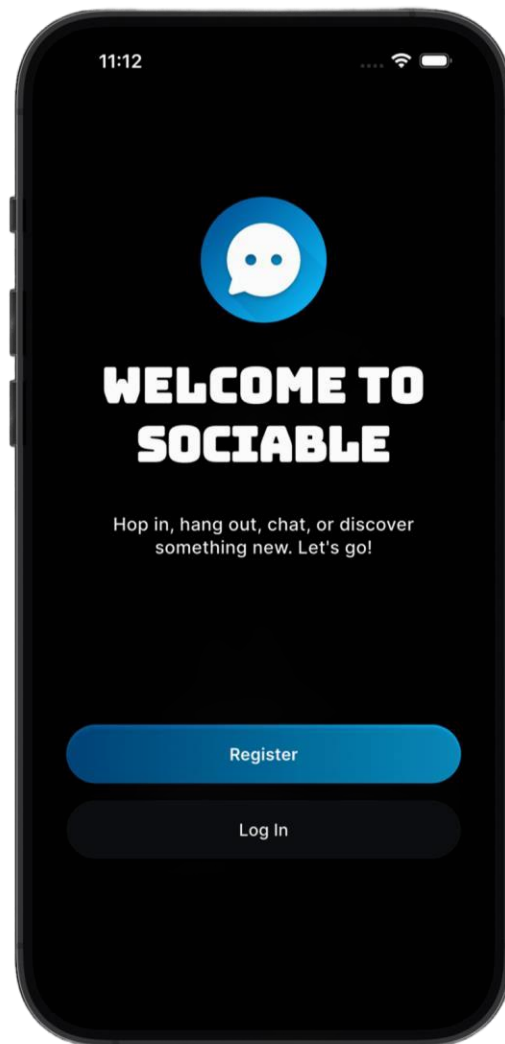
STT	Field	Type	Description
1	createdAt	timestamp	Time when the conversation was created

2	lastMessage	string	Content of the last message in the conversation
3	lastMessageSender	string	ID of the user who sent the last message
4	lastMessageTime	timestamp	Time when the last message was sent
5	lastReadTimestamp	timestamp	Last time the current user read the conversation
6	Participants	array	List of user IDs involved in the conversation
7	Type	string	Conversation type: private, group, server
8	unreadCount	number	Number of unread messages
9	updatedAt	timestamp	Time when the conversation was last updated

CHAPTER 3: SYSTEM IMPLEMENTATION AND DEPLOYMENT

3.1. Application building results

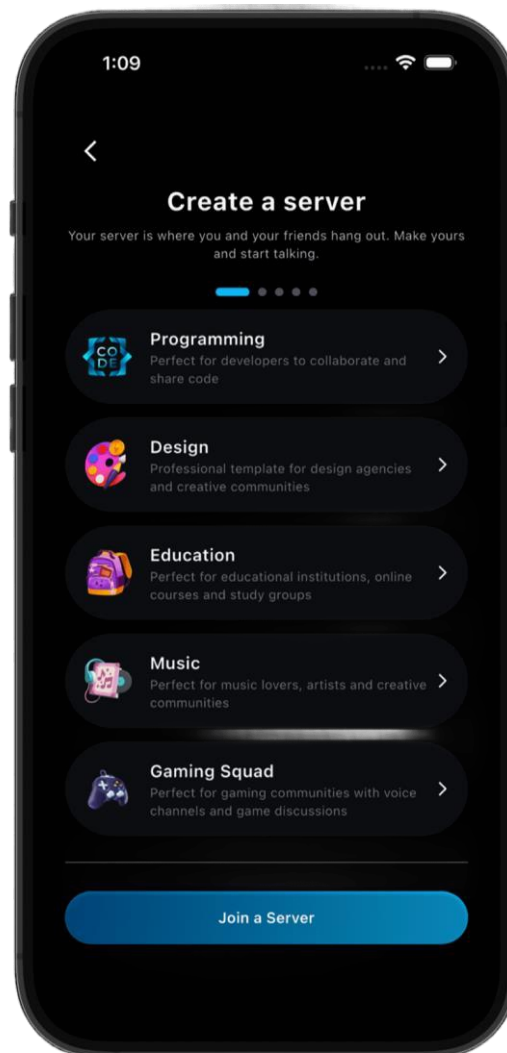
3.1.1. sociable splash screen



Describe:

- This is the Welcome screen of the Sociable app.
- Shows the logo icon in the middle with the words “WELCOME TO SOCIABLE”.
- Below is the description: “Hop in, hang out, chat, or discover something new. Let's go!”
- There are two function buttons:
- Register: Register a new account.
- Log In: Log in to the app.

3.1.2. Create a server



Describe:

This image shows the server creation interface in Sociable Users can choose the type of server they want to create such as:

- Programming – for programmers to share source code.
- Design – for design teams and creative communities.

- Education – for learning, courses, and research groups.
- Music – for music lovers and artists.
- Gaming Squad – for gaming communities with voice channels and game discussions.
- Below is the “Join a Server” button to join an existing server.

3.1.3. Profile friends



Describe:

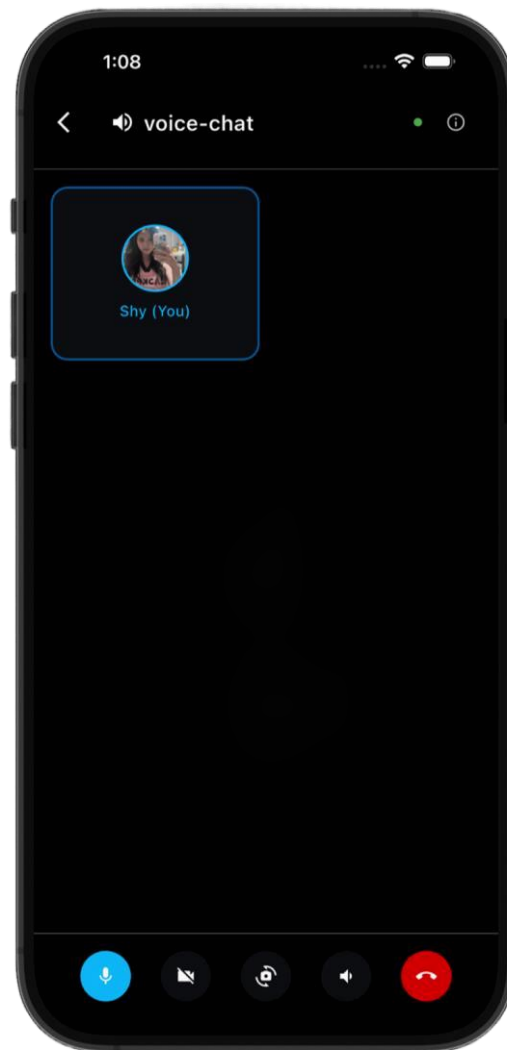
This image is the user profile interface in Discord.

Users can view account information such as:

- Avatar photo, display name (Apple), and gender tag (she/her)
- Role (Gamer) and Discord username (@shy11)

- Function buttons: Message, Voice Call, Video Call
- The Friends since section shows the date of friendship (September 19, 2025)

3.1.4. Voice channel



Describe:

This image shows the voice chat interface in Sociable.

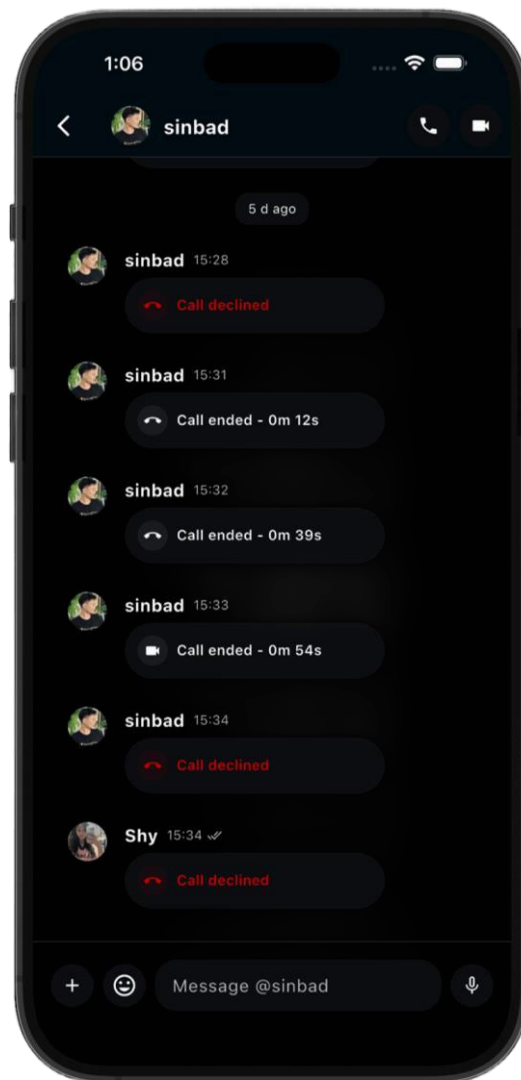
The user is participating in a voice-chat channel.

At the bottom of the screen are the function buttons:

- Mute/unmute mic

- Mute speaker/audio device
- End call

3.1.5. two-way texting screen

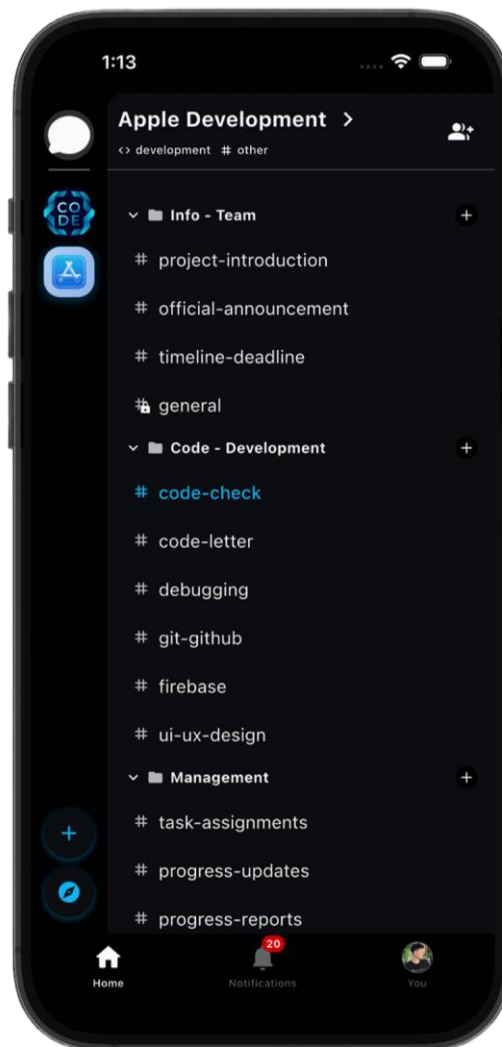


Describe:

This image shows the private messaging interface in Sociable between two users sinbad and Shy. The screen displays the call history including:

- Rejected calls show the text Call declined in red.
- Ended calls show the text Call ended with the duration such as 0m12s, 0m39s, 0m54s.
- Above is the option to make a voice or video call, and below is the message input box to send to sinbad.

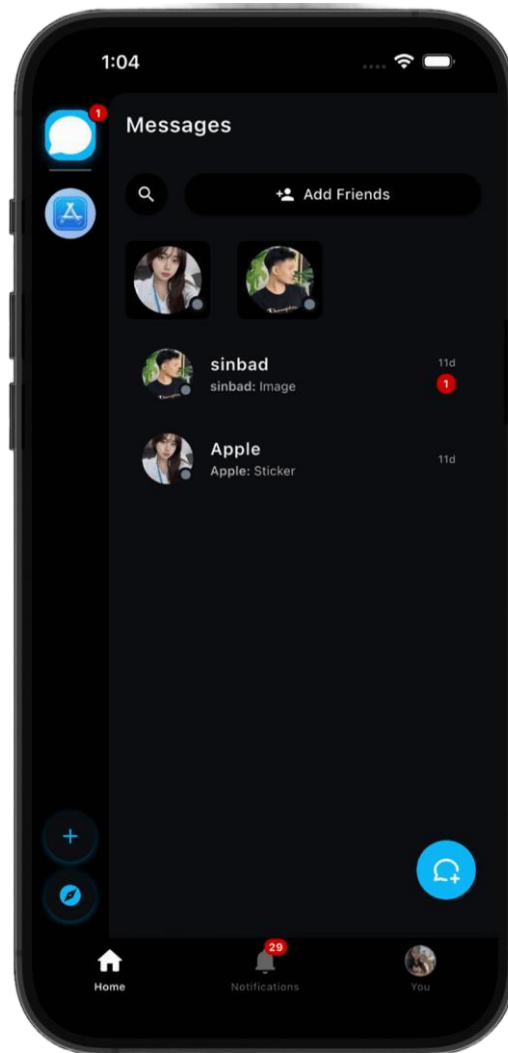
3.1.6. server screen



Describe:

Server workflow in Sociable – channels

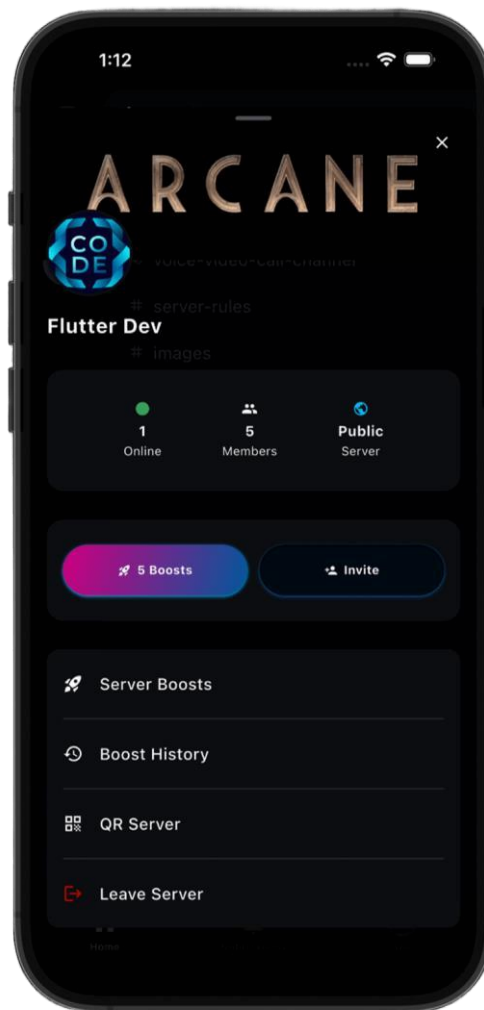
3.1.7. Chat area



Describe:

- This image is the main screen when chatting privately with friends, on the left is the server list
- The front is the screen showing the list of friends who have recently contacted
- Above is the list of active friends
- Add Friends is used to search for friends or find friends to make friends

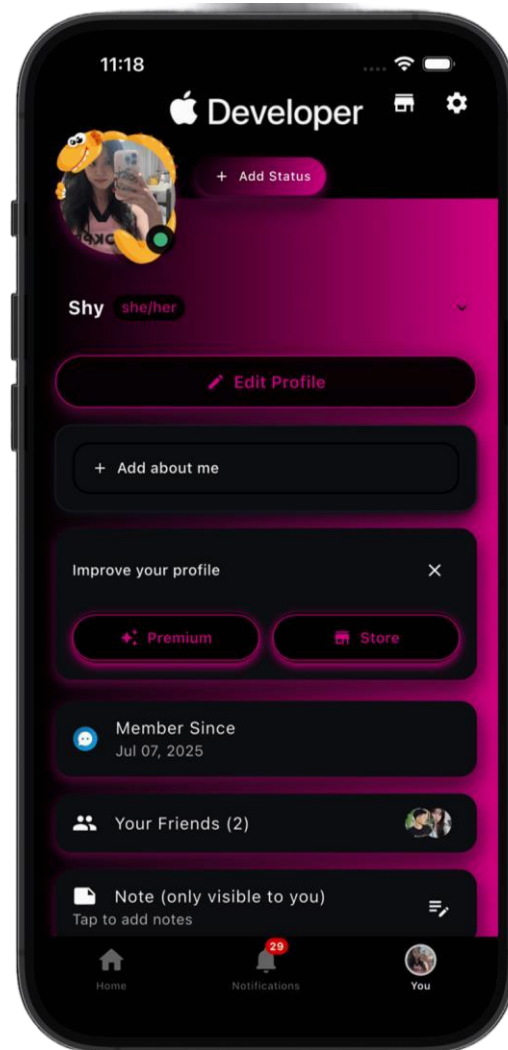
3.1.8. “Server” info



Describe:

- This image is the interface of a server including the number of active people and the number of people in the server
- Boost is used to upgrade the server to use stickers or other rights such as banners to send more files.
- Qr is used to scan to join the server and leave Server is used to exit the server

3.1.9. User profile



Describe:

- Shows the user profile page.
- Has a profile picture, username (Shy) and pronouns (she/her).
- Allows adding statuses with the “+ Add Status” button.
- Has an Edit Profile button to edit information.
- Add about me to add a personal description.

- Improve your profile with Premium and Store buttons.
- Shows the date of joining (Member Since: Jul 07, 2025).
- Shows a list of friends (Your Friends (2)).
- Has a Note section (only visible to you) to write personal notes.

CHAPTER 4: CONCLUSION AND DEVELOPMENT DIRECTIONS

4.1. App Conclusion Review

Sociable is a comprehensive communication platform that enables users to chat, make voice and video calls, and build their own communities through flexible, easy-to-manage servers and topic-based channels.

The application provides an ideal space for studying, working in groups, playing games, or socializing with friends.

Sociable aims to enhance community connection, improve communication efficiency, and foster a friendly, creative, and interactive environment for all users.

4.2. General conclusion

4.2.1. Results achieved

After the development phase, the project has successfully completed around **90% of its core features**, including:

- Conducting full system analysis and designing functional relationship diagrams.
- Building a real-time, scalable **Firestore database** architecture.
- Designing a user-friendly and mobile-optimized **Flutter UI/UX**.
- Implementing key modules: authentication, server and channel management, realtime messaging, user profile management, voice and video calling, and basic rolebased permissions.
- Ensuring stable performance across **both iOS and Android platforms**.

4.2.2. Limit

Due to limited time and resources, some **advanced and administrative features** have not yet been fully implemented, such as:

- Advanced server analytics and moderation tools.

- Detailed permission management and community statistics.
 - Extended notification and customization settings for power users.
- Despite these constraints, Sociable has achieved a highly functional and stable foundation, sufficient for real-world use and future scalability.

4.3. Development direction

Future development directions include:

- Completing remaining advanced features and improving stability through continuous testing.
- Enhancing **server management tools**, **role customization**, and **community moderation systems**.
- Expanding cross-platform support to **web and desktop**.
- Introducing new features to **increase user engagement** and **explore monetization opportunities**, such as premium memberships or exclusive content access.